CHAPTER 1. MANAGEMENT SCIENCE

Management science is the application of a scientific approach to solving management problems in order to help managers make better decisions. As implied by this definition, management science encompasses a number of mathematically oriented techniques that have either been developed within the field of management science or been adapted from other disciplines, such as the natural sciences, mathematics, statistics, and engineering.

This text provides an introduction to the techniques that make up management science and demonstrates their applications to management problems.

Management science is a scientific approach to solving management problems.

The applications of management science techniques are widespread, and they have been frequently credited with increasing the efficiency and productivity of business firms.

In various surveys of businesses, many indicate that they use management science techniques, and most rate the results to be very good.

Management science (also referred to as operations research, quantitative methods, quantitative analysis, and decision sciences) is part of the fundamental curriculum of most programs in business.

Management science can be used in a variety of organizations to solve many different types of problems including services, government, military, business and industry, and health care.

Second, in this text all of the modeling techniques and solution methods are mathematically based. In some instances the manual, mathematical solution approach is shown because it helps to understand how the modeling techniques are applied to different problems.

However, a computer solution (we use POM-QM SW) is possible for each of the modeling techniques in this text, and in many cases the computer solution is emphasized.

Finally, as the various management science techniques are presented, keep in mind that management science is more than just a collection of techniques. Management science also involves the philosophy of approaching a problem in a logical manner (i.e., a scientific approach). The logical, consistent, and systematic approach to problem solving can be as useful (and valuable) as the knowledge of the mechanics of the mathematical techniques themselves. This understanding is especially important for those readers who do not always see the immediate benefit of studying mathematically oriented disciplines such as management science.

Management science encompasses a logical approach to problem solving.

The Management Science Approach to Problem Solving:

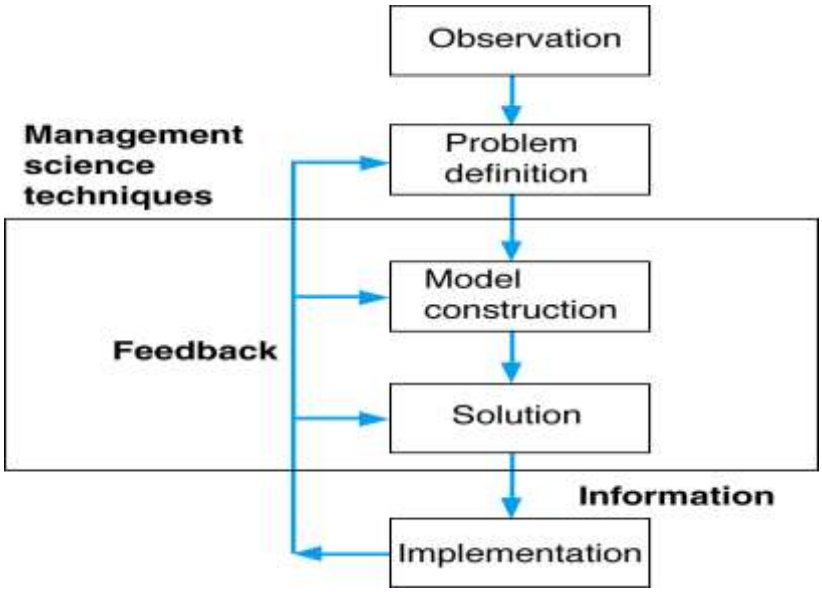


Figure 1.1. The management science process

This approach, as shown in *Figure 1.1*, follows a generally recognized and ordered series of steps:

(1) observation, (2) definition of the problem, (3) model construction, (4) model solution, and (5) implementation of solution results. We will analyze each of these steps individually.

The steps of the scientific method are (1) observation, (2) problem definition, (3) model construction, (4) model solution, and (5) implementation.

Observation

The first step in the management science process is the identification of a problem that exists in the system (organization).

The system must be continuously and closely observed so that problems can be identified as soon as they occur or are anticipated.

Problems are not always the result of a crisis that must be reacted to but, instead, frequently involve an anticipatory or planning situation. The person who normally identifies a problem is the manager because the managers work in places where problems might occur. However, problems can often be identified by a management scientist, a person skilled in the techniques of management science and trained to identify problems, who has been hired specifically to solve problems using management science techniques

A management scientist is a person skilled in the application of management science techniques.

Definition of the Problem

Once it has been determined that a problem exists, the problem must be clearly and concisely defined. Improperly defining a problem can easily result in no solution or an inappropriate solution. Therefore, the limits of the problem and the degree to which it pervades other units of the organization must be included in the problem definition. Because the existence of a problem implies that the objectives of the firm are not being met in some way, the goals (or objectives) of the organization must also be clearly defined. A stated objective helps to focus attention on what the problem actually is.

Model Construction

A management science ***model*** is an abstract representation of an existing problem situation. It can be in the form of a graph or chart, but most frequently a management science model consists of a set of mathematical relationships. These mathematical relationships are made up of numbers and symbols.

A model is an abstract mathematical representation of a problem situation.

As an example, consider a business firm that sells a product. The product costs $5 to produce and sells for $20. A model that computes the total profit that will accrue from the items sold is

$$Z = \$20x - 5x$$

A variable is a symbol used to represent an item that can take on any value.

In this equation x represents the number of units of the product that are sold, and Z represents the total profit that results from the sale of the product. The symbols x and Z are variables. The term ***variable*** is used because no set numeric value has been specified for these items. The number of units sold, x, and the profit, Z, can be any amount (within limits); they can vary. These two variables can be further distinguished. Z is a dependent variable because its value is dependent on the number of units sold; x is an independent variable because the number of units sold is not dependent on anything else (in this equation).

Parameters are known, constant values that are often coefficients of variables in equations.

The numbers $20 and $5 in the equation are referred to as ***parameters***. Parameters are constant values that are generally coefficients of the variables (symbols) in an

equation. Parameters usually remain constant during the process of solving a specific problem. The parameter values are derived from ***data*** (i.e., pieces of information) from the problem environment. Sometimes the data are readily available and quite accurate. For example, presumably the selling price of $20 and product cost of $5 could be obtained from the firm's accounting department and would be very accurate. However, sometimes data are not as readily available to the manager or firm, and the parameters must be either estimated or based on a combination of the available data and estimates. In such cases, the model is only as accurate as the data used in constructing the model.

Data are pieces of information from the problem environment.

The equation as a whole is known as a ***functional relationship*** (also called function and relationship). The term is derived from the fact that profit, Z, is a function of the number of units sold, x, and the equation relates profit to units sold.

A model is a functional relationship that includes variables, parameters, and equations.

Because only one functional relationship exists in this example, it is also the model. In this case the relationship is a model of the determination of profit for the firm. However, this model does not really replicate a problem. Therefore, we will expand our example to create a problem situation.

Let us assume that the product is made from steel and that the business firm has 100 pounds of steel available. If it takes 4 pounds of steel to make each unit of the product, we can develop an additional mathematical relationship to represent steel usage:

$4x = 100$ *lb. of steel*

This equation indicates that for every unit produced, 4 of the available 100 pounds of steel will be used. Now our model consists of two relationships:

$Z = \$20X - \$5X$
$4X = 100$

We say that the profit equation in this new model is an ***objective function***, and the resource equation is a ***constraint***. In other words, the objective of the firm is to achieve as much profit, Z, as possible, but the firm is constrained from achieving an infinite profit by the limited amount of steel available. To signify this distinction between the two relationships in this model, we will add the following notations:

Maximize $Z = \$20X - \$5X$
subject to
$4X = 100$

This model now represents the manager's problem of determining the number of units to produce. You will recall that we defined the number of units to be produced as x. Thus, when we determine the value of x, it represents a potential (or recommended) decision for the manager. Therefore, x is also known as a ***decision variable***. The next step in the management science process is to solve the model to determine the value of the decision variable.

Model Solution

A management science technique usually applies to a specific model type.

Once models have been constructed in management science, they are solved using the management science techniques presented in this text. A management science solution technique usually applies to a specific type of model. Thus, the model type and solution method are both part of the management science technique. We are able to say that a model is solved because the model represents a problem. When we refer to model solution, we also mean problem solution.

For the example model developed in the previous section,

maximize Z = $20X - $5X
subject to 4X = 100

the solution technique is simple algebra. Solving the constraint equation for x, we have

4x = 100
or x = 25 units

Substituting the value of 25 for x into the profit function results in the total profit:

Z = $20x - $5x
= (20)(25) – (5)(25)
= $375

Thus, if the manager decides to produce 25 units of the product and all 25 units sell, the business firm will receive $375 in profit. Note, however, that the value of the decision variable does not constitute an actual decision; rather, it is information that serves as a recommendation or guideline, helping the manager make a decision.

A management science solution can be either a recommended decision or information that helps a manager make a decision.

Implementation

The final step in the management science process for problem solving described in Figure 1.1 is implementation. ***Implementation*** is the actual use of the model once it has been developed or the solution to the problem the model was developed to solve. This is a critical but often overlooked step in the process. It is not always a given that once a model is developed or a solution found, it is automatically used. Frequently the person responsible for putting the model or solution to use is not the same person who developed the model and, thus, the user may not fully understand how the model works or exactly what it is supposed to do. Individuals are also sometimes hesitant to change the normal way they do things or to try new things. In this situation the model and solution may get pushed to the side or ignored altogether if they are not carefully explained and their benefit fully demonstrated. If the management science model and solution are not implemented, then the effort and resources used in their development have been wasted.

Model Building: Break-Even Analysis

In the previous section we gave a brief, general description of how management science models are formulated and solved, using a simple algebraic example. In this section we will continue to explore the process of building and solving management science models, using ***break-even analysis***, also called profit analysis. Break-even analysis is a good topic to expand our discussion of model building and solution because it is straightforward, relatively familiar to most people, and not overly complex. In addition, it provides a convenient means to demonstrate the different ways management science models can be solved mathematically (by hand), graphically, and with a computer.

The purpose of break-even analysis is to determine the number of units of a product (i.e., the volume) to sell or produce that will equate total revenue with total cost. The point where total revenue equals total cost is called the break-even point, and at this point profit is zero. The break-even point gives a manager a point of reference in determining how many units will be needed to ensure a profit.

Components of Break-Even Analysis

The three components of break-even analysis are volume, cost, and profit. Volume is the level of sales or production by a company. It can be expressed as the number of units (i.e., quantity) produced and sold, as the dollar volume of sales, or as a percentage of total capacity available.

Two type of costs are typically incurred in the production of a product: fixed costs and variable costs. ***Fixed costs*** are generally independent of the volume of units produced and sold. That is, fixed costs remain constant, regardless of how many units of product are produced within a given range. Fixed costs can include such items as rent on plant and equipment, taxes, staff and management salaries, insurance, advertising, depreciation, heat and light, plant maintenance, and so on. Taken together, these items result in total fixed costs.

Fixed costs are independent of volume and remain constant.

Variable costs are determined on a per-unit basis. Thus, total variable costs depend on the number of units produced. Variable costs include such items as raw materials and resources, direct labor, packaging, material handling, and freight.

Variable costs depend on the number of items produced.

Total variable costs are a function of the volume and the variable cost per unit. This relationship can be expressed mathematically as

total variable cost = vc_v

where c_v = variable cost per unit and v = volume (number of units) sold.

The total cost of an operation is computed by summing total fixed cost and total variable cost, as follows:

total cost = total fixed cost + total variable cost

or

$TC = c_f + vc_v$

where c_f = fixed cost.

Total cost (TC) equals the fixed cost (c_f) plus the variable cost per unit (c_v) multiplied by volume (v).

As an example, consider Western Clothing Company, which produces denim jeans. The company incurs the following monthly costs to produce denim jeans:

Fixed Costs: c_f = \$10000

Variable Costs: c_v = \$8 per pair

If we arbitrarily let the monthly sales volume, v, equal 400 pairs of denim jeans, the total cost is

$TC = c_f + vc_v$ = \$10,000 + (400)(8) = \$13,200

The third component in our break-even model is profit. Profit is the difference between ***total revenue*** and total cost. Total revenue is the volume multiplied by the price per unit,

total revenue = vp

where p = price per unit.

Profit is the difference between total revenue (volume multiplied by price) and total cost.

For our clothing company example, if denim jeans sell for $23 per pair and we sell 400 pairs per month, then the total monthly revenue is

total revenue = vp = (400)(23) = $9,200

Now that we have developed relationships for total revenue and total cost, profit (Z) can be computed as follows:

Profit (Z) = difference between total revenue vp (p = unit price) and total cost, i.e.

$$Z = vp - c_f - vc_v$$

Computing the Break-Even Point

For our clothing company example, we have determined total revenue and total cost to be $9,200 and $13,200, respectively. With these values, there is no profit but, instead, a loss of $4,000:

total profit = total revenue total cost = $9,200 13,200 = $4,000

We can verify this result by using our total profit formula,

$Z = vp\ c_f\ vc_v$

and the values v = 400, p = $23, c_f = $10,000, and c_v = $8:

$Z = vp - c_f - vc_v$

= - $4000

Obviously, the clothing company does not want to operate with a monthly loss of $4,000 because doing so might eventually result in bankruptcy. If we assume that price is static because of market conditions and that fixed costs and the variable cost per unit are not subject to change, then the only part of our model that can be varied is volume. Using the modeling terms we developed earlier in this chapter, price, fixed costs, and variable costs are parameters, whereas the volume, v, is ***a decision variable***. In break-even analysis we want to compute the value of v that will result in zero profit.

The break-even point is the volume (v) that equates total revenue with total cost where profit is zero.

At the ***break-even point,*** where total revenue equals total cost, the profit, Z, equals zero. Thus, if we let profit, Z, equal zero in our total profit equation and solve for v, we can determine the break-even volume:

$Z = vp - c_f - vc_v$

$15v=10000$

$V=666.7$ *pairs of jeans*

In other words, if the company produces and sells 666.7 pairs of jeans, the profit (and loss) will be zero and the company will break even. This gives the company a point of reference from which to determine how many pairs of jeans it needs to produce and sell in order to gain a profit (subject to any capacity limitations). For example, a sales volume of 800 pairs of denim jeans will result in the following monthly profit:

$Z = vp - c_f - vc_v$

= $2000

In general, the break-even volume can be determined using the following formula:

$Z = vp - c_f - vc_v$

$vp - c_f - vc_v = 0$

$v = c_f/(p - c_v)$

For our example,

$v = c_f/(p - c_v)$

=10000/23-8= 666.7 pairs of jeans

Graphical Solution

It is possible to represent many of the management science models in this text graphically and use these graphical models to solve problems.

Graphical models also have the advantage of providing a "picture" of the model that can sometimes help us understand the modeling process better than the mathematics alone can.

We can easily graph the break-even model for our Western Clothing Company example because the functions for total cost and total revenue are linear.

That means we can graph each relationship as a straight line on a set of coordinates, as shown in Figure 1.2.

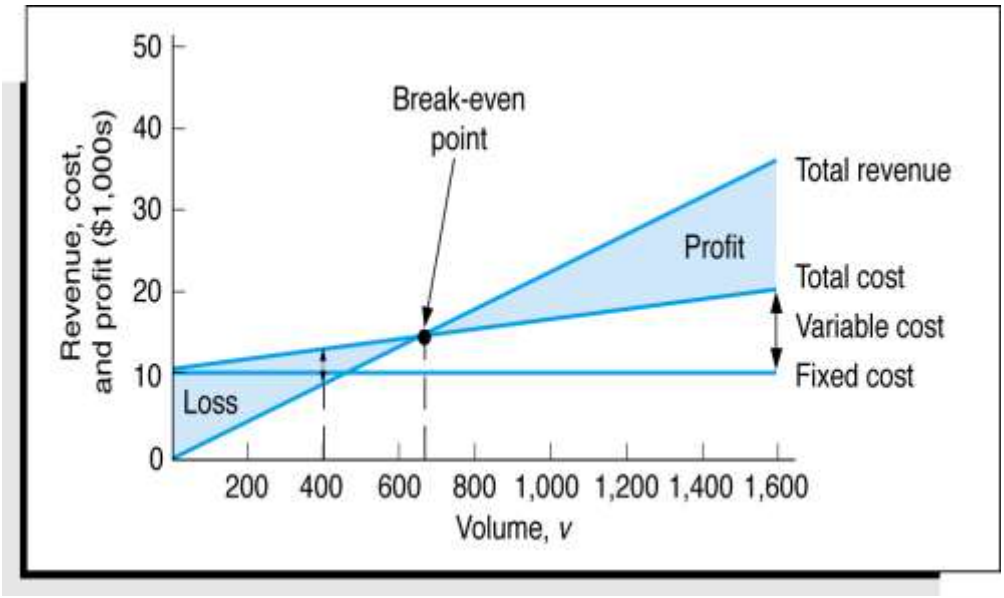


Figure 1.2. Break-even model

In Figure 1.2, the fixed cost, c_f, has a constant value of $10,000, regardless of the volume. The total cost line, TC, represents the sum of variable cost and fixed cost. The total cost line increases because variable cost increases as the volume increases. The total revenue line also increases as volume increases, but at a faster rate than total cost. The point where these two lines intersect indicates that total revenue equals total cost. The volume, v, that corresponds to this point is the break-even volume. The break-even volume in Figure 1.2 is 666.7 pairs of denim jeans.

Sensitivity Analysis

We have now developed a general relationship for determining the break-even volume, which was the objective of our modeling process. This relationship enables us to see how the level of profit (and loss) is directly affected by changes in volume. However, when we developed this model, we assumed that our parameters, fixed and variable costs and price, were constant. In reality such parameters are frequently uncertain and can rarely be assumed to be constant, and changes in any of the parameters can affect the model solution. The study of changes on a management science model is called ***sensitivity analysis*** that is, seeing how sensitive the model is to changes.

Sensitivity analysis can be performed on all management science models in one form or another. In fact, sometimes companies develop models for the primary purpose of experimentation to see how the model will react to different changes the company is contemplating or that management might expect to occur in the future. As a demonstration of how sensitivity analysis works, we will look at the effects of some changes on our break-even model.

The first thing we will analyze is price. As an example, we will increase the price for denim jeans from $23 to $30. As expected, this increases the total revenue, and it

therefore reduces the break-even point from 666.7 pairs of jeans to 454.5 pairs of jeans:

$v = c_f/(p - c_v)$
= 454.5 pairs of jeans

In general, an increase in price lowers the break-even point, all other things held constant.

The effect of the price change on break-even volume is illustrated in Figure 1.3.

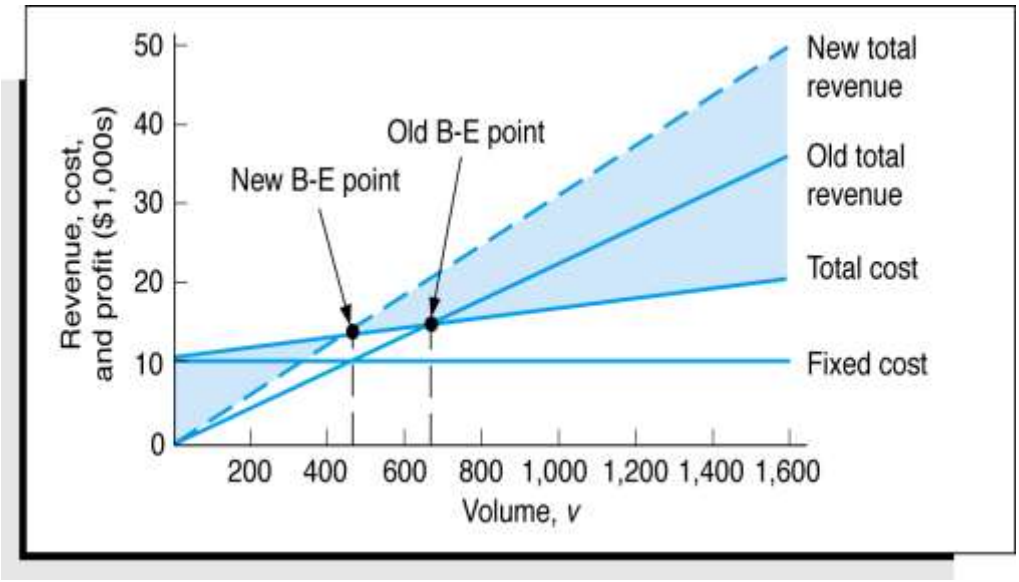


Figure 1.3. Break-even model with an increase in price

Although a decision to increase price looks inviting from a strictly analytical point of view, it must be remembered that the lower break-even volume and higher profit are possible but not guaranteed.

A higher price can make it more difficult to sell the product. Thus, a change in price often must be accompanied by corresponding increases in costs, such as those for advertising, packaging, and possibly production (to enhance quality).

However, even such direct changes as these may have little effect on product demand because price is often sensitive to numerous factors, such as the type of market, monopolistic elements, and product differentiation.

When we increased price, we mentioned the possibility of raising the quality of the product to offset a potential loss of sales due to the price increase.

For example, suppose the stitching on the denim jeans is changed to make the jeans more attractive and stronger. This change results in an increase in variable costs of $4 per pair of jeans, thus raising the variable cost per unit, c_v, to $12 per pair. This change (in conjunction with our previous price change to $30) results in a new break-even volume:

$v = c_f/(p - c_v)$
=555.5 pairs of jeans

In general, an increase in variable costs will decrease the break-even point, all other things held constant.

This new break-even volume and the change in the total cost line that occurs as a result of the variable cost change are shown in Figure 1.4.

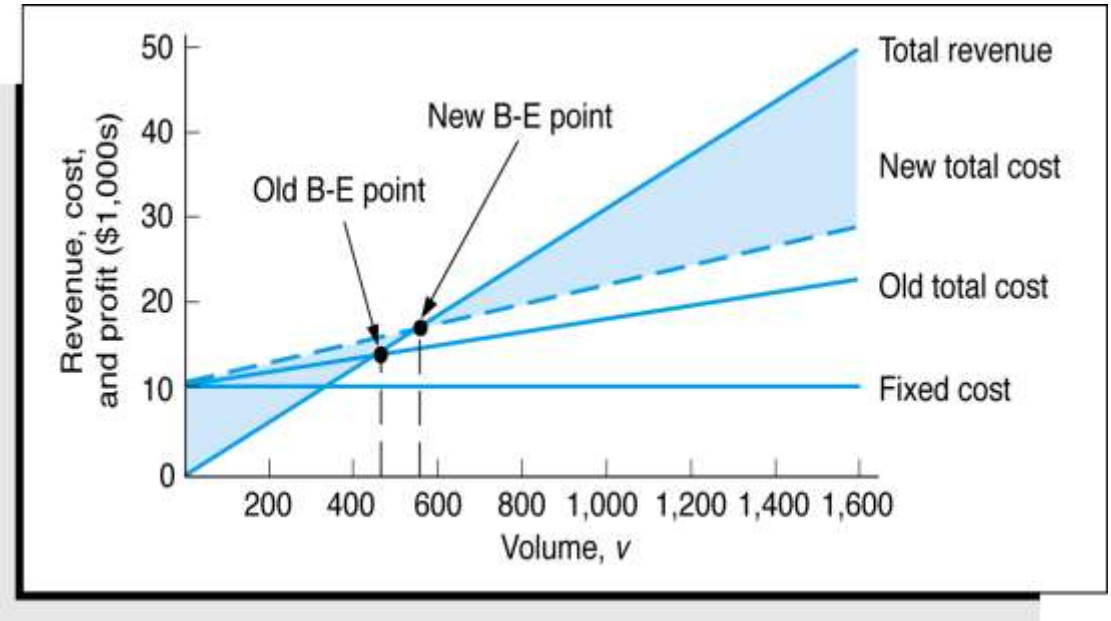


Figure 1.4. Break-even model with an increase in variable cost

Next let's consider an increase in advertising expenditures to offset the potential loss in sales resulting from a price increase.

An increase in advertising expenditures is an addition to fixed costs. For example, if the clothing company increases its monthly advertising budget by $3,000, then the total fixed cost, c_f, becomes $13,000.

Using this fixed cost, as well as the increased variable cost per unit of $12 and the increased price of $30, we compute the break-even volume as follows:

$v = c_f/(p - c_v)$
=722.2 pairs of jeans

In general, an increase in fixed costs will increase the break-even point, all other things held constant.

This new break-even volume, representing changes in price, fixed costs, and variable costs, is illustrated in Figure 1.5. Notice that the break-even volume is now higher than the original volume of 666.7 pairs of jeans, as a result of the increased costs necessary to offset the potential loss in sales. This indicates the necessity to analyze the effect of a change in one of the break-even components on the whole break-even

model. In other words, generally it is not sufficient to consider a change in one model component without considering the overall effect.

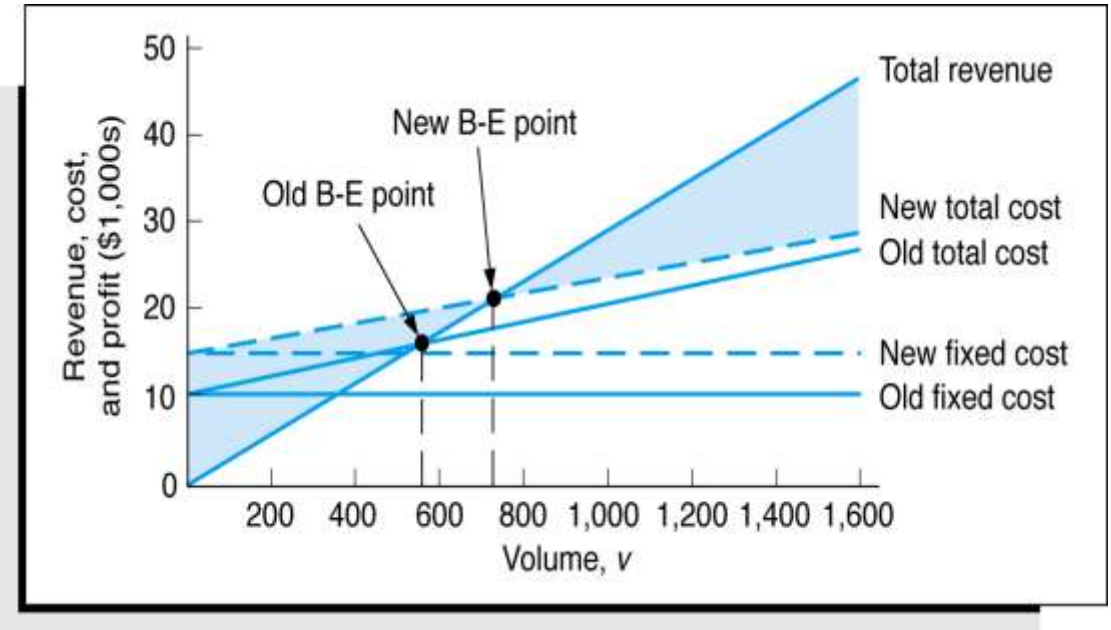


Figure 1.5

Computer Solution

Throughout the text we will demonstrate how to solve management science models on the computer by POM QM for Windows (LAB), a general-purpose quantitative methods software package by Howard Weiss. QM for Windows has program modules to solve almost every type of management science problem you will encounter in this book. There are a number of similar quantitative methods software packages available on the market, with similar characteristics and capabilities as QM for Windows. In most cases you simply input problem data (i.e., model parameters) into a model template, click on a solve button, and the solution appears in a Windows format. QM for Windows is included on the CD that accompanies this text.

QM for Windows

You begin using QM for Windows by clicking on the "Module" button on the toolbar at the top of the main window that appears when you start the program. This will pull down a window with a list of all the model solution modules available in QM for Windows.

Clicking on the "Break-even Analysis" module will access a new screen for typing in the problem title.

Clicking again will access a screen with input cells for the model parametersthat is, fixed cost, variable cost, and price (or revenue).

Next, clicking on the "Solve" button at the top of the screen will provide the solution to the Western Clothing Company example, as shown in Exhibit 1.4.

QM for Windows

File Edit View Module Format Tools Window Help

Edit

Arial

Volume for volume analysis

0

Instruction

There are more results available in additional windows. These may be opened by using the WINDOW option in the Main Menu.

Breakeven/Cost-Volume Analysis Results

Western Clothing Company Example Solution

	Cost Type	Costs	Revenues
Fixed Costs	Fixed	10,000.	xxxxxx
Variable costs	Variable	8.	xxxxxx
Revenue per unit	Variable	xxxxxx	23.
BREAKEVEN POINTS	Units	Dollars	
Costs vs Revenues	666.6667	15,333.33	

Exhibit 1.4.

You can also get the graphical model and solution for this problem by clicking on "Window" at the top of the solution screen and selecting the menu item for a graph of the problem. The break-even graph for the Western Clothing example is shown in Exhibit 1.5.

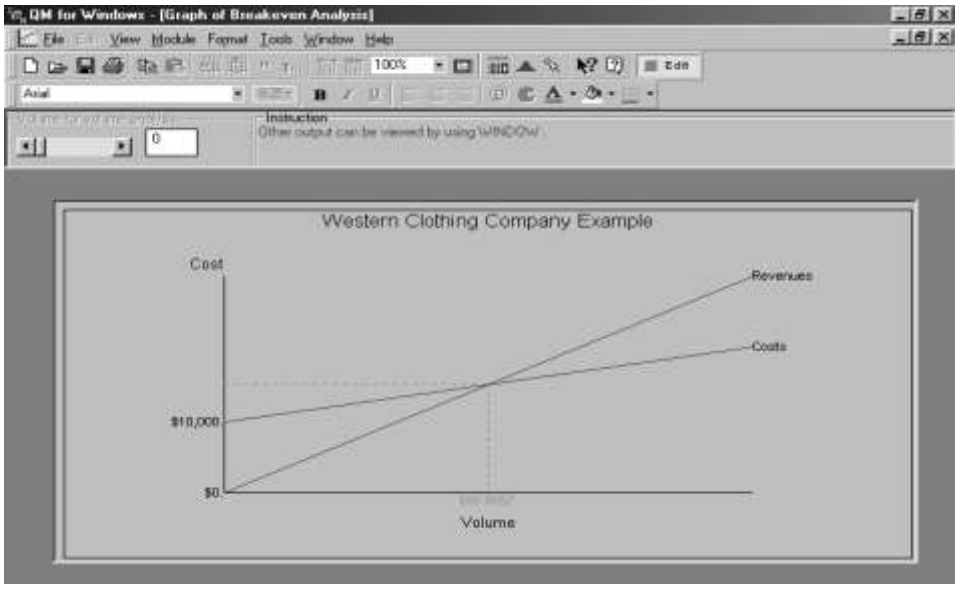


Exhibit 1.5.

Management Science Modeling Techniques

This text focuses primarily on two of the five steps of the management science process described in Figure 1.1 model construction and solution.

These are the two steps that use the management science technique. In a textbook, it is difficult to show how an unstructured real-world problem is identified and defined because the problem must be written out.

However, once a problem statement has been given, we can show how a model is constructed and a solution is derived.

The techniques presented in this text can be loosely classified into four categories, as shown in Figure 1.6.

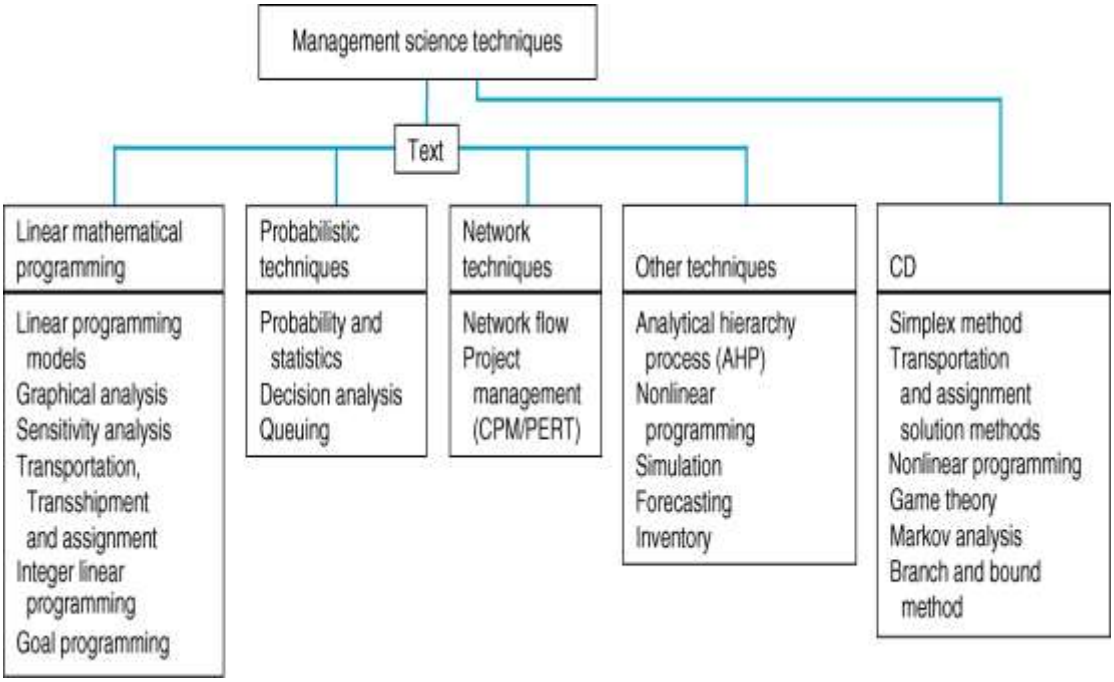


Figure 1.6. Classification of management science techniques

Linear Mathematical Programming - clear objective; restrictions on resources and requirements; parameters known with certainty.

Probabilistic Techniques - results contain uncertainty.

Network Techniques - model often formulated as diagram; deterministic or probabilistic.

Forecasting and Inventory Analysis Techniques - probabilistic and deterministic methods in demand forecasting and inventory control.

Other Techniques - variety of deterministic and probabilistic methods for specific types of problems

Business Use of Management Science

Some application areas:

- Project Planning
- Capital Budgeting
- Inventory Analysis
- Production Planning
- Scheduling

Interfaces - Applications journal published by Institute for Operations Research and Management Sciences (INFORMS)

Management Science Models in Decision Support Systems

Historically management science models have been applied to the solution of specific types of problems; for example, a waiting line model is used to analyze a specific waiting line system at a store or bank.

However, the evolution of computer and information technology has enabled the development of expansive computer systems that combine several management science models and solution techniques in order to address more complex, interrelated organizational problems.

A ***decision support system (DSS)*** is a computer-based system that helps decision makers address complex problems that cut across different parts of an organization and operations.

A DSS is normally interactive, combining various databases and different management science models and solution techniques with a user interface that enables the decision maker to ask questions and receive answers.

In its simplest form any computer-based software program that helps a decision maker make a decision can be referred to as a DSS. For example, the QM for Windows model shown in Exhibit 1.4 can realistically be called a DSS.

Alternatively enterprise wide DSSs can encompass many different types of models and large data warehouses, and they can serve many decision makers in an organization.

They can provide decision makers with interrelated information and analyses about almost anything in a company.

Figure 1.7 illustrates the basic structure of a DSS with a database component, a modeling component, and a user interface with the decision maker.

As noted earlier, each of these components can be small and singular, with one analytical model linked to a database, or they can be very large and complex, linking many models and large databases.

A DSS can be primarily a data-oriented system, or it can be a model-oriented system. A new type of DSS, called an online analytical processing system, or OLAP, focuses on the use of analytical techniques such as management science models and statistics for decision making.

A desktop DSS for a single user can be a spreadsheet program such as Excel to develop specific solutions to individual problems.

Expert Choice is another example of a desktop DSS that uses the analytical hierarchy process (AHP) to structure complex problems by establishing decision criteria, developing priorities, and ranking decision alternatives.

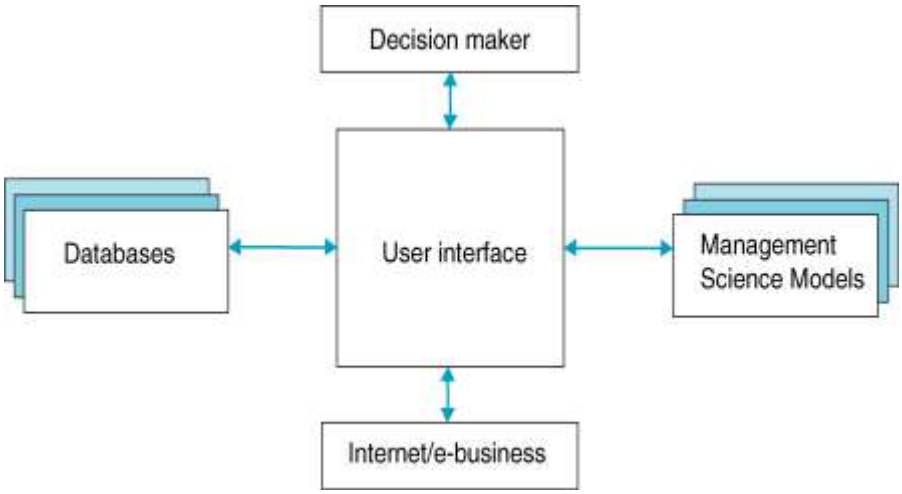


Figure 1.7

On the other end of the DSS spectrum, an enterprise resource planning (ERP) system is software that can connect the components and functions of an entire company.

It can transform data, such as individual daily sales, directly into information that supports immediate decisions in other parts of the company, such as ordering, manufacturing, inventory, and distribution.

A large-scale DSS such as an ERP system in a company might include a forecasting model to analyze sales data and help determine future product demand; an inventory model to determine how much inventory to keep on hand; a linear programming model to determine how much material to order and product to produce, and when to produce it; a transportation model to determine the most cost-effective method of distributing a product to customers; and a network flow model to determine the best delivery routes.

All these different management science models and the data necessary to support them can be linked in a single enterprise wide DSS that can provide many decisions to many different decision makers.

Problems

1. The Willow Furniture Company produces tables. The fixed monthly cost of production is $8,000, and the variable cost per table is $65. The tables sell for $180 apiece.

a. For a monthly volume of 300 tables, determine the total cost, total revenue, and profit.
b. Determine the monthly break-even volume for the Willow Furniture Company.

2. Pastureland Dairy makes cheese, which it sells at local supermarkets. The fixed monthly cost of production is $4,000, and the variable cost per pound of cheese is $0.21. The cheese sells for $0.75 per pound; however, the dairy is considering raising the price to $0.95 per pound. The dairy currently produces and sells 9,000 pounds of cheese per month, but if it raises its price per pound, sales will decrease to 5,700 pounds per month. Should the dairy raise the price?

3. The General Store at State University is an auxiliary bookstore located near the dormitories that sells academic supplies, toiletries, sweatshirts and T-shirts, magazines, packaged food items, and canned soft drinks and fruit drinks. The manager of the store has noticed that several pizza delivery services near campus make frequent deliveries. The manager is therefore considering selling pizza at the store. She could buy premade frozen pizzas and heat them in an oven. The cost of the oven and freezer would be $27,000. The frozen pizzas cost $3.75 each to buy from a distributor and to prepare (including labor and a box). To be competitive with the local delivery services, the manager believes she should sell the pizzas for $8.95 apiece. The manager needs to write up a proposal for the university's director of auxiliary services.

   a. Determine how many pizzas would have to be sold to break even.
   b. If The General Store sells 20 pizzas per day, how many days would it take to break even?
   c. The manager of the store anticipates that once the local pizza delivery services start losing business, they will react by cutting prices. If after a month (30 days) the manager has to lower the price of a pizza to $7.95 to keep demand at 20 pizzas per day, as she expects, what will the new break-even point be, and how long will it take the store to break even?

T#1

The College of Business at Tech is planning to begin an online MBA program. The initial start-up cost for computing equipment, facilities, course development, and staff recruitment and development is $350,000. The college plans to charge tuition of $18,000 per student per year. However, the university administration will charge the college $12,000 per student for the first 100 students enrolled each year for administrative costs and its share of the tuition payments.

a. How many students does the college need to enroll in the first year to break even?
b. If the college can enroll 75 students the first year, how much profit will it make?
c. The college believes it can increase tuition to $24,000, but doing so would reduce enrollment to 35. Should the college consider doing this?

Solution:

CHAPTER 2. LINEAR PROGRAMMING: MODEL FORMULATION AND GRAPHICAL SOLUTION

Many major decisions faced by a manager of a business focus on the best way to achieve the objectives of the firm, subject to the restrictions placed on the manager by the operating environment.

These restrictions can take the form of limited resources, such as time, labor, energy, material, or money; or they can be in the form of restrictive guidelines, such as a recipe for making cereal or engineering specifications.

One of the most frequent objectives of business firms is to gain the most profit possible or, in other words, to maximize profit.

The objective of individual organizational units within a firm (such as a production or packaging department) is often to minimize cost.

When a manager attempts to solve a general type of problem by seeking an objective that is subject to restrictions, the management science technique called **linear programming** is frequently used.

Objectives of a business frequently are to maximize profit or minimize cost.

Linear programming is a model that consists of linear relationships representing a firm's decision(s), given an objective and resource constraints.

There are three steps in applying the linear programming technique.

First, the problem must be identified as being solvable by linear programming.
Second, the unstructured problem must be formulated as a mathematical model.
Third, the model must be solved by using established mathematical techniques.

The linear programming technique derives its name from the fact that the functional relationships in the mathematical model are linear, and the solution technique consists of predetermined mathematical steps that is, a program.

In this chapter we will concern ourselves with the formulation of the mathematical model that represents the problem and then with solving this model by using a graph.

Model Formulation

A linear programming model consists of certain common components and characteristics.

The model components include decision variables, an objective function, and model constraints, which consist of decision variables and parameters.

Decision variables are mathematical symbols that represent levels of activity by the firm.

For example, an electrical manufacturing firm desires to produce x_1 radios, x_2 toasters, and x_3 clocks, where x_1, x_2, and x_3 are symbols representing unknown variable quantities of each item. The final values of x_1, x_2, and x_3, as determined by the firm, constitute a decision (e.g., the equation $x_1 = 100$ radios is a decision by the firm to produce 100 radios).

Decision variables are mathematical symbols that represent levels of activity.

The **objective function** is a linear mathematical relationship that describes the objective of the firm in terms of the decision variables.

The objective function always consists of either maximizing or minimizing some value (e.g., maximize the profit or minimize the cost of producing radios).

The objective function is a linear relationship that reflects the objective of an operation.

The model constraints are also linear relationships of the decision variables; they represent the restrictions placed on the firm by the operating environment.

The restrictions can be in the form of limited resources or restrictive guidelines.

For example, only 40 hours of labor may be available to produce radios during production.

The actual numeric values in the objective function and the constraints, such as the 40 hours of available labor, are **parameters**.

A constraint is a linear relationship that represents a restriction on decision making.

Parameters are numerical values that are included in the objective functions and constraints.

The next section presents an example of how a linear programming model is formulated. Although this example is simplified, it is realistic and represents the type

of problem to which linear programming can be applied. In the example, the model components are distinctly identified and described. By carefully studying this example, you can become familiar with the process of formulating linear programming models.

A Maximization Model Example

Beaver Creek Pottery Company is a small crafts operation run by a Native American tribal council.

The company employs skilled artisans to produce clay bowls and mugs with authentic Native American designs and colors.

The two primary resources used by the company are special pottery clay and skilled labor. Given these limited resources, the company desires to know how many bowls and mugs to produce each day in order to maximize profit.

This is generally referred to as a product mix problem type.

The two products have the following resource requirements for production and profit per item produced (i.e., the model parameters):

	Resource Requirements		
Product	Labor (Hr./Unit)	Clay (Lb./Unit)	Profit ($/Unit)
Bowl	1	4	40
Mug	2	3	50

There are 40 hours of labor and 120 pounds of clay available each day for production. We will formulate this problem as a linear programming model by defining each component of the model separately and then combining the components into a single model.

The steps in this formulation process are summarized as follows:

Summary of LP Model Formulation Steps

Step 1. Define the decision variables

How many bowls and mugs to produce

Step 2. Define the objective function

Maximize profit

Step 3. Define the constraints

The resources (clay and labor) available

A linear programming model consists of decision variables, an objective function, and constraints.

Decision Variables

The decision confronting management in this problem is how many bowls and mugs to produce.

The two decision variables represent the number of bowls and mugs to be produced on a daily basis.

The quantities to be produced can be represented symbolically as

x_1 = number of bowls to produce

x_2 = number of mugs to produce

The Objective Function

The objective of the company is to maximize total profit.

The company's profit is the sum of the individual profits gained from each bowl and mug. Profit derived from bowls is determined by multiplying the unit profit of each bowl, \$40, by the number of bowls produced, x_1.

Likewise, profit derived from mugs is derived from the unit profit of a mug, \$50, multiplied by the number of mugs produced, x_2.

Thus, total profit, which we will define symbolically as Z, can be expressed mathematically as $\$40x_1 + \$50x_2$.

By placing the term maximize in front of the profit function, we express the objective of the firmto maximize total profit:

maximize $Z = \$40x_1 + 50x_2$
where

Z = total profit per day

$\$40x_1$ = profit from bowls

$\$50x_2$ = profit from mugs

Model Constraints

In this problem two resources are used for production labor and clay both of which are limited.

Production of bowls and mugs requires both labor and clay.

For each bowl produced, 1 hour of labor is required. Therefore, the labor used for the production of bowls is $1x_1$ hours.

Similarly, each mug requires 2 hours of labor; thus, the labor used to produce mugs every day is $2x_2$ hours.

The total labor used by the company is the sum of the individual amounts of labor used for each product:

$1x_1 + 2x_2$

However, the amount of labor represented by $1x_1 + 2x_2$ is limited to 40 hours per day; thus, the complete labor constraint is

$1x_1 + 2x_2 \leq 40\ hr.$

The "less than or equal to" (≤) inequality is employed instead of an equality (=) because the 40 hours of labor is a maximum limitation that can be used, not an amount that must be used.

This constraint allows the company some flexibility; the company is not restricted to using exactly 40 hours but can use whatever amount is necessary to maximize profit, up to and including 40 hours.

This means that it is possible to have idle, or excess, capacity (i.e., some of the 40 hours may not be used).

The constraint for clay is formulated in the same way as the labor constraint.

Because each bowl requires 4 pounds of clay, the amount of clay used daily for the production of bowls is $4x_1$ pounds; and because each mug requires 3 pounds of clay, the amount of clay used daily for mugs is $3x_2$.

Given that the amount of clay available for production each day is 120 pounds, the material constraint can be formulated as

$4x_1 + 3x_2 \leq 120\ lb.$

A final restriction is that the number of bowls and mugs produced must be either zero or a positive value because it is impossible to produce negative items.

These restrictions are referred to as nonnegativity constraints and are expressed mathematically as

Nonnegativity constraints restrict the decision variables to zero or positive values.

$x_1 \geq 0, x_2 \geq 0$

The complete linear programming model for this problem can now be summarized as follows:

$Maximize\ Z = \$40x_1 + \$50x_2$

$subject\ to: \quad 1x_1 + 2x_2 \leq 40$

$4x_2 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$

The solution of this model will result in numeric values for x_1 and x_2 that will maximize total profit, Z.

As one possible solution, consider $x_1 = 5$ bowls and $x_2 = 10$ mugs.

First, we will substitute this hypothetical solution into each of the constraints in order to make sure that the solution does not require more resources than the constraints show are available:

1(5) + 2(10) = 25 < 40 hours, within constraint

and

4(5) + 3(10) = 70 < 120 pounds, within constraint

Because neither of the constraints is violated by this hypothetical solution, we say the solution is **feasible** (i.e., it is possible).

Substituting these solution values in the objective function gives Z = 40(5) + 50(10) = $700. However, for the time being, we do not have any way of knowing whether $700 is the maximum profit.

A feasible solution does not violate any of the constraints.

Now consider a solution of $x_1 = 10$ bowls and $x_2 = 20$ mugs. This solution results in a profit of

$x_1 = 10$ *bowls*

$x_2 = 20$ *mugs*

$Z = \$1400$

Although this is certainly a better solution in terms of profit, it is **infeasible** (i.e., not possible) because it violates the resource constraint for labor:

$1(10) + 2(20) = 50 > 40$

An infeasible problem violates at least one of the constraints.

The solution to this problem must maximize profit without violating the constraints. The solution that achieves this objective is $x_1 = 24$ bowls and $x_2 = 8$ mugs, with a corresponding profit of $1,360. The determination of this solution is shown using the graphical solution approach in the following section.

Graphical Solutions of Linear Programming Models

Following the formulation of a mathematical model, the next stage in the application of linear programming to a decision-making problem is to find the solution of the model.

A common solution approach is to solve algebraically the set of mathematical relationships that form the model either manually or using a computer program, thus determining the values for the decision variables.

However, because the relationships are linear, some models and solutions can be illustrated graphically.

Graphical solutions are limited to linear programming problems with only two decision variables.

The graphical method is realistically limited to models with only two decision variables, which can be represented on a graph of two dimensions.

Models with three decision variables can be graphed in three dimensions, but the process is quite cumbersome, and models of four or more decision variables cannot be graphed at all.

Although the graphical method is limited as a solution approach, it is very useful at this point in our presentation of linear programming in that it gives a picture of how a solution is derived.

Graphs can provide a clearer understanding of how the computer and mathematical solution approaches presented in subsequent chapters work and, thus, a better understanding of the solutions.

The graphical method provides a picture of how a solution is obtained for a linear programming problem.

Graphical Solution of a Maximization Model

The product mix model will be used to demonstrate the graphical interpretation of a linear programming problem.

Recall that the problem describes Beaver Creek Pottery Company's attempt to decide how many bowls and mugs to produce daily, given limited amounts of labor and clay. The complete linear programming model was formulated as

Maximize $Z = \$40x_1 + 50x_2$
Subject to
$1x_1 + 2x_2 \leq 40$
$4x_2 + 3x_2 \leq 120$
$x_1, x_2 \geq 0$
where
x_1 = *number of bowls produced*
x_2 = *number of mugs produced*

Figure 2.2 is a set of coordinates for the decision variables x_1 and x_2, on which the graph of our model will be drawn.

Note that only the positive quadrant is drawn (i.e., the quadrant where x_1 and x_2 will always be positive) because of the nonnegativity constraints, $x_1 \geq 0$ and $x_2 \geq 0$.

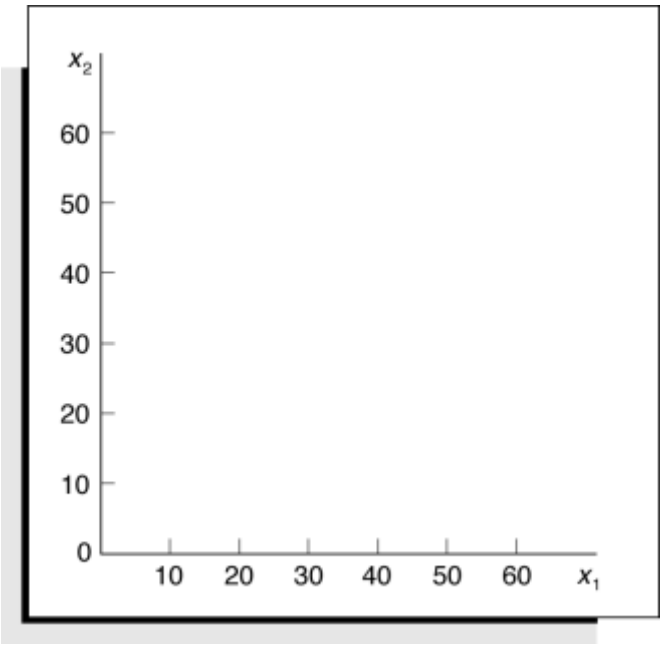


Figure 2.2

The first step in drawing the graph of the model is to plot the constraints on the graph. This is done by treating both constraints as equations (or straight lines) and plotting each line on the graph. Let's consider the labor constraint line first:

$x_1 + 2x_2 = 40$

Constraint lines are plotted as equations.

A simple procedure for plotting this line is to determine two points that are on the line and then draw a straight line through the points.

One point can be found by letting $x_1 = 0$ and solving for x_2:

$0 + 2x_2 = 40$

$x_2 = 20$

Thus, one point is at the coordinates $x_1 = 0$ and $x_2 = 20$. A second point can be found by letting $x_2 = 0$ and solving for x_1:

$x_1 + 2(0) = 40$

$x_1 = 40$

Now we have a second point, $x_1 = 40$, $x_2 = 0$. The line on the graph representing this equation is drawn by connecting these two points, as shown in Figure 2.3.

However, this is only the graph of the constraint line and does not reflect the entire constraint, which also includes the values that are less than or equal to (≤) this line. The area representing the entire constraint is shown in Figure 2.4.

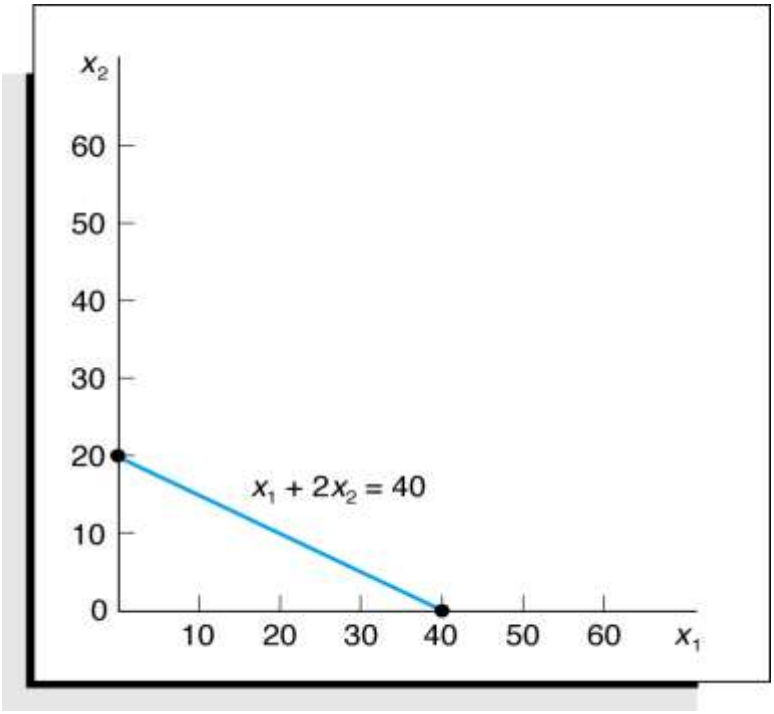


Figure 2.3. Graph of the labor constraint line

To test the correctness of the constraint area, we check any two points one inside the constraint area and one outside.

For example, check point A in Figure 2.4, which is at the intersection of $x_1 = 10$ and $x_2 = 10$. Substituting these values into the following labor constraint,

$10+2(10) \leq 40$

$30 \leq 40$ hr

Shows that point A is indeed within the constraint area, as these values for x_1 and x_2 yield a quantity that does not exceed the limit of 40 hours.

Next, we check point B at $x_1 = 40$ and $x_2 = 30$:

$40+2(30) \leq 40$

$100 \neq 40$ hr

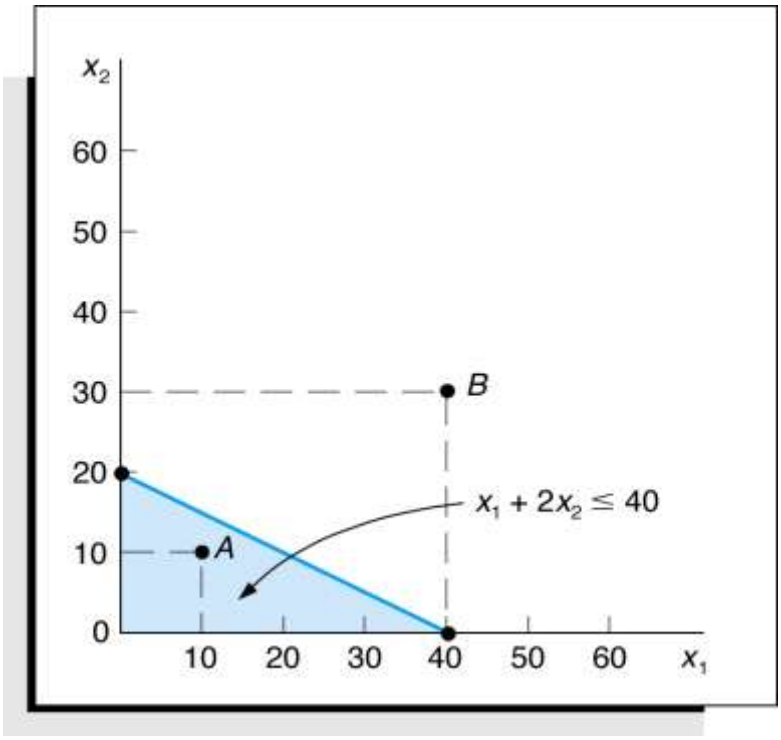


Figure 2.4. The labor constraint area

Point B is obviously outside the constraint area because the values for x_1 and x_2 yield a quantity (100) that exceeds the limit of 40 hours.

We draw the line for the clay constraint the same way as the one for the labor constraint by finding two points on the constraint line and connecting them with a straight line. First, let $x_1 = 0$ and solve for x_2:

$4(0) + 3x_2 = 120$

$x_2 = 40$

Performing this operation results in a point, $x_1 = 0$, $x_2 = 40$. Next, we let $x_2 = 0$ and then solve for x_1:

$4\,x_1 + 3(0) = 120$

$x_1 = 30$

This operation yields a second point, $x_1 = 30$, $x_2 = 0$.

Plotting these points on the graph and connecting them with a line gives the constraint line and area for clay, as shown in Figure 2.5.

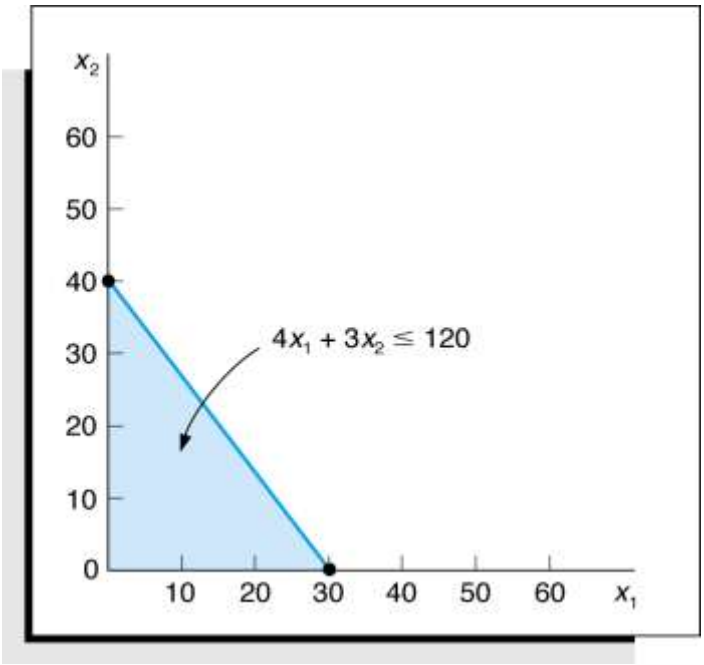


Figure 2.5. The constraint area for clay

Combining the two individual graphs for both labor and clay (Figures 2.4 and 2.5) produces a graph of the model constraints, as shown in Figure 2.6.

The shaded area in Figure 2.6 is the area that is common to both model constraints. Therefore, this is the only area on the graph that contains points (i.e., values for x_1 and x_2) that will satisfy both constraints simultaneously.

For example, consider the points R, S, and T in Figure 2.7. Point R satisfies both constraints; thus, we say it is a feasible solution point.

Point S satisfies the clay constraint ($4x_1 + 3x_2 \leq 120$) but exceeds the labor constraint; thus, it is infeasible. Point T satisfies neither constraint; thus, it is also infeasible

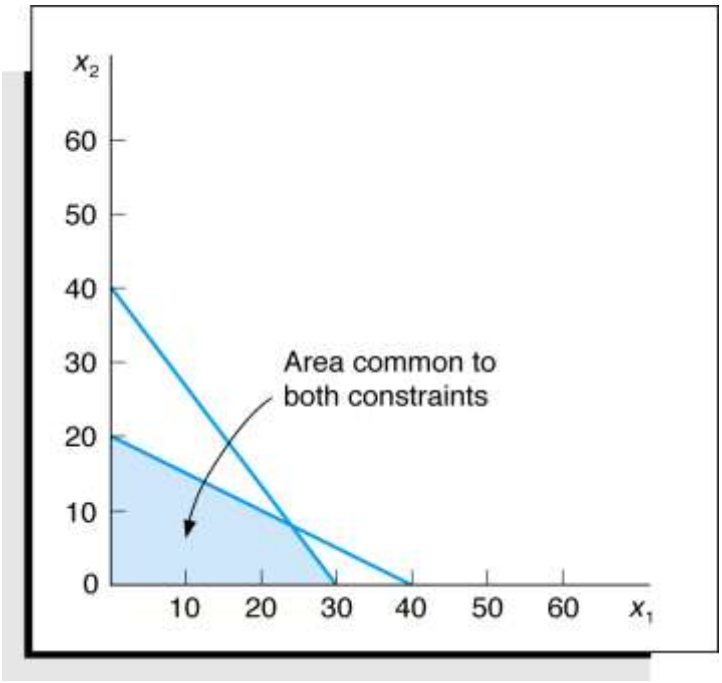


Figure 2.6. Graph of both model constraints

The shaded area in Figure 2.7 is referred to as the feasible solution area because all the points in this area satisfy both constraints.

Some point within this feasible solution area will result in maximum profit for Beaver Creek Pottery Company.

The next step in the graphical solution approach is to locate this point.

The feasible solution area is an area on the graph that is bounded by the constraint equations.

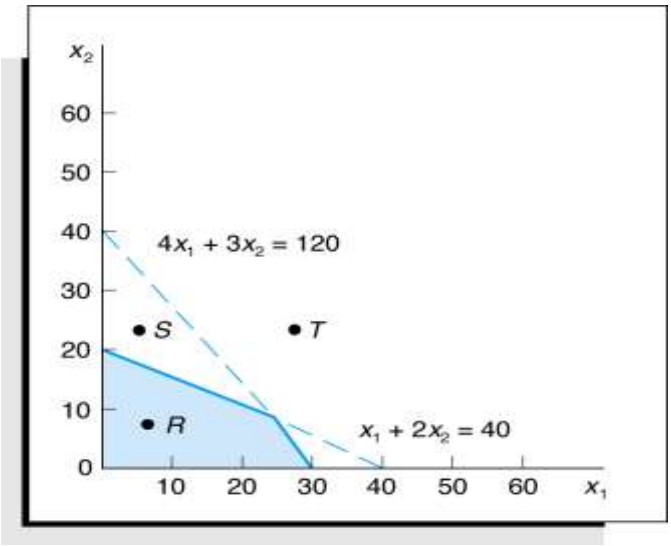


Figure 2.7. The feasible solution area constraints

The Optimal Solution Point

The second step in the graphical solution method is to locate the point in the feasible solution area that will result in the greatest total profit.

To begin the solution analysis, we first plot the objective function line for an arbitrarily selected level of profit. For example, if we say profit, Z, is \$800, the objective function is

$\$800 = 40x_1 + 50x_2$

Plotting this line just as we plotted the constraint lines results in the graph shown in Figure 2.8.

Every point on this line is in the feasible solution area and will result in a profit of \$800 (i.e., every combination of x_1 and x_2 on this line will give a Z value of \$800).

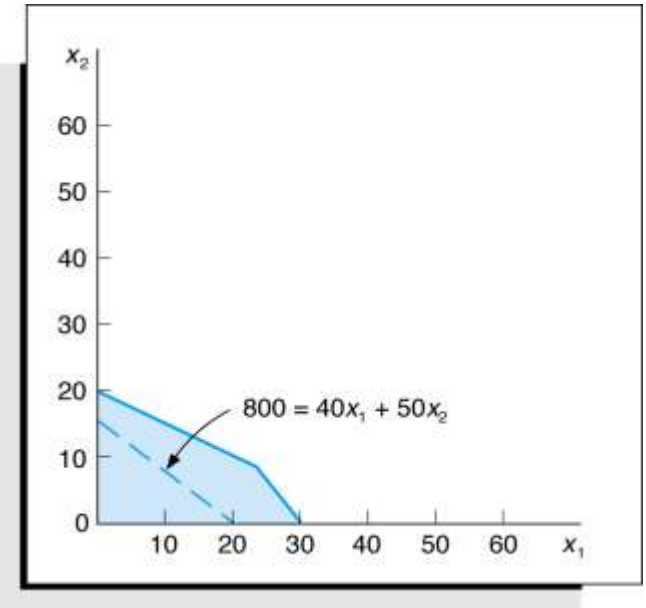


Figure 2.8. Objective function line for Z = \$800

However, let us see whether an even greater profit will still provide a feasible solution. For example, consider profits of \$1,200 and \$1,600, as shown in Figure 2.9.

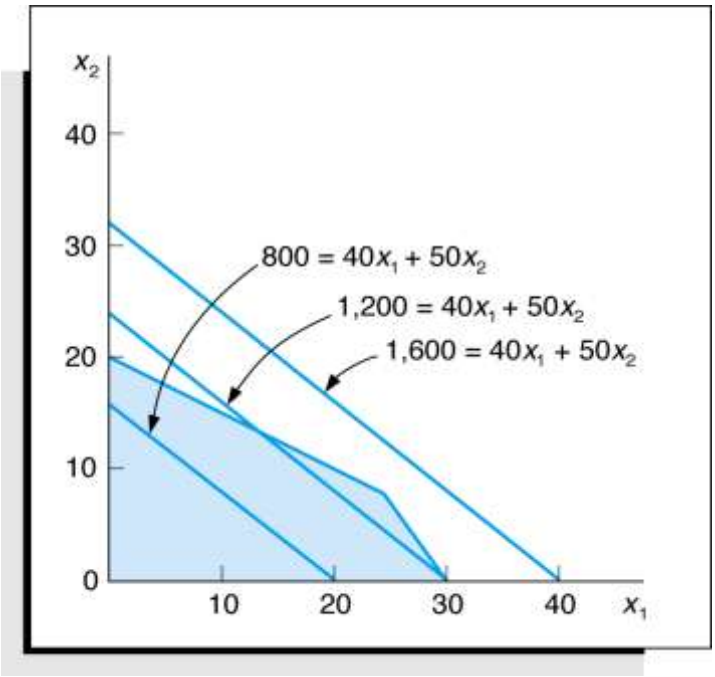


Figure 2.9. Alternative objective function lines for profits, Z, of $800, $1,200, and $1,600

A portion of the objective function line for a profit of $1,200 is outside the feasible solution area, but part of the line remains within the feasible area.

Therefore, this profit line indicates that there are feasible solution points that give a profit greater than $800.

Now let us increase profit again, to $1,600. This profit line, also shown in Figure 2.9, is completely outside the feasible solution area.

The fact that no points on this line are feasible indicates that a profit of $1,600 is not possible.

Because a profit of $1,600 is too great for the constraint limitations, as shown in Figure 2.9, the question of the maximum profit value remains.

We can see from Figure 2.9 that profit increases as the objective function line moves away from the origin (i.e., the point $x_1 = 0$, $x_2 = 0$).

Given this characteristic, the maximum profit will be attained at the point where the objective function line is farthest from the origin and is still touching a point in the feasible solution area. This point is shown as point B in Figure 2.10.

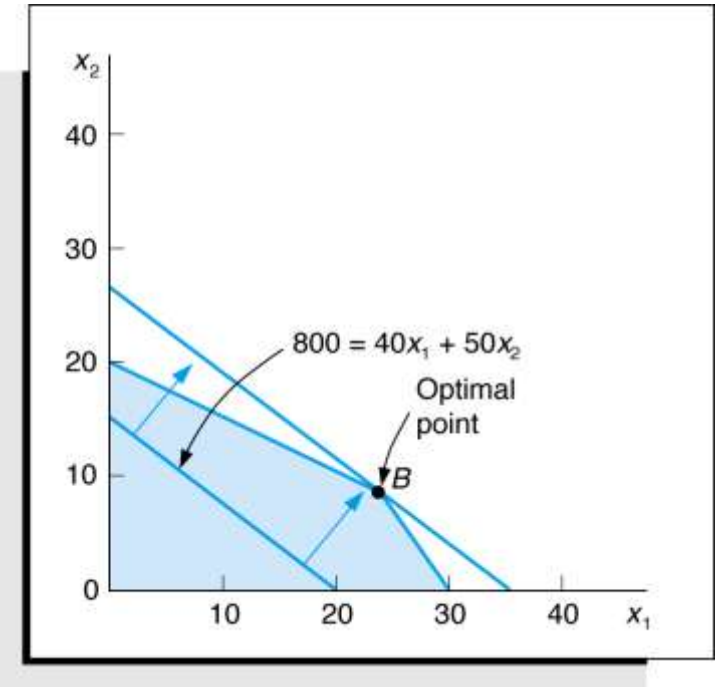


Figure 2.10. Identification of optimal solution point

To find point B, we place a straightedge parallel to the objective function line \$800 = $40x_1 + 50x_2$ in Figure 2.10 and move it outward from the origin as far as we can without losing contact with the feasible solution area. Point B is referred to as the *optimal (i.e., best) solution.*

The optimal solution is the best feasible solution.

The Solution Values

The third step in the graphical solution approach is to solve for the values of x_1 and x_2 once the optimal solution point has been found.

It is possible to determine the x_1 and x_2 coordinates of point B in Figure 2.10 directly from the graph, as shown in Figure 2.11. The graphical coordinates corresponding to point B in Figure 2.11 are $x_1 = 24$ and $x_2 = 8$.

This is the optimal solution for the decision variables in the problem. However, unless an absolutely accurate graph is drawn, it is frequently difficult to determine the correct solution directly from the graph.

A more exact approach is to determine the solution values mathematically once the optimal point on the graph has been determined.

The mathematical approach for determining the solution is described in the following pages. First, however, we will consider a few characteristics of the solution.

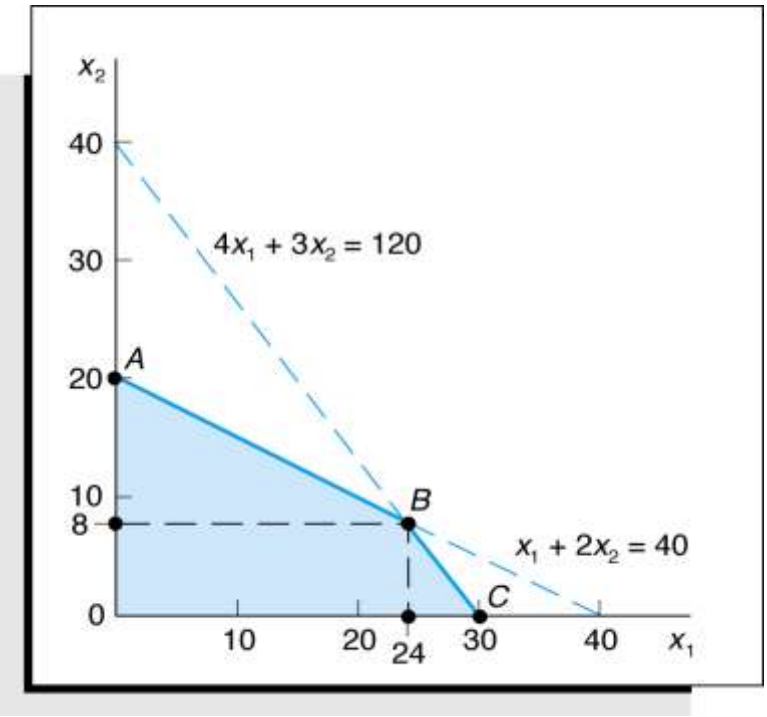


Figure 2.11. Optimal solution coordinates

In Figure 2.10, as the objective function was increased, the last point it touched in the feasible solution area was on the boundary of the feasible solution area.

The solution point is always on this boundary because the boundary contains the points farthest from the origin (i.e., the points corresponding to the greatest profit). This characteristic of linear programming problems reduces the number of possible solution points considerably, from all points in the solution area to just those points on the boundary. However, the number of possible solution points is reduced even more by another characteristic of linear programming problems.

The optimal solution point is the last point the objective function touches as it leaves the feasible solution area.

The solution point will be on the boundary of the feasible solution area and at one of the corners of the boundary where two constraint lines intersect. (The graphical axes, you will recall, are also constraints because $x_1 \geq 0$ and $x_2 \geq 0$.) These corners (points A, B, and C in Figure 2.11) are protrusions, or extremes, in the feasible solution area; they are called **extreme points**.

It has been proven mathematically that the optimal solution in a linear programming model will always occur at an extreme point. Therefore, in our sample problem the possible solution points are limited to the three extreme points, A, B, and C.

The optimal extreme point is the extreme point the objective function touches last as it leaves the feasible solution area, as shown in Figure 2.10.

Extreme points are corner points on the boundary of the feasible solution area.

From the graph shown in Figure 2.10, we know that the optimal solution point is B. Because point B is formed by the intersection of two constraint lines, as shown in Figure 2.11, these two lines are equal at point B. Thus, the values of x_1 and x_2 at that intersection can be found by solving the two equations simultaneously.

Constraint equations are solved simultaneously at the optimal extreme point to determine the variable solution values.

First, we convert both equations to functions of x_1:

$x_1 + 2x_2 = 40$
$x_1 = 40 - 2x_2$
and
$4x_1 + 3x_2 = 120$
$4x_1 = = 120 - 3x_2$
$x_1 = 30 - (3x_2/4)$

Now we let x_1 in the first equation equal x_1 in the second equation,

$40 - 2x_2 = 30 - (3x_2/4)$

and solve for x_2*:*

$x_2 = 8$

Substituting $x_2 = 8$ into either one of the original equations gives a value for x_1:

$x_1 = 24$

Thus, the optimal solution at point B in Figure 2.11 is $x_1 = 24$ and $x_2 = 8$. Substituting these values into the objective function gives the maximum profit,

$Z = \$1360$

In terms of the original problem, the solution indicates that if the pottery company produces 24 bowls and 8 mugs, it will receive $1,360, the maximum daily profit possible (given the resource constraints).

Given that the optimal solution will be at one of the extreme corner points, A, B, or C, we can also find the solution by testing each of the three points to see which results in the greatest profit, rather than by graphing the objective function and seeing which point it last touches as it moves out of the feasible solution area. Figure 2.12 shows the solution values for all three points, A, B, and C, and the amount of profit, Z, at each point.

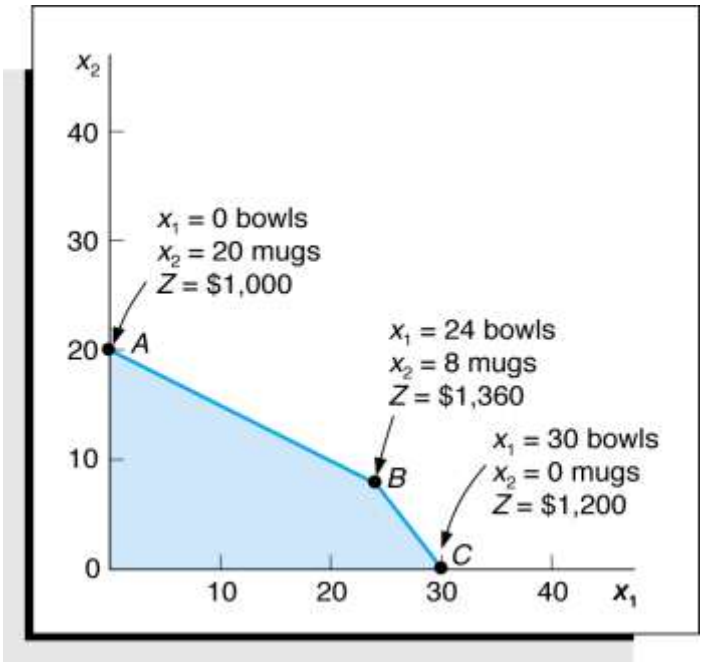


Figure 2.12. Solutions at all corner points

As indicated in the discussion of Figure 2.10, point B is the optimal solution point because it is the last point the objective function touches before it leaves the solution area.

In other words, the objective function determines which extreme point is optimal. This is because the objective function designates the profit that will accrue from each combination of x_1 and x_2 values at the extreme points.

If the objective function had had different coefficients (i.e., different x_1 and x_2 profit values), one of the extreme points other than B might have been optimal.

Let's assume for a moment that the profit for a bowl is \$70 instead of \$40, and the profit for a mug is \$20 instead of \$50. These values result in a new objective function, $Z = \$70x_1 + 20x_2$.

If the model constraints for labor or clay are not changed, the feasible solution area remains the same, as shown in Figure 2.13.

However, the location of the objective function in Figure 2.13 is different from that of the original objective function in Figure 2.10.

The reason for this change is that the new profit coefficients give the linear objective function a new slope.

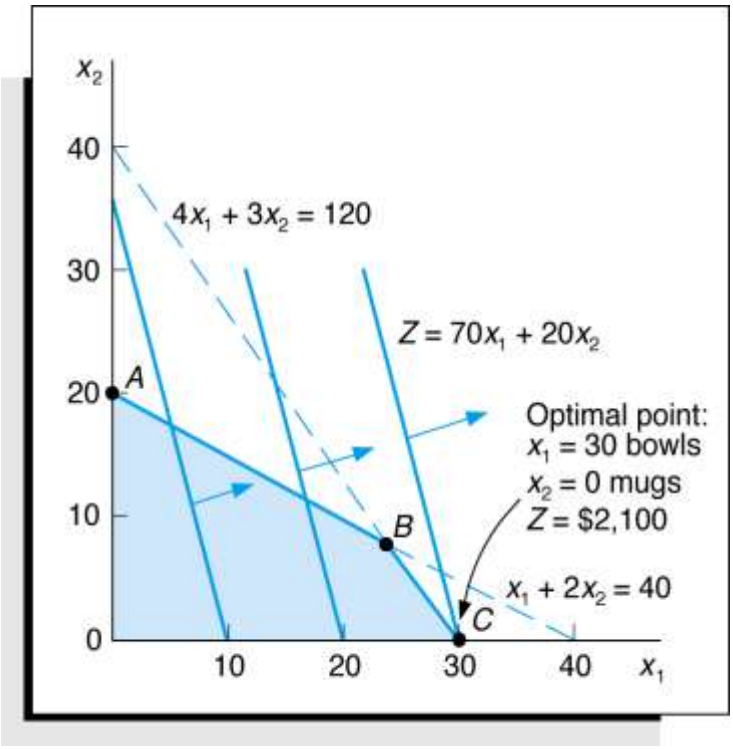


Figure 2.13. The optimal solution with $Z = 70x_1 + 20x_2$

The **slope** can be determined by transforming the objective function into the general equation for a straight line, y = a + bx, where y is the dependent variable, a is the y intercept, b is the slope, and x is the independent variable. For our sample objective function, x_2 is the dependent variable corresponding to y (i.e., it is on the vertical axis), and x_1 is the independent variable. Thus, the objective function can be transformed into the general equation of a line as follows:

LECTURE NOTES:

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The slope is computed as the "rise" over the "run."

This transformation identifies the slope of the new objective function as 7/2 (the minus sign indicates that the line slopes downward). In contrast, the slope of the original objective function was 4/5.

If we move this new objective function out through the feasible solution area, the last extreme point it touches is point C.

Simultaneously solving the constraint lines at point C results in the following solution:

$x_1 = 30$
$4x_1 + 3x_2 = 120$
And
$x_2 = 40 - (4x_1/3)$
$x_2 = 0$

Thus, the optimal solution at point C in Figure 2.13 is $x_1 = 30$ bowls, $x_2 = 0$ mugs, and $Z = \$2,100$ profit.

Altering the objective function coefficients results in a new solution.

This brief example of the effects of altering the objective function highlights two useful points.

First, the optimal extreme point is determined by the objective function, and an extreme point on one axis of the graph is as likely to be the optimal solution as is an extreme point on a different axis.

Second, the solution is sensitive to the values of the coefficients in the objective function.

If the objective function coefficients are changed, as in our example, the solution may change. Likewise, if the constraint coefficients are changed, the solution space and solution points may change also.

This information can be of consequence to the decision maker trying to determine how much of a product to produce.

Sensitivity analysis the use of linear programming to evaluate the effects of changes in model parameters is discussed in Chapter 3.

Sensitivity analysis is used to analyze changes in model parameters.

It should be noted that some problems do not have a single extreme point solution. For example, when the objective function line parallels one of the constraint lines, an entire line segment is bounded by two adjacent corner points that are optimal; there is no single extreme point on the objective function line. In this situation there are *multiple optimal solutions*. This and other irregular types of solution outcomes in linear programming are discussed at the end of this chapter.

Multiple optimal solutions can occur when the objective function is parallel to a constraint line.

Slack Variables

Once the optimal solution was found at point B in Figure 2.12, simultaneous equations were solved to determine the values of x_1 and x_2.

Recall that the solution occurs at an extreme point where constraint equation lines intersect with each other or with the axis.

Thus, the model constraints are considered as equations (=) rather than ≤ or ≥ inequalities.

There is a standard procedure for transforming ≤ inequality constraints into equations. This transformation is achieved by adding a new variable, called a *slack variable,* to each constraint. For the pottery company example, the model constraints are

$x_1 + 2x_2 \leq 40$ *hr. of labor*
$4x_1 + 3x_2 \leq 120$ *lb. of clay*

A slack variable is added to a ≤ constraint to convert it to an equation (=).

The addition of a unique slack variable, s_1, to the labor constraint and s_2 to the constraint for clay results in the following equations:

$x_1 + 2x_2 + s_1 = 40$ *hr. of labor*
$4x_1 + 3x_2 + s_2 = 120$ *lb. of clay*

A slack variable represents unused resources.

The slack variables in these equations, s_1 and s_2, will take on any value necessary to make the left-hand side of the equation equal to the right-hand side. For example, consider a hypothetical solution of $x_1 = 5$ and $x_2 = 10$. Substituting these values into the foregoing equations yields

Lecture Notes:

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In this example, $x_1 = 5$ bowls and $x_2 = 10$ mugs represent a solution that does not make use of the total available amount of labor and clay. In the labor constraint, 5

bowls and 10 mugs require only 25 hours of labor. This leaves 15 hours that are not used. Thus, s_1 represents the amount of unused labor, or slack.

In the clay constraint, 5 bowls and 10 mugs require only 50 pounds of clay. This leaves 70 pounds of clay unused. Thus, s_2 represents the amount of unused clay. In general, slack variables represent the amount of unused resources.

The ultimate instance of unused resources occurs at the origin, where $x_1 = 0$ and $x_2 = 0$. Substituting these values into the equations yields

$x_1 + 2x_2 + s_1 = 40$
$s_1 = 40$ *hr. of label*
&
$4x_2 + 3x_2 + s_2 = 120$
$s_2 = 120$ *lb.of clay*

Because no production takes place at the origin, all the resources are unused; thus, the slack variables equal the total available amounts of each resource: $s_1 = 40$ hours of labor and $s_2 = 120$ pounds of clay.

What is the effect of these new slack variables on the objective function? The objective function for our example represents the profit gained from the production of bowls and mugs,

$Z = \$40x_1 + \$50x_2$

The coefficient \$40 is the contribution to profit of each bowl; \$50 is the contribution to profit of each mug. What, then, do the slack variables s_1 and s_2 contribute? They contribute nothing to profit because they represent unused resources. Profit is made only after the resources are put to use in making bowls and mugs. Using slack variables, we can write the objective function as

A slack variable contributes nothing to the objective function value.

maximize $Z = \$40x_1 + \$50x_2 + 0s_1 + 0s_2$

As in the case of decision variables (x_1 and x_2), slack variables can have only nonnegative values because negative resources are not possible. Therefore, for this model formulation, x_1, x_2, s_1, and $s_2 \geq 0$.

The complete linear programming model can be written in what is referred to as standard form with slack variables as follows:

$Max\ Z = 40x_1 + 50x_2 + s_1 + s_2$
subject to: $1x_1 + 2x_2 + s_1 = 40$
$4x_2 + 3x_2 + s_2 = 120$
$x_1, x_2, s_1, s_2 \geq 0$
Where:
x_1 = *number of bowls*
x_2 = *number of mugs*

s_1, s_2 are slack variables

The solution values, including the slack at each solution point, are summarized as follows:

Solution Summary with Slack

Point	Solution Values	Z	Slack
A	$x_1 = 0$ bowls, $x_2 = 20$ mugs	\$1,000	$s_1 = 0$ hr.; $s_2 = 60$ lb.
B	$x_1 = 24$ bowls, $x_2 = 8$ mugs	\$1,360	$s_1 = 0$ hr.; $s_2 = 0$ lb.
C	$x_1 = 30$ bowls, $x_2 = 0$ mugs	\$1,200	$s_1 = 10$ hr.; $s_2 = 0$ lb.

Figure 2.14 shows the graphical solution of this example, with slack variables included at each solution point.

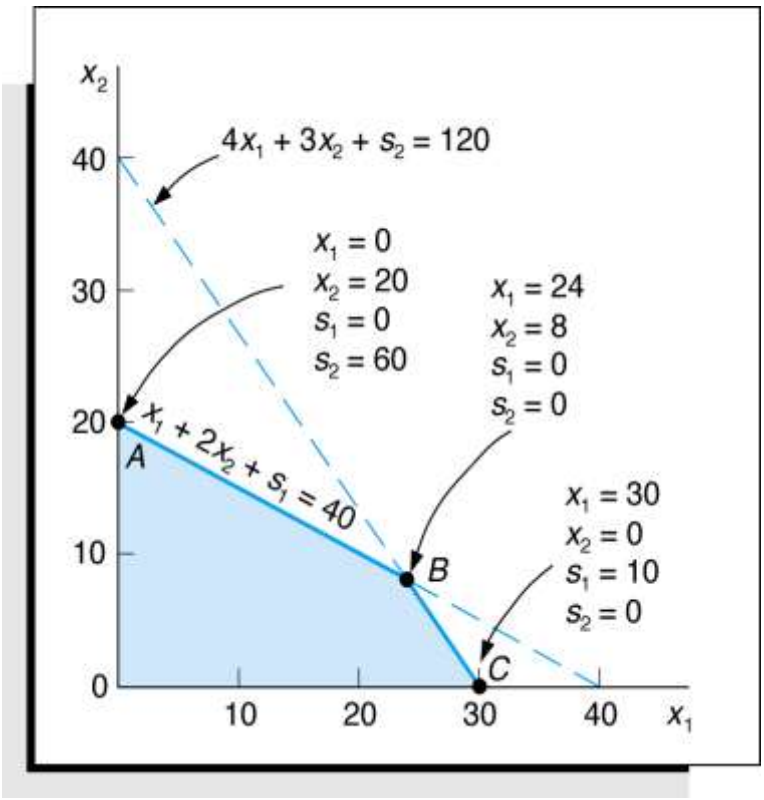


Figure 2.14. Solutions at points A, B, and C with slack

Summary of the Graphical Solution Steps

The steps for solving a graphical linear programming model are summarized here:

1. Plot the model constraints as equations on the graph; then, considering the inequalities of the constraints, indicate the feasible solution area.
2. Plot the objective function; then move this line out from the origin to locate the optimal solution point.
3. Solve simultaneous equations at the solution point to find the optimal solution values.

Or

2. Solve simultaneous equations at each corner point to find the solution values at each point.
3. Substitute these values into the objective function to find the set of values that results in the maximum Z value.

A Minimization Model Example

As mentioned at the beginning of this chapter, there are two types of linear programming problems: maximization problems (like the Beaver Creek Pottery Company example) and minimization problems.

A minimization problem is formulated the same basic way as a maximization problem, except for a few minor differences. The following sample problem will demonstrate the formulation of a minimization model.

A farmer is preparing to plant a crop in the spring and needs to fertilize a field. There are two brands of fertilizer to choose from, Super-gro and Crop-quick. Each brand yields a specific amount of nitrogen and phosphate per bag, as follows:

Chemical Contribution

Brand	NITROGEN (LB./BAG)	PHOSPHATE (LB./BAG)
Super-gro	2	4
Crop-quick	4	3

The farmer's field requires at least 16 pounds of nitrogen and 24 pounds of phosphate. Super-gro costs $6 per bag, and Crop-quick costs $3. The farmer wants to know how many bags of each brand to purchase in order to minimize the total cost of fertilizing. This scenario is illustrated in Figure 2.15.

The steps in the linear programming model formulation process are summarized as follows:

Summary of LP Model Formulation Steps

Step 1. Define the decision variables

How many bags of Super-gro and Crop-quick to buy

Step 2. Define the objective function

Minimize cost

Step 3. Define the constraints

The field requirements for nitrogen and phosphate

Decision Variables

This problem contains two decision variables, representing the number of bags of each brand of fertilizer to purchase:

x_1 = bags of Super-gro

x_2 = bags of Crop-quick

The Objective Function

The farmer's objective is to minimize the total cost of fertilizing. The total cost is the sum of the individual costs of each type of fertilizer purchased. The objective function that represents total cost is expressed as

minimize $Z = \$6x_1 + 3x_2$

where

$\$6x_1$ = cost of bags of Super-gro

$\$3x_2$ = cost of bags of Crop-quick

Model Constraints

The requirements for nitrogen and phosphate represent the constraints of the model. Each bag of fertilizer contributes a number of pounds of nitrogen and phosphate to the field. The constraint for nitrogen is

$2x_1 + 4x_2 \geq 16$ *lb.*

where

$2x_1$ = *the nitrogen contribution (lb.) per bag of Super-gro*

$4x_2$ = *the nitrogen contribution (lb.) per bag of Crop-quick*

Rather than a ≤ (less than or equal to) inequality, as used in the Beaver Creek Pottery Company model, this constraint requires a ≥ (greater than or equal to) inequality. This is because the nitrogen content for the field is a minimum requirement specifying that at least 16 pounds of nitrogen be deposited on the farmer's field. If a minimum cost solution results in more than 16 pounds of nitrogen on the field, that is acceptable; however, the amount cannot be less than 16 pounds.

The constraint for phosphate is constructed like the constraint for nitrogen:

$4x_1 + 3x_2 \geq 24$ *lb.*

With this example we have shown two of the three types of linear programming model constraints, ≤ and ≥ . The third type is an exact equality, =. This type specifies that a constraint requirement must be exact. For example, if the farmer had said that the phosphate requirement for the field was exactly 24 pounds, the constraint would have been

The three types of linear programming constraints are ≤, =, and ≥ .

$4x_1 + 3x_2 = 24$ *lb.*

As in our maximization model, there are also nonnegativity constraints in this problem to indicate that negative bags of fertilizer cannot be purchased:

$x_1, x_2 \geq 0$

The complete model formulation for this minimization problem is

Minimize $Z = \$6x_1 + \$3x_2$

subject to: $2x_1 + 4x_2 \geq 16$

$4x_2 + 3x_2 \geq 24$

$x_1, x_2 \geq 0$

Graphical Solution of a Minimization Model

We follow the same basic steps in the graphical solution of a minimization model as in a maximization model. The fertilizer example will be used to demonstrate the graphical solution of a minimization model.

The first step is to graph the equations of the two model constraints, as shown in Figure 2.16. Next, the feasible solution area is chosen, to reflect the ≥ inequalities in the constraints, as shown in Figure 2.17.

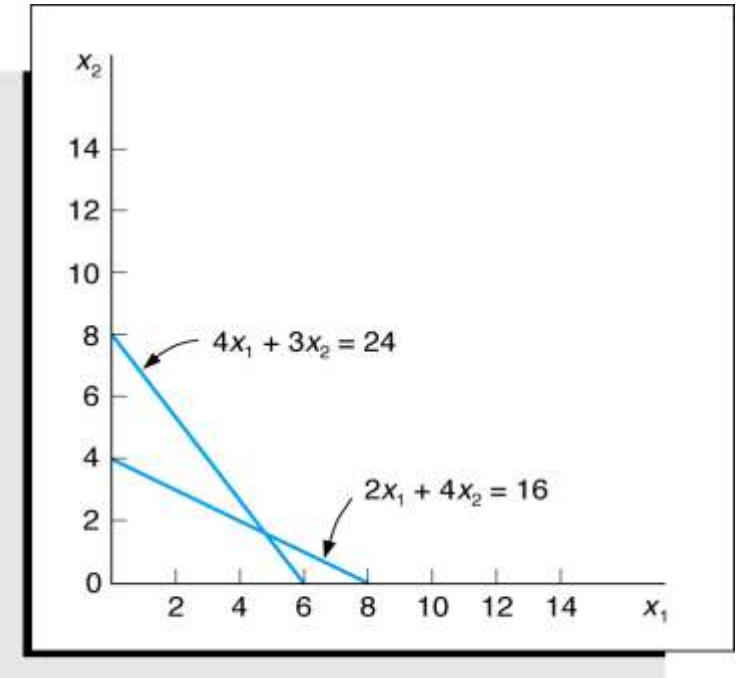


Figure 2.16. Constraint lines for fertilizer model

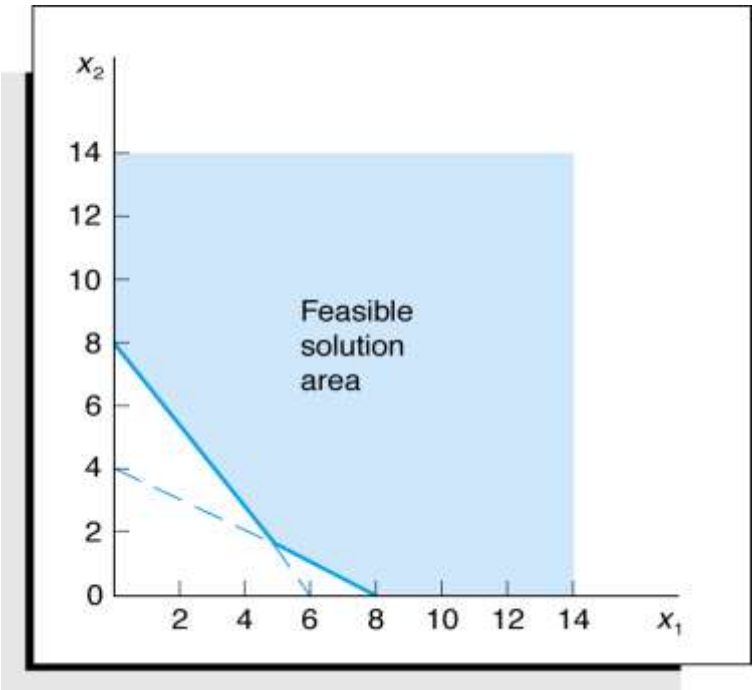


Figure 2.17. Feasible solution area

After the feasible solution area has been determined, the second step in the graphical solution approach is to locate the optimal point. Recall that in a maximization problem, the optimal solution is on the boundary of the feasible solution area that

contains the point(s) farthest from the origin. The optimal solution point in a minimization problem is also on the boundary of the feasible solution area; however, the boundary contains the point(s) closest to the origin (zero being the lowest cost possible).

The optimal solution of a minimization problem is at the extreme point closest to the origin.

As in a maximization problem, the optimal solution is located at one of the extreme points of the boundary.

In this case the corner points represent extremities in the boundary of the feasible solution area that are closest to the origin. Figure 2.18 shows the three corner pointsA, B, and Cand the objective function line.

As the objective function edges toward the origin, the last point it touches in the feasible solution area is A.

In other words, point A is the closest the objective function can get to the origin without encompassing infeasible points. Thus, it corresponds to the lowest cost that can be attained.

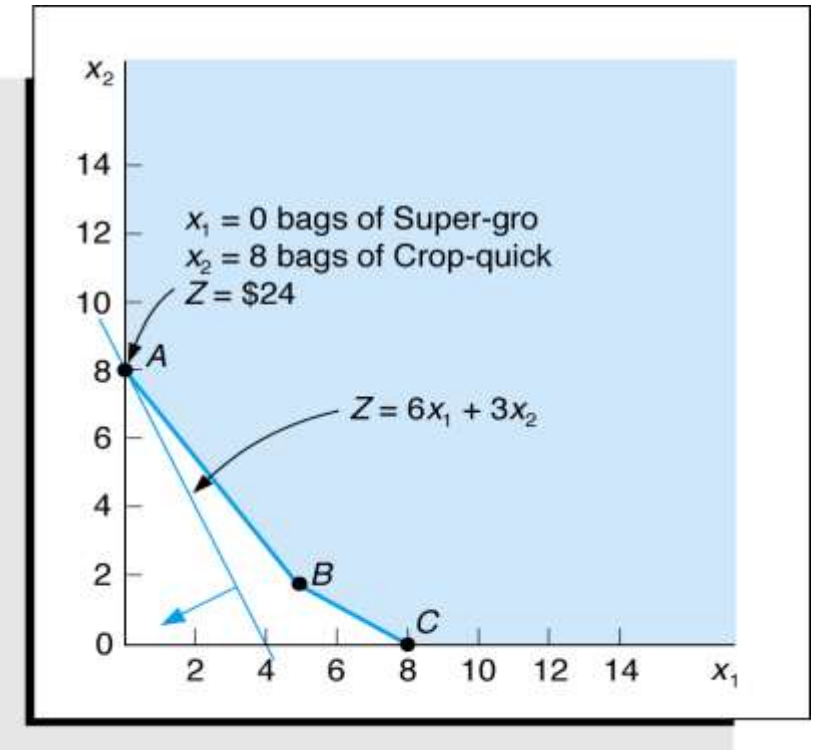


Figure 2.18. The optimal solution point

The final step in the graphical solution approach is to solve for the values of x_1 and x_2 at point A. Because point A is on the x_2 axis, $x_1 = 0$; thus,

$x_2=8$

Given that the optimal solution is $x_1 = 0$, $x_2 = 8$, the minimum cost, Z, is

$Z = \$6x_1 + \$3x_2$

$Z = \$24$

This means the farmer should not purchase any Super-gro but, instead, should purchase eight bags of Crop-quick, at a total cost of $24.

Surplus Variables

Greater than or equal to constraints cannot be converted to equations by adding slack variables, as with ≤ constraints. Recall our fertilizer model, formulated as

Minimize $Z = \$6x_1 + \$3x_2$

subject to: $2x_1 + 4x_2 \geq 16$

$4x_2 + 3x_2 \geq 24$

$x_1, x_2 \geq 0$

where

x_1 = *bags of Super-gro fertilizer*

x_2 = *bags of Crop-quick fertilizer*

Z = farmer's total cost ($) of purchasing fertilizer

Because this problem has ≥ constraints as opposed to the ≤ constraints of the Beaver Creek Pottery Company maximization example, the constraints are converted to equations a little differently.

A surplus variable is subtracted from a ≥ constraint to convert it to an equation (=).

Instead of adding a slack variable with a ≥ constraint, we subtract *a surplus variable*. Whereas a slack variable is added and reflects unused resources, a surplus variable is subtracted and reflects the excess above a minimum resource requirement level. Like a slack variable, a surplus variable is represented symbolically by s_1 and must be nonnegative.

For the nitrogen constraint, the subtraction of a surplus variable gives

A surplus variable represents an excess above a constraint requirement level.

$2x_1 + 4x_2\ s_1 = 16$

The surplus variable s_1 transforms the nitrogen constraint into an equation.

As an example, consider the hypothetical solution

$x_1 = 0$

$x_2 = 10$

Substituting these values into the previous equation yields

$2x_1 + 4x_2 - s_1 = 16$

$s_1 = 24$ *lb. of nitrogen*

In this equation s_1 can be interpreted as the extra amount of nitrogen above the minimum requirement of 16 pounds that would be obtained by purchasing 10 bags of Crop-quick fertilizer.

In a similar manner, the constraint for phosphate is converted to an equation by subtracting a surplus variable, s_2:

$4x_1 + 3x_2\ s_2 = 24$

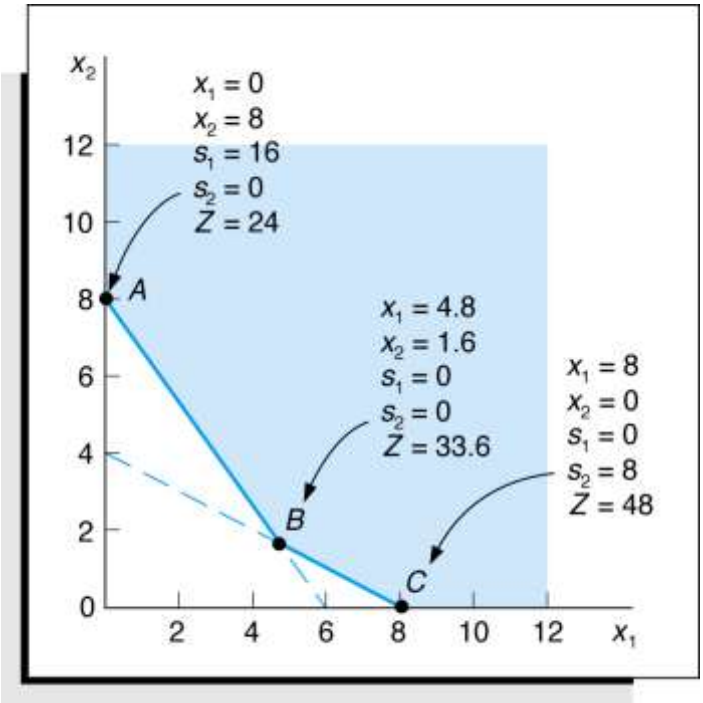


Figure 2.19. Graph of the fertilizer example

As is the case with slack variables, surplus variables contribute nothing to the overall cost of a model. For example, putting additional nitrogen or phosphate on the field will not affect the farmer's cost; the only thing affecting cost is the number of bags of

fertilizer purchased. As such, the standard form of this linear programming model is summarized as

Minimize $Z = \$6x_1 + \$3x_2 + 0s_1 + 0s_2$

subject to: $2x_1 + 4x_2 - s_1 = 16$

$4x_2 + 3x_2 - s_2 = 24$

$x_1, x_2, s_1, s_2 \geq 0$

Figure 2.19 shows the graphical solutions for our example, with surplus variables included at each solution point.

Irregular Types of Linear Programming Problems

The basic forms of typical maximization and minimization problems have been shown in this chapter. However, there are several special types of atypical linear programming problems. Although these special cases do not occur frequently, they will be described so that you can recognize them when they arise. These special types include problems with more than one optimal solution, infeasible problems, and problems with unbounded solutions.

For some linear programming models, the general rules do not always apply.

Multiple Optimal Solutions

Consider the Beaver Creek Pottery Company example, with the objective function changed from $Z = 40x_1 + 50x_2$ to $Z = 40x_1 + 30x_2$:

Maximize $Z=\$40x_1 + 30x_2$

subject to: $1x_1 + 2x_2 \leq 40$

$4x_2 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$

Where:

x_1 = *number of bowls*

x_2 = *number of mugs*

The graph of this model is shown in Figure 2.20. The slight change in the objective function makes it now parallel to the constraint line, $4x_1 + 3x_2 = 120$. Both lines now

have the same slope of 4/3. Therefore, as the objective function edge moves outward from the origin, it touches the whole line segment BC rather than a single extreme corner point before it leaves the feasible solution area. This means that every point along this line segment is optimal (i.e., each point results in the same profit of Z = $1,200). The endpoints of this line segment, B and C, are typically referred to as the alternate optimal solutions. It is understood that these points represent the endpoints of a range of optimal solutions.

Alternate optimal solutions are at the endpoints of the constraint line segment that the objective function parallels.

Multiple optimal solutions provide greater flexibility to the decision maker.

The pottery company, therefore, has several options in deciding on the number of bowls and mugs to produce. Multiple optimal solutions can benefit the decision maker because the number of decision options is enlarged. The multiple optimal solutions (along the line segment BC in Figure 2.20) allow the decision maker greater flexibility. For example, in the case of Beaver Creek Pottery Company, it may be easier to sell bowls than mugs; thus, the solution at point C, where only bowls are produced, would be more desirable than the solution at point B, where a mix of bowls and mugs is produced.

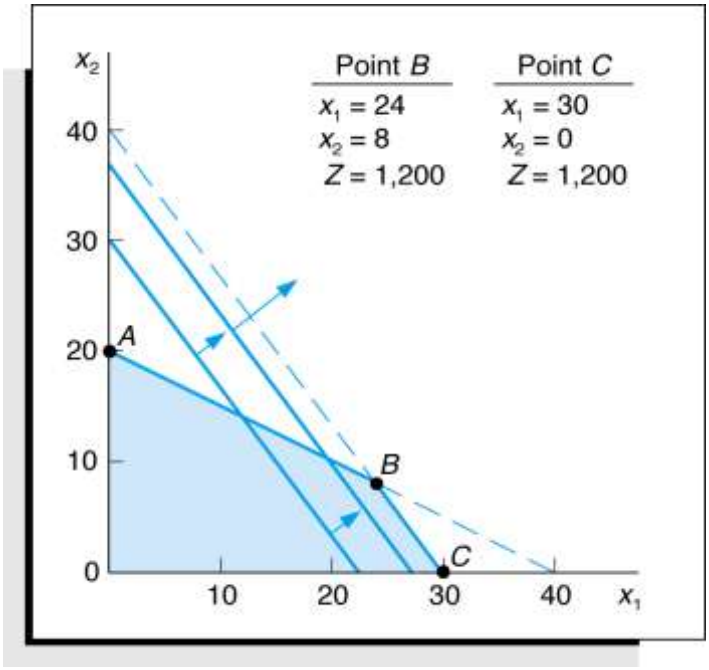


Figure 2.20. Graph of the Beaver Creek Pottery Company example with multiple optimal solutions

An Infeasible Problem

In some cases a linear programming problem has no feasible solution area; thus, there is no solution to the problem. An example of an infeasible problem is formulated next and depicted graphically in Figure 2.21:

Maximize $Z = 5x_1 + 3x_2$

subject to: $4x_1 + 2x_2 \leq 8$

$x_1 \geq 4$

$x_2 \geq 6$

$x_1, x_2 \geq 0$

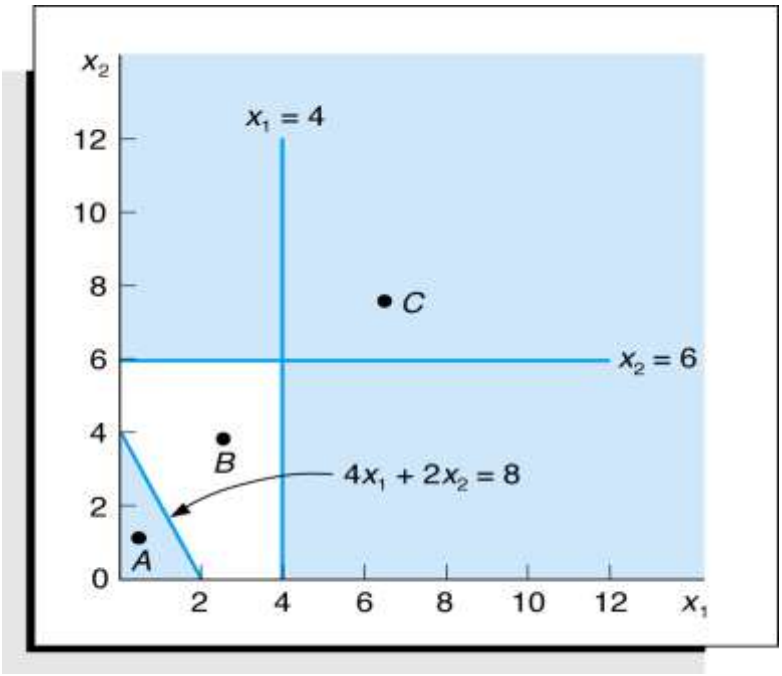


Figure 2.21. Graph of an infeasible problem

An infeasible problem has no feasible solution area; every possible solution point violates one or more constraints.

Point A in Figure 2.21 satisfies only the constraint $4x_1 + 2x_2 \leq 8$, whereas point C satisfies only the constraints $x_1 \geq 4$ and $x_2 \geq 6$. Point B satisfies none of the constraints. The three constraints do not overlap to form a feasible solution area. Because no point satisfies all three constraints simultaneously, there is no solution to the problem. Infeasible problems do not typically occur, but when they do, they are usually a result of errors in defining the problem or in formulating the linear programming model.

An Unbounded Problem

In some problems the feasible solution area formed by the model constraints is not closed. In these cases it is possible for the objective function to increase indefinitely without ever reaching a maximum value because it never reaches the boundary of the feasible solution area.

In an unbounded problem the objective function can increase indefinitely without reaching a maximum value.

An example of this type of problem is formulated next and shown graphically in Figure 2.22:

Maximize $Z = 4x_1 + 2x_2$

subject to: $x_1 \geq 4$

$x_2 \leq 2$

$x_1, x_2 \geq 0$

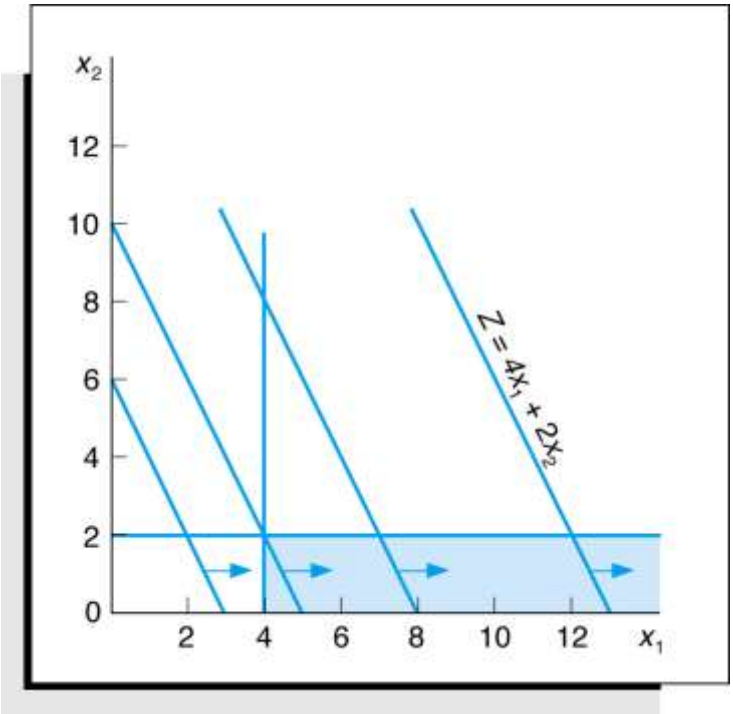


Figure 2.22. An unbounded problem

In Figure 2.22 the objective function is shown to increase without bound; thus, a solution is never reached.

The solution space is not completely closed in.

Unlimited profits are not possible in the real world; an unbounded solution, like an infeasible solution, typically reflects an error in defining the problem or in formulating the model.

Characteristics of Linear Programming Problems

Now that we have had the opportunity to construct several linear programming models, let's review the characteristics that identify a linear programming problem.

The components of a linear programming model are an objective function, decision variables, and constraints.

A linear programming problem requires a choice between alternative courses of action (i.e., a decision). The decision is represented in the model by decision variables. A typical choice task for a business firm is deciding how much of several different products to produce, as in the Beaver Creek Pottery Company example presented earlier in this chapter. Identifying the choice task and defining the decision variables is usually the first step in the formulation process because it is quite difficult to construct the objective function and constraints without first identifying the decision variables.

The problem encompasses an objective that the decision maker wants to achieve. The two most frequently encountered objectives for a business are maximizing profit and minimizing cost.

A third characteristic of a linear programming problem is that restrictions exist, making unlimited achievement of the objective function impossible. In a business firm these restrictions often take the form of limited resources, such as labor or material; however, the sample models in this chapter exhibit a variety of problem restrictions. These restrictions, as well as the objective, must be definable by mathematical functional relationships that are linear. Defining these relationships is typically the most difficult part of the formulation process.

Properties of Linear Programming Models

In addition to encompassing only linear relationships, a linear programming model also has several other implicit properties, which have been exhibited consistently throughout the examples in this chapter. The term linear not only means that the functions in the models are graphed as a straight line; it also means that the relationships exhibit proportionality. In other words, the rate of change, or slope, of the function is constant; therefore, changes of a given size in the value of a decision variable will result in exactly the same relative changes in the functional value.

Proportionality means the slope of a constraint or objective function line is constant.

Linear programming also requires that the objective function terms and the constraint terms be additive. For example, in the Beaver Creek Pottery Company model, the total profit (Z) must equal the sum of profits earned from making bowls ($\$40x_1$) and mugs ($\$50x_2$). Also, the total resources used must equal the sum of the resources used for each activity in a constraint (e.g., labor).

The terms in the objective function or constraints are additive.

Another property of linear programming models is that the solution values (of the decision variables) cannot be restricted to integer values; the decision variables can take on any fractional value. Thus, the variables are said to be continuous or divisible, as opposed to integer or discrete. For example, although decision variables

representing bowls or mugs or airplanes or automobiles should realistically have integer (whole number) solutions, the solution methods for linear programming will not necessarily provide such solutions. This is a property that will be discussed further as solution methods are presented in subsequent chapters.

The values of decision variables are continuous or divisible.

The final property of linear programming models is that the values of all the model parameters are assumed to be constant and known with certainty. In real situations, however, model parameters are frequently uncertain because they reflect the future as well as the present, and future conditions are rarely known with certainty.

All model parameters are assumed to be known with certainty.

Then:

* To summarize, a linear programming model has the following general properties: linearity, proportionality, additivity, divisibility, and certainty

* The two example problems in this chapter were formulated as linear programming models in order to demonstrate the modeling process. These problems were similar in that they concerned achieving some objective subject to a set of restrictions or requirements. Linear programming models exhibit certain common characteristics:

- An objective function to be maximized or minimized
- A set of constraints
- Decision variables for measuring the level of activity
- Linearity among all constraint relationships and the objective function

The graphical approach to the solution of linear programming problems is not a very efficient means of solving problems. For one thing, drawing accurate graphs is tedious. Moreover, the graphical approach is limited to models with only two decision variables. However, the analysis of the graphical approach provides valuable insight into linear programming problems and their solutions.

In the graphical approach, once the feasible solution area and the optimal solution point have been determined from the graph, simultaneous equations are solved to determine the values of x_1 and x_2 at the solution point.

Solved problems:

1. Moore's Meatpacking Company produces a hot dog mixture in 1,000-pound batches. The mixture contains two ingredients chicken and beef. The cost per pound of each of these ingredients is as follows:

Ingredient	Cost/lb.
Chicken	$3
Beef	$5

Each batch has the following recipe requirements:

a. At least 500 pounds of chicken
b. At least 200 pounds of beef

The ratio of chicken to beef must be at least 2 to 1. The company wants to know the optimal mixture of ingredients that will minimize cost. Formulate a linear programming model for this problem.

Solution

Step 1:

Identify decision variables.

x_1 = lb of chicken in mixture (1000 lb.)

x_2 = lb of beef in mixture (1000 lb.)

Step 2:

Formulate the objective function.

Minimize $Z = \$3x_1 + \$5x_2$

where Z = cost per 1,000-lb batch

$\$3x_1$ = cost of chicken

$\$5x_2$ = cost of beef

Step 3:

Establish Model Constraints

$x_1 + x_2 = 1{,}000$ lb

$x_1 \geq 500$ lb of chicken

$x_2 \geq 200$ lb of beef

$x_1/x_2 \geq 2/1$ or $x_1 - 2x_2 \geq 0$

$x_1, x_2 \geq 0$

The Model:

$$\text{Minimize } Z = \$3x_1 + 5x_2$$
$$\text{subject to: } x_1 + x_2 = 1{,}000 \text{ lb}$$
$$x_1 \geq 50$$
$$x_2 \geq 200$$
$$x_1 - 2x_2 \geq 0$$
$$x_1, x_2 \geq 0$$

2. Solve the following model graphically:

$$\text{Maximize } Z = 4x_1 + 5x_2$$

$$\text{subject to: } x_1 + 2x_2 \leq 10$$

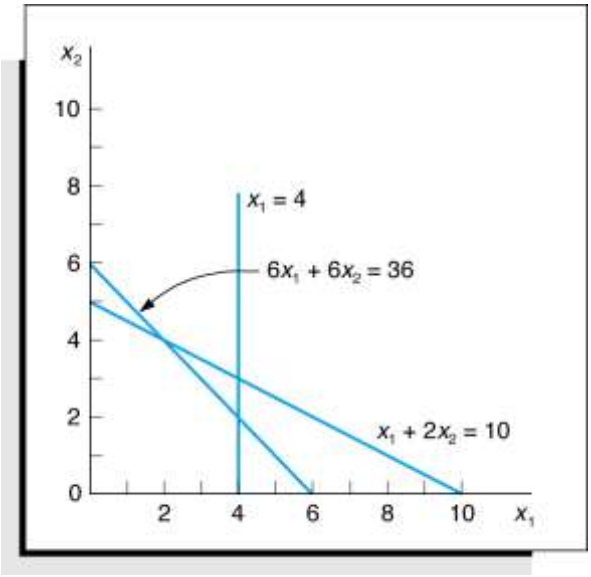
$$6x_1 + 6x_2 \leq 36$$

$$x_1 \leq 4$$

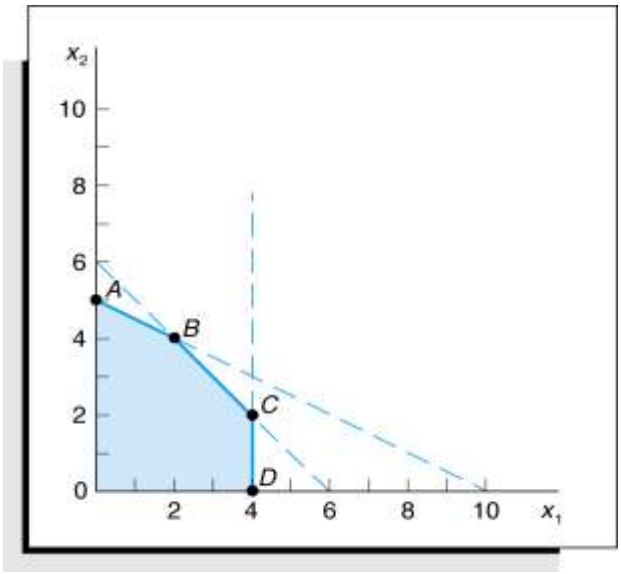
$$x_1, x_2 \geq 0$$

Solution

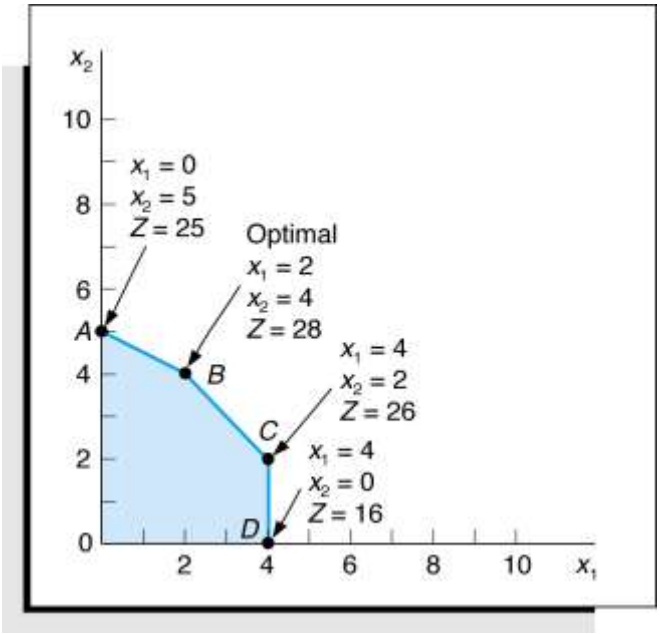
Step 1: Plot the constraints as equations



Step 2: Determine the feasible solution space



Step 3 and 4: Determine the solution points and optimal solution



Problems:

1. A company produces two products that are processed on two assembly lines. Assembly line 1 has 100 available hours, and assembly line 2 has 42 available hours. Each product requires 10 hours of processing time on line 1, while on line 2 product 1 requires 7 hours and product 2 requires 3 hours. The profit for product 1 is $6 per unit, and the profit for product 2 is $4 per unit.

a. Formulate a linear programming model for this problem.
b. Solve this model by using graphical analysis.

2. The Pinewood Furniture Company produces chairs and tables from two resourceslabor and wood. The company has 80 hours of labor and 36 pounds of wood available each day. Demand for chairs is limited to 6 per day. Each chair requires 8 hours of labor and 2 pounds of wood, whereas a table requires 10 hours of labor and 6 pounds of wood. The profit derived from each chair is $400 and from each table, $100. The company wants to determine the number of chairs and tables to produce each day in order to maximize profit.

a. Formulate a linear programming model for this problem.
b. Solve this model by using graphical analysis.

3. A jewelry store makes necklaces and bracelets from gold and platinum. The store has 18 ounces of gold and 20 ounces of platinum. Each necklace requires 3 ounces of gold and 2 ounces of platinum, whereas each bracelet requires 2 ounces of gold and 4 ounces of platinum. The demand for bracelets is no more than four. A necklace earns $300 in profit and a bracelet, $400. The store wants to determine the number of necklaces and bracelets to make in order to maximize profit.

a. Formulate a linear programming model for this problem.
b. Solve this model by using graphical analysis.

4. Universal Claims Processors processes insurance claims for large national insurance companies. Most claim processing is done by a large pool of computer operators, some of whom are permanent and some of whom are temporary. A permanent operator can process 16 claims per day, whereas a temporary operator can process 12 per day, and on average the company processes at least 450 claims each day. The company has 40 computer workstations. A permanent operator generates about 0.5 claims with errors each day, whereas a temporary operator averages about 1.4 defective claims per day. The company wants to limit claims with errors to 25 per day. A permanent operator is paid $64 per day, and a temporary operator is paid $42 per day. The company wants to determine the number of permanent and temporary operators to hire in order to minimize costs.

a. Formulate a linear programming model for this problem.

b. Solve this model by using graphical analysis.

5. Solve the following linear programming model graphically:

$$\text{Minimize } Z = 3x_1 + 6x_2$$
$$\text{subject to: } 3x_1 + 2x_2 \leq 18$$
$$x_1 + x_2 \leq 5$$
$$x_1 \leq 4$$
$$x_2 \leq 7$$
$$x_2 / x_1 \leq 7/8$$
$$x_1, x_2 \geq 0$$

6. Solve the following linear programming model graphically:

$$\text{Minimize } Z = 8X1 + 2X2$$

S.T.
$$2X1 - 6X2 \leq 12$$
$$5X1 + 4X2 \geq 40$$
$$X1 + 2X2 \geq 12$$
$$X2 \leq 6$$
$$X1, X2 \geq 0$$

7. Landm's Restaurant has an ice-cream counter where it sells two main products, ice cream and frozen yogurt, each in a variety of flavors. The restaurant makes one order for ice cream and yogurt each week, and the store has enough freezer space for 115 gallons total of both products. A gallon of frozen yogurt costs $0.75 and a gallon of ice cream costs $0.93, and the restaurant budgets $90 each week for these products. The manager estimates that each week the restaurant sells at least twice as much ice cream as frozen yogurt. Profit per gallon of ice cream is $4.15, and profit per gallon of yogurt is $3.60.

a. Formulate a linear programming model for this problem.
b. Solve this model by using graphical analysis.

8. A manufacturing firm produces two products. Each product must undergo an assembly process and a finishing process. It is then transferred to the warehouse, which has space for only a limited number of items. The firm has 80 hours available for assembly and 112 hours for finishing, and it can store a maximum of 10 units in the warehouse. Each unit of product 1 has a profit of $30 and requires 4 hours to assemble and 14 hours to finish. Each unit of product 2 has a profit of $70 and requires 10 hours to assemble and 8 hours to finish. The firm wants to determine the quantity of each product to produce in order to maximize profit.

a. Formulate a linear programming model for this problem.
b. Solve this model by using graphical analysis.

9. Professor Mohamed teaches two sections of business statistics, which combined will result in 120 final exams to be graded. Professor Mohamed has two graduate assistants. Ahmed and Amr, who will grade the final exams. There is a 3-day period between the time the exam is administered and when final grades must be posted. During this period Ahmed has 12 hours available and Amr has 10 hours available to grade the exams. It takes Ahmed an average of 7.2 minutes to grade an exam, and it takes Amr 12 minutes to grade an exam; however, Ahmed 's exams will have errors that will require Professor Mohamed to ultimately regrade 10% of the exams, while only 6% of Amr's exams will require regrading. Professor Mohamed wants to know how many exams to assign to each graduate assistant to grade in order to minimize the number of exams to regrade.

a. Formulate a linear programming model for this problem.
b. Solve this model by using graphical analysis.

CHAPTER 3. LINEAR PROGRAMMING: COMPUTER SOLUTION AND SENSITIVITY ANALYSIS

Chapter 2 demonstrated how a linear programming model is formulated and how a solution can be derived from a graph of the model.

The fact that this solution method is limited to problems with only two decision variables restricts its usefulness as a general solution technique.

In this chapter we will show how linear programming problems can be solved using several personal computer software packages (we use only POMQM).

We will also describe how to use a computer solution result to experiment with a linear programming model to see what effect parameter changes have on the optimal solution, referred to as sensitivity analysis.

Computer Solution

When linear programming was first developed in the 1940s, virtually the only way to solve a problem was by using a lengthy manual mathematical solution procedure called the **simplex method**. However, during the next six decades, as computer technology evolved, the computer was used more and more to solve linear programming models. The mathematical steps of the simplex method were simply programmed in prewritten software packages designed for the solution of linear programming problems. The ability to solve linear programming problems quickly and cheaply on the computer, regardless of the size of the problem, popularized linear programming and expanded its use by businesses. There are currently dozens of software packages with linear programming capabilities. Many of these are general-purpose management science or quantitative methods packages with linear programming modules, among many other modules for other techniques. There are also numerous software packages that are devoted exclusively to linear programming and its derivatives. These packages are generally cheap, efficient, and easy to use.

The simplex method is a procedure involving a set of mathematical steps to solve linear programming problems.

In the next few sections we demonstrate how to solve linear programming problems by using QM for Windows, a typical general-purpose quantitative methods software package.

QM for Windows

We will demonstrate how to use QM for Windows by solving our Beaver Creek Pottery Company example. The linear programming module in QM for Windows is accessed by clicking on "Module" at the top of the initial window. This will bring down a menu with all the program modules available in QM for Windows, as shown in Exhibit 3.7. By clicking on "Linear Programming," a window for this program will

come up on the screen, and by clicking on "File" and then "New," a screen for inputting problem dimensions will appear. Exhibit 3.8 shows the data entry screen with the model type and the number of constraints and decision variables for our Beaver Creek Pottery Company example.

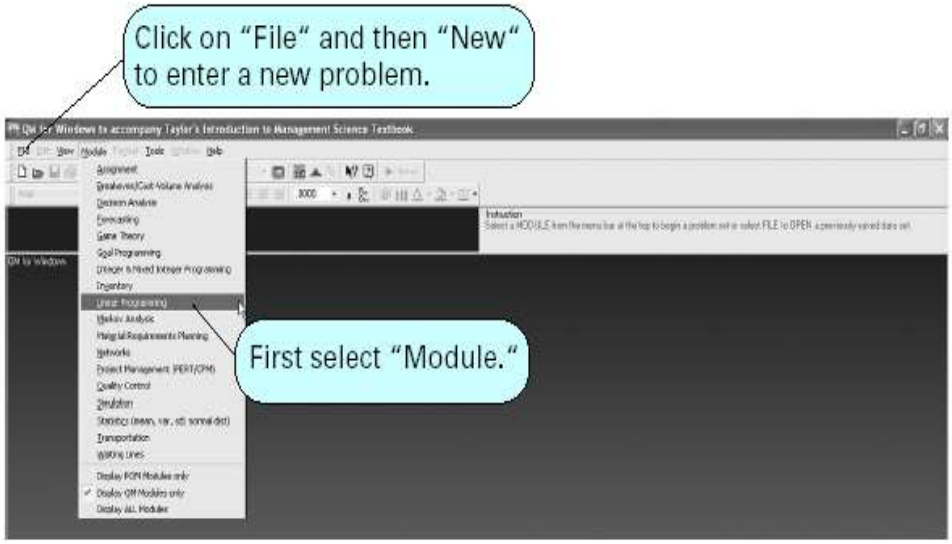


Exhibit 3.7

Exhibit 3.8

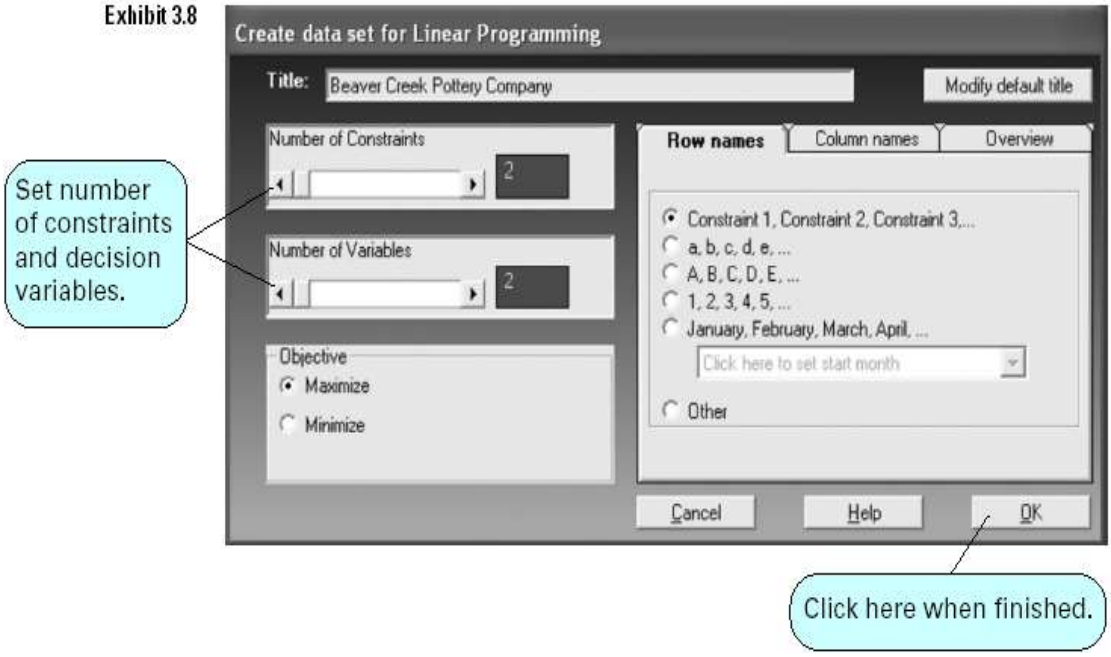


Exhibit 3.9 shows the data table for our example, with the model parameters, including the objective function and constraint coefficients, and the constraint

quantity values. Notice that we also customized the row headings for the constraints by typing in "Labor (hrs)" and "Clay (lbs)." Once the model parameters have been input, click on "Solve" to get the model solution, as shown in Exhibit 3.10. It is not necessary to put the model into standard form by adding slack variables.

Exhibit 3.9

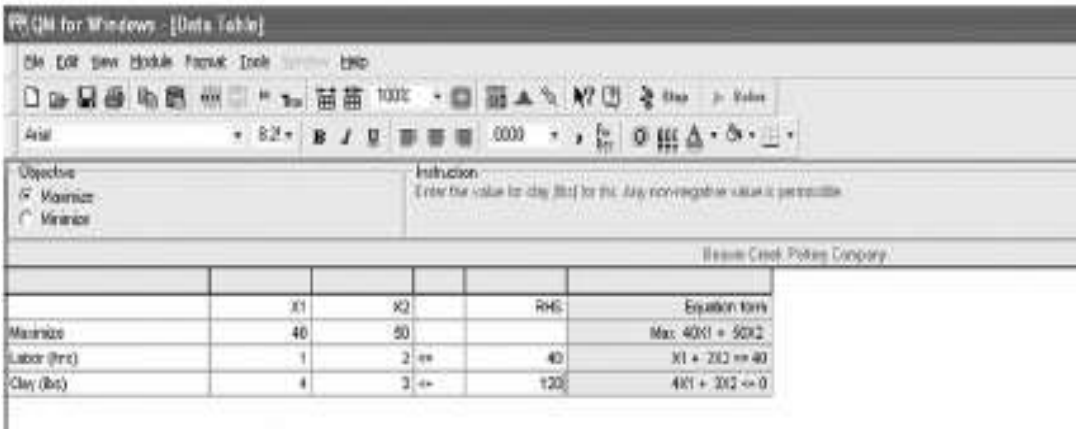
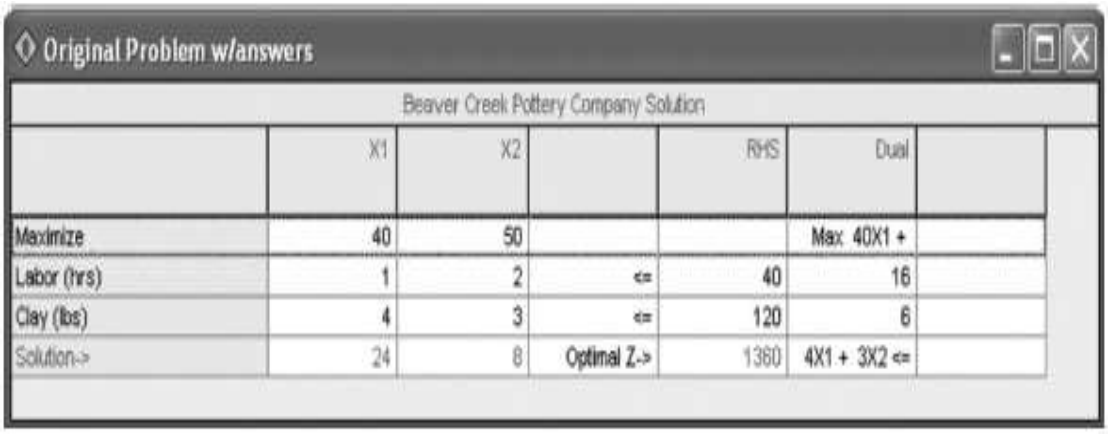


Exhibit 3.10



Notice the values 16 and 6 under the column labeled "Dual" for the "Labor" and "Clay" rows. These dual values are the marginal values of labor and clay in our problem. This is useful information that is provided in addition to the normal model solution values when you solve a linear programming model.

the marginal value is the dollar amount the company would be willing to pay for one additional unit of a resource. For example, the dual value of 16 for the labor constraint means that if 1 additional hour of labor could be obtained by the company, it would increase profit by $16. Likewise, if 1 additional pound of clay could be obtained, it would increase profit by

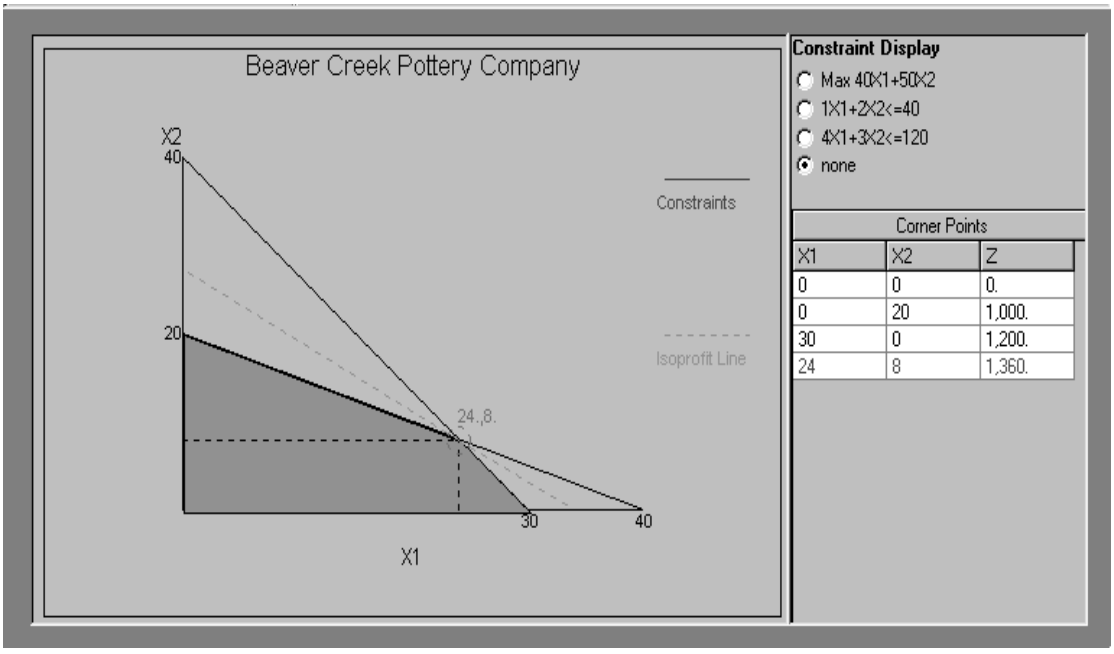
$6. Thus, the company would be willing to pay up to $16 for 1 more hour of labor and $6 for 1 more pound of clay. The dual value is not the purchase price of one of these resources; it is the maximum amount the company would pay to get more of the resource. These dual values are helpful to the company in making decisions about acquiring additional resources.

T*hen :*

The marginal value is the dollar amount one would be willing to pay for one additional resource unit.

QM for Windows provides solution results in several different formats, including a graphical solution by clicking on "Window" and then selecting "Graph." Exhibit 3.11 shows the graphical solution for our Beaver Creek Pottery Company example.

Exhibit 3.11



Hint:

Lab section (go to the lab for more illustration)

Linear Programming Problem: Standard Form

- Standard form requires all variables in the constraint equations to appear on the left of the inequality (or equality) and all numeric values to be on the right-hand side.

- Examples:
  - $x_3 \geq x_1 + x_2$ must be converted to $x_3 - x_1 - x_2 \geq 0$
  - $x_1/(x_2 + x_3) \geq 2$ becomes $x_1 \geq 2(x_2 + x_3)$

  and then $x_1 - 2x_2 - 2x_3 \geq 0$

Sensitivity Analysis

- Sensitivity analysis determines the effect on the optimal solution of changes in parameter values of the objective function and constraint equations.
- Changes may be reactions to anticipated uncertainties in the parameters or to new or changed information concerning the model.

Changes in Objective Function Coefficients

The first model parameter change we will analyze is a change in an objective function coefficient. We will use our now-familiar Beaver Creek Pottery Company example to illustrate this change:

Maximize $Z = \$40x_1 + \$50x_2$

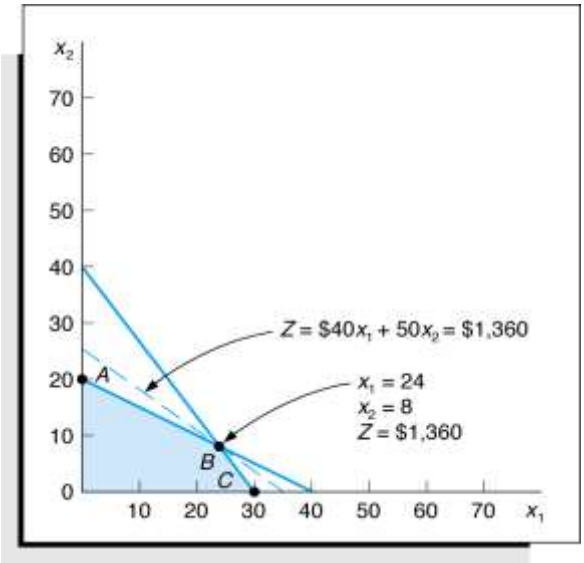
subject to: $1x_1 + 2x_2 \leq 40$

$4x_2 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$

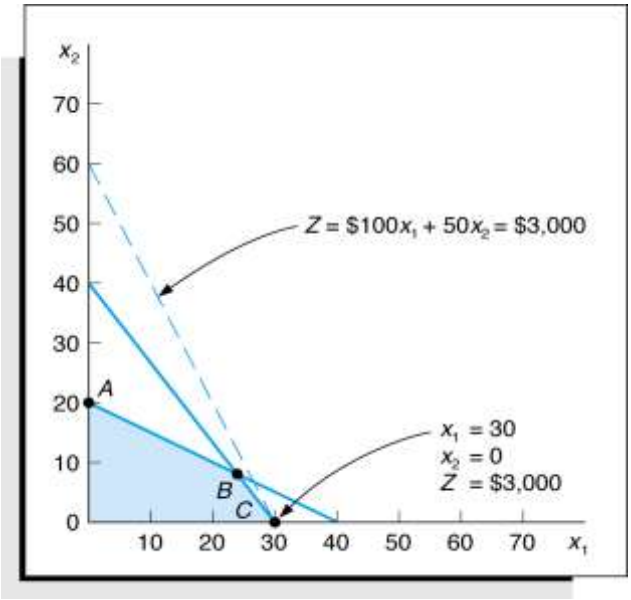
The graphical solution for this problem is shown in Figure 3.1.

Figure 3.1. Optimal solution point



In Figure 3.1 the optimal solution point is shown to be at point B ($x_1 = 24$ and $x_2 = 8$), which is the last point the objective function, denoted by the dashed line, touches as it leaves the feasible solution area. However, what if we changed the profit of a bowl, x_1, from \$40 to \$100? How would that affect the solution identified in Figure 3.1? This change is shown in Figure 3.2.

Figure 3.2. Changing the objective function x_1 coefficient

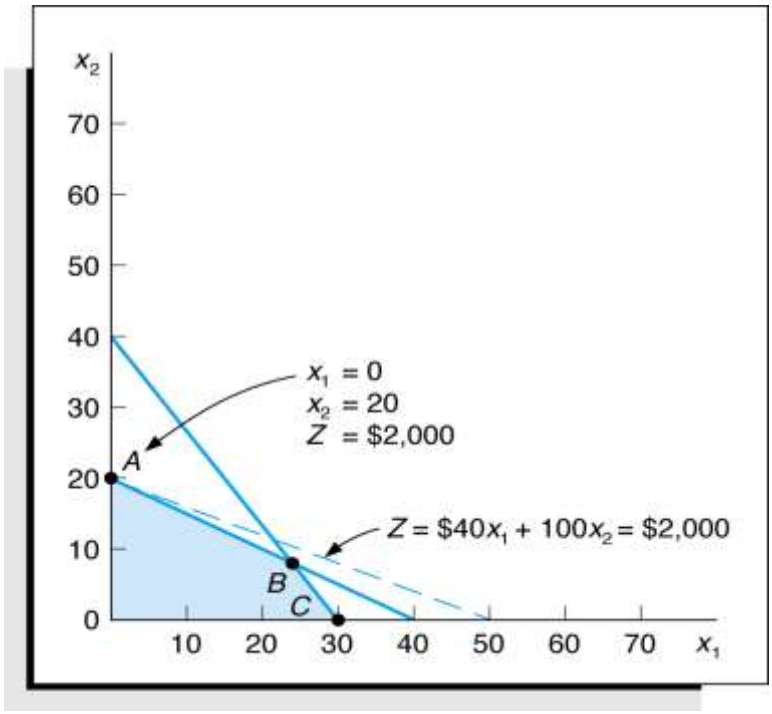


Increasing profit for a bowl (i.e., the x_1 coefficient) from \$40 to \$100 makes the objective function line steeper, so much so that the optimal solution point changes from point B to point C. Alternatively, if we increased the profit for a mug, the x_2 coefficient, from \$50 to \$100, the objective function line would become flatter, to the extent that point A would become optimal, with

$x_1 = 0$, $x_2 = 20$, and Z = \$2,000

This is shown in Figure 3.3.

Figure 3.3. Changing the objective function x_2 coefficient



The sensitivity range for an objective coefficient is the range of values over which the current optimal solution point will remain optimal.

The objective of sensitivity analysis in this case is to determine the range of values for a specific objective function coefficient over which the optimal solution point, x_1 and x_2, will remain optimal. For example, the coefficient of x_1 in the objective function is originally \$40, but at some value greater than \$40, point C will become optimal, and at some value less than \$40, point A will become optimal. The focus of sensitivity analysis is to determine those two values, referred to as the sensitivity range for the x_1 coefficient, which we will designate as c_1.

For our simple example, we can look at the graph in Figure 3.1 and determine the sensitivity range for the x_1 coefficient. The slope of the objective function is currently 4/5, determined as follows:

$Z = 40x_1 + 50x_2$

or

$50x_2 = Z - 40x_1$

and

$x_2 = Z/50 - 4x_1/5$

The objective function is now in the form of the equation of a straight line, y = a + bx, where the intercept, a, equals Z/50 and the slope, b, is 4/5.

If the slope of the objective function increases to 4/3, the objective function line becomes exactly parallel to the constraint line,

$4x_1 + 3x_2 = 120$

and point C becomes optimal (along with B). The slope of this constraint line is 4/3, so we ask ourselves what objective function coefficient for x_1 will make the objective function slope equal 4/3? The answer is determined as follows, where c_1 is the objective function coefficient for x_1:

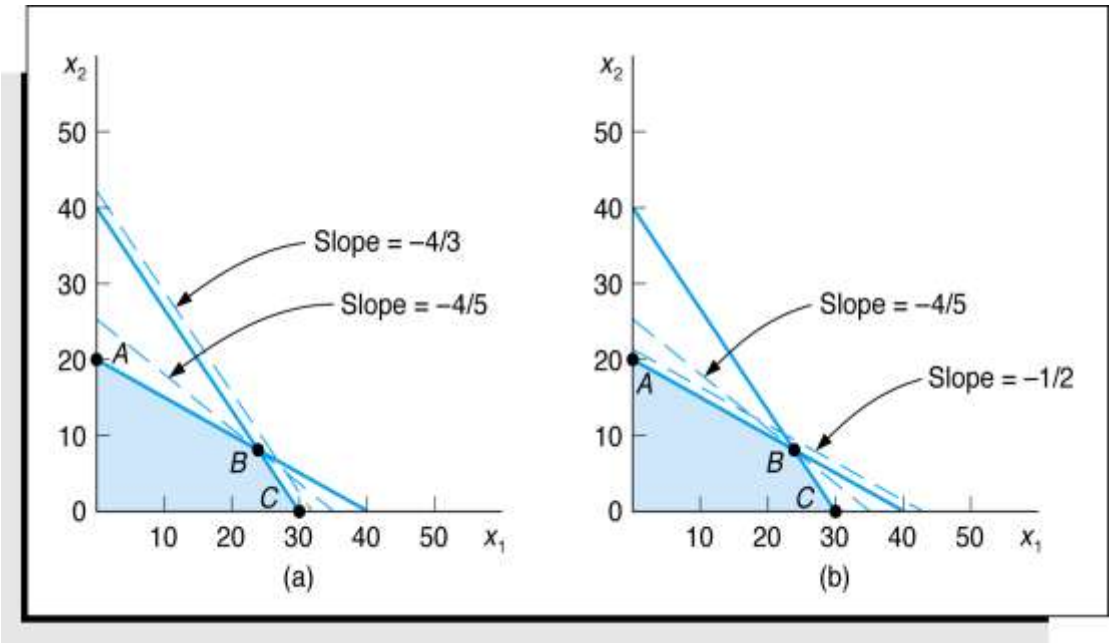
$-c_1/50 = -4/3$

$-3c_1 = -200$

$C_1 = \$66.67$

If the coefficient of x_1 is 66.67, then the objective function will have a slope of 66.67/50 or 4/3. This is illustrated in Figure 3.4(a).

Figure 3.4. Determining the sensitivity range for c_i



We have determined that the upper limit of the sensitivity range for c_1, the x_1 coefficient, is 66.67. If profit for a bowl increases to exactly \$66.67, the solution

points will be both B and C. If the profit for a bowl is more than $66.67, point C will be the optimal solution point.

The lower limit for the sensitivity range can be determined by observing Figure 3.4(b). In this case, if the objective function line slope decreases (becomes flatter) from 4/5 to the same slope as the constraint line,

$x_1 + 2x_2 = 40$

then point A becomes optimal (along with B). The slope of this constraint line is 1/2, that is, $x_2 = 20 (1/2)x_1$. In order to have an objective function slope of 1/2, the profit for a bowl would have to decrease to $25, as follows:

Lecture notes:

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For the manager this is useful information. Changing the production schedule in terms of how many items are produced can have a number of ramifications in an operation. Packaging, logistical, and marketing requirements for the product might need to be altered. However, with the preceding sensitivity range, the manager knows how much the profit, and hence the selling price and costs, can be altered without resulting in a change in production.

Performing the same type of graphical analysis will provide the sensitivity range for the x_2 objective function coefficient, c_2. This range is $30 \leq c_2 \leq 80$. This means that the profit for a mug can vary between $30 and $80, and the optimal solution point, B, will not change. However, for this case and the range for c_1, the sensitivity range generally applies only if one coefficient is varied and the other held constant. Thus, when we say that profit for a mug can vary between $30 and $80, this is true only if c_1 remains constant.

Simultaneous changes can be made in the objective function coefficients as long as the changes taken together do not change the optimal solution point. However, determining the effect of these simultaneous changes is overly complex and time-consuming using graphical analysis. In fact, using graphical analysis is a tedious way to perform sensitivity analysis in general, and it is impossible when the linear programming model contains three or more variables, thus requiring a three-dimensional graph. However, Excel and QM for Windows provide sensitivity analysis

for linear programming problems as part of their standard solution output. Determining the effect of simultaneous changes in model parameters and performing sensitivity analysis in general are much easier and more practical using the computer. Later in this chapter we will show the sensitivity analysis output for Excel and QM for Windows.

However, before moving on to computer-generated sensitivity analysis, we want to look at one more aspect of the sensitivity ranges for objective function coefficients. Recall that the model for our fertilizer minimization model from Chapter 2 is

Task #1

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Hints:

Sensitivity ranges:

$4 \leq c_1 \leq \infty$

$0 \leq c_2 \leq 4.5$

Objective Function Coefficient Ranges with the Computer

Objective Function Coefficient Ranges Beaver Creek Example Sensitivity Report

The sensitivity ranges for the objective function coefficients for our example are presented in QM for Windows

. Notice that this output provides the upper and lower limits of the sensitivity ranges for both variables, x_1 and x_2.

Ranging

Beaver Creek Pottery Company Solution					
Variable	Value	Reduced Cost	Original Val	Lower Bound	Upper Bound
X1	24.	0.	40.	25.	66.67
X2	8.	0.	50.	30.	80.
Constraint	Dual Value	Slack/Surplus	Original Val	Lower Bound	Upper Bound
Labor (hrs)	16.	0.	40.	30.	80.
Clay (lbs)	6.	0.	120.	60.	160.

Changes in Constraint Quantity Values

The second type of sensitivity analysis we will discuss is the sensitivity ranges for the constraint quantity valuesthat is, the values to the right of the inequality signs in the constraints. For our Beaver Creek Pottery Company model,

Maximize $Z = \$40x_1 + \$50x_2$

subject to: $1x_1 + 2x_2 \leq 40$

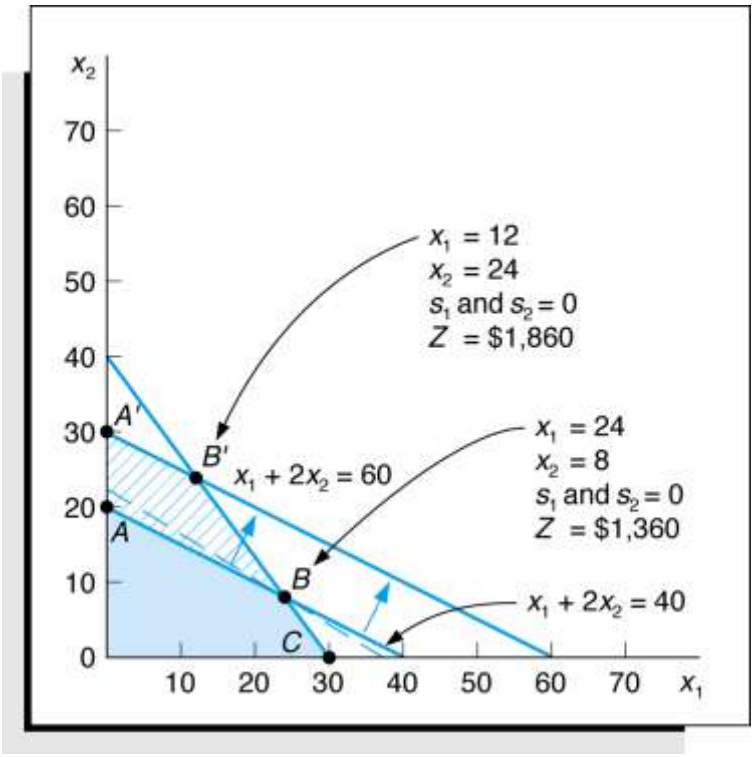
$4x_2 + 3x_2 \leq 120$

$x_1, x_2 \geq 0$

the constraint quantity values are 40 and 120.

Consider a change in which the manager of the pottery company can increase the labor hours from 40 to 60. The effect of this change in the model is graphically displayed in Figure 3.6.

Figure 3.6. Increasing the labor constraint quantity



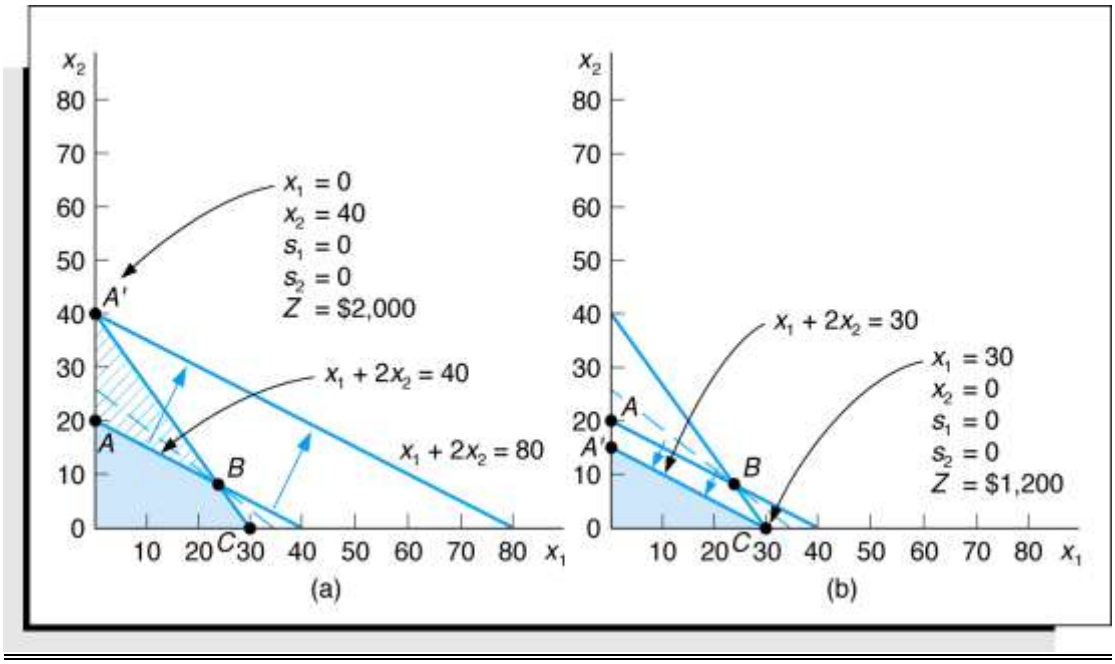
Increasing the available labor hours from 40 to 60 causes the feasible solution space to change. It was originally OABC, and now it is OA'B'C. B' is the new optimal solution, instead of B. However, the important consideration in this type of sensitivity analysis is that the solution mix (or variables that do not have zero values), including slack variables, did not change, even though the values of x_1 and x_2 did change (from $x_1 = 24$, $x_2 = 8$ to $x_1 = 12$, $x_2 = 24$). The focus of sensitivity analysis for constraint

quantity values is to determine the range over which the constraint quantity values can change without changing the solution variable mix, specifically including the slack variables.

The sensitivity range for a right-hand-side value is the range of values over which the quantity values can change without changing the solution variable mix, including slack variables.

If the quantity value for the labor constraint is increased from 40 to 80 hours, the new solution space is OA'C, and a new solution variable mix occurs at A', as shown in Figure 3.7(a). Whereas at the original optimal point, B, both x_1 and x_2 are in the solution, at the new optimal point, A', only x_2 is produced (i.e., $x_1 = 0$, $x_2 = 40$, $s_1 = 0$, $s_2 = 0$).

Figure 3.7. Determining the sensitivity range for labor quantity

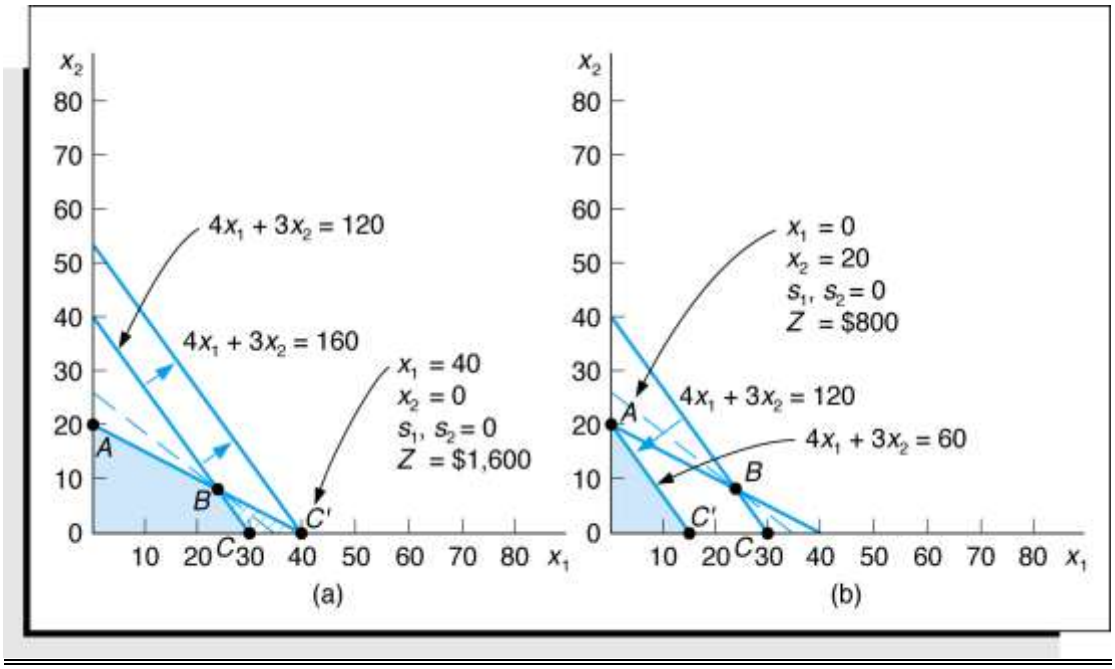


Thus, the upper limit of the sensitivity range for the quantity value for the first constraint, which we will refer to as q_1, is 80 hours. At this value the solution mix changes such that bowls are no longer produced. Furthermore, as q_1 increases past 80 hours, s_1 increases (i.e., slack hours are created). Similarly, if the value for q_1 is decreased to 30 hours, the new feasible solution space is OA'C, as shown in Figure 3.7(b). The new optimal point is at C, where no mugs (x_2) are produced. The new solution is $x_1 = 30$, $x_2 = 0$, $s_1 = 0$, $s_2 = 0$, and $Z = \$1{,}200$. Again, the variable mix is changed. Summarizing, the sensitivity range for the constraint quantity value for labor hours is

$30 \leq q_1 \leq 80$ hr.

The sensitivity range for clay can be determined graphically in a similar manner. If the quantity value for the clay constraint, $4x_1 + 3x_2 \leq 120$, is increased from 120 to 160, shown in Figure 3.8(a), then a new solution space, OAC', results, with a new optimal point, C'. Alternatively, if the quantity value is decreased from 120 to 60, as shown in Figure 3.8(b), the new solution space is OAC', and the new optimal point is A ($x_1 = 0$, $x_2 = 20$, $s_1 = 0$, $s_2 = 0$, $Z = \$800$).

Figure 3.8. Determining the sensitivity range for clay quantity



Summarizing, the sensitivity ranges for q_1 and q_2 are

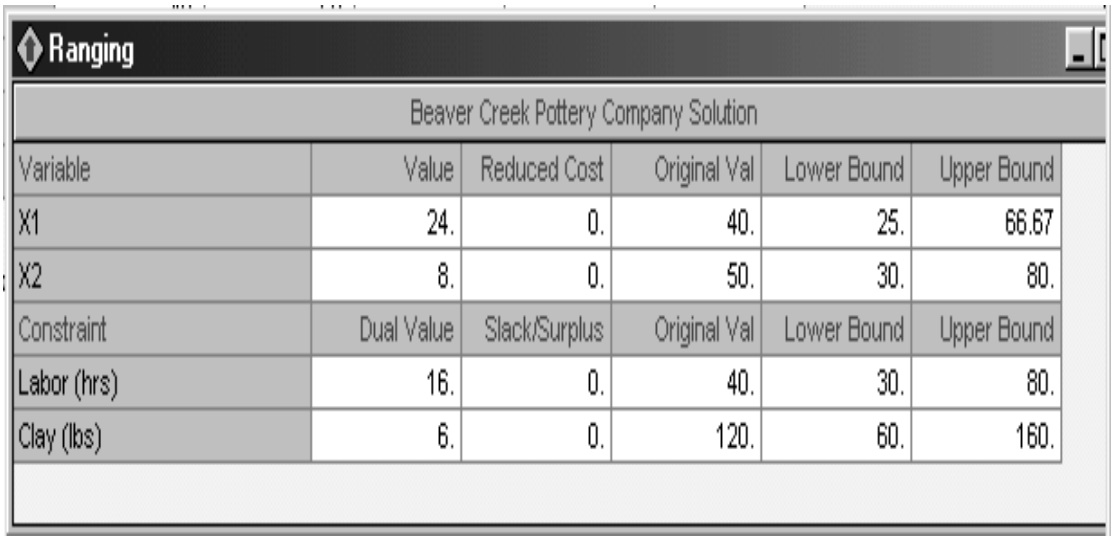
$30 \leq q_1 \leq 80$ hr.

$60 \leq q_2 \leq 160$ lb.

As was the case with the sensitivity ranges for the objective function coefficient, these sensitivity ranges are valid for only one q_i value and assumes all other q_i values are held constant. However, simultaneous changes can occur, as long as they do not change the variable mix.

These ranges for constraint quantity values provide useful information for the manager, especially regarding production scheduling and planning. If resources are reduced at the pottery company, then at some point one of the products will no longer be produced, and the support facilities and apparatus for that production will not be needed, or extra hours of resources will be created that are not needed. A similar result, albeit a better one, will occur if resources are increased because profit will be more than with a reduction in resources.

Constraint Quantity Value Ranges with QM for Windows



Ranging

Beaver Creek Pottery Company Solution					
Variable	Value	Reduced Cost	Original Val	Lower Bound	Upper Bound
X1	24.	0.	40.	25.	66.67
X2	8.	0.	50.	30.	80.
Constraint	Dual Value	Slack/Surplus	Original Val	Lower Bound	Upper Bound
Labor (hrs)	16.	0.	40.	30.	80.
Clay (lbs)	6.	0.	120.	60.	160.

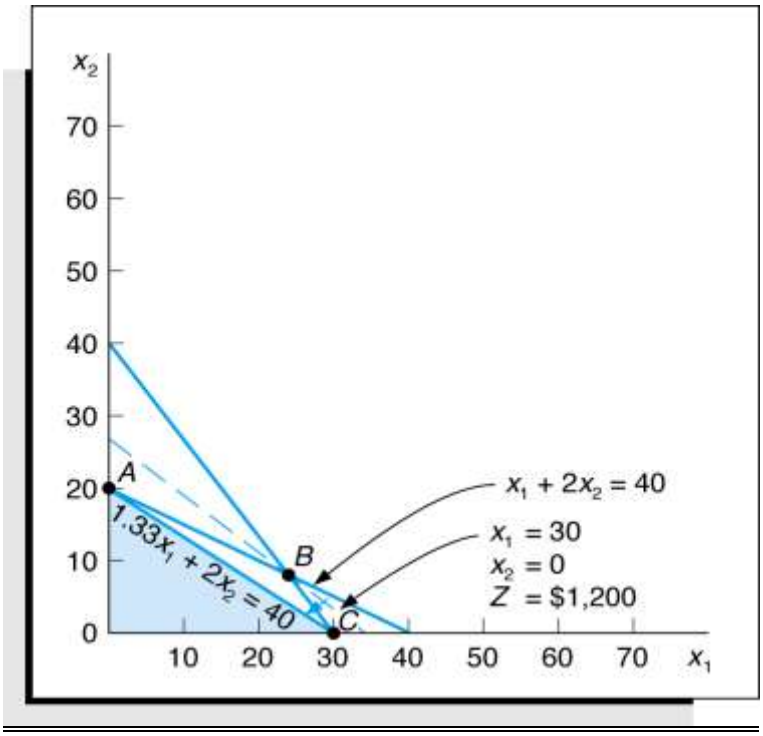
Other Forms of Sensitivity Analysis

Excel and QM for Windows provide sensitivity analysis ranges for objective function coefficients and constraint quantity values as part of the standard solution output. However, there are other forms of sensitivity analysis, including changing individual constraint parameters, adding new constraints, and adding new variables.

Other forms of sensitivity analysis include changing constraint parameter values, adding new constraints, and adding new variables.

For instance, in our Beaver Creek Pottery example, if a new, less-experienced artisan were hired to make pottery, it might take this individual 1.33 hours to produce a bowl instead of 1 hour. Thus, the labor constraint would change from $x_1 + 2x_2 \leq 40$ to $1.33x_1 + 2x_2 \leq 40$. This change is illustrated in Figure 3.9.

Figure 3.9. Changing the x_1 coefficient in the labor constraint



Note that a change in the coefficient for x_1 in the labor constraint rotates the constraint line, changing the solution space from OABC to OAC. It also results in a new optimal solution point, C, where $x_1 = 30$, $x_2 = 0$, and $Z = \$1,200$. Then 1.33 hours would be the logical upper limit for this constraint coefficient. However, as we pointed out, this type of sensitivity analysis for constraint variable coefficients is not typically provided in the standard linear programming computer output. As a result, the most logical way to ascertain the effect of this type of change is simply to run the computer program with different values.

Other types of sensitivity analysis are to add a new constraint to the model or to add a new variable. For example, suppose the Beaver Creek Pottery Company added a third constraint for packaging its pottery, as follows:

$0.20x_1 + 0.10x_2 \leq 5$ hr.

This would require the model to be solved again with the new constraint. This new constraint does, in fact, change the optimal solution

If a new variable is added to the model, this would also require that the model be solved again to determine the effect of the change. For example, suppose the pottery company was contemplating producing a third product, cups. It can secure no additional resources, and the profit for a cup is estimated to be \$30. This change is reflected in the following model reformulation:

Maximize $Z = \$40x_1 + 50x_2 + 30x_3$

subject to:

$$x_1 + 2x_2 + 1.2x_3 \leq 40 \text{ hr of labor}$$

$$4x_1 + 3x_2 + 2x_3 \leq 120 \text{ lb of clay}$$

$$x_1, x_2, x_3 \geq 0$$

Solving this new formulation with the computer will show that this prospective change will have no effect on the original solution that is, the model is not sensitive to this change. The estimated profit from cups was not enough to offset the production of bowls and mugs, and the solution remained the same.

Shadow Prices

We briefly discussed dual values (also called shadow prices) earlier in this chapter, in our discussion of QM for Windows. You will recall that a dual value was defined as the marginal value of one additional unit of resource. We need to mention shadow prices again at this point in our discussion of sensitivity analysis because decisions are often made regarding resources by considering the marginal value of resources in conjunction with their sensitivity ranges.

The shadow price (or marginal value) for labor is $16 per hour, and the shadow price for clay is $6 per pound. This means that for every additional hour of labor that can be obtained, profit will increase by $16. If the manager of the pottery company can secure more labor at $16 per hour, how much more can be obtained before the optimal solution mix will change and the current shadow price is no longer valid? The answer is at the upper limit of the sensitivity range for the labor constraint value. A maximum of 80 hours of labor can be used before the optimal solution mix changes. Thus, the manager can secure 40 more hours, the allowable increase shown in the Excel sensitivity output for the labor constraint. If 40 extra hours of labor can be obtained, what is its total value? The answer is ($16/hr.)(40 hr.) = $640. In other words, profit is increased by $640 if 40 extra hours of labor can be obtained.

where increasing the labor constraint from 40 to 80 hours has increased profit from $1,360 to $2,000, or $640

Looking back to Figure 3.8(a), this is the solution point at C'. Increasing the labor hours to more than 80 hours (the upper limit of the sensitivity range) will not result in an additional increase in profit or a new solution; it will only result in slack hours of labor.

Alternatively, what would be the effect on profit if one of the Native American artisans were sick one day during the week, and the available labor hours decreased from 40 to 32? Profit in this case would decrease by $16 per hour, or a total amount of $128. Thus, total profit would fall from $1,360 to $1,232.

Similarly, if the pottery company could obtain only 100 pounds of clay instead of its normal weekly allotment of 120 pounds, what would be the effect on profit? Profit

would decrease by $6 per pound for 20 pounds, or a total of $120. This would result in an overall reduction in profit from $1,360 to $1,240.

Thus, another piece of information that is provided by the sensitivity ranges for the constraint quantity values is the range over which the shadow price remains valid. When q_i increases past the upper limit of the sensitivity range or decreases below the lower limit, the shadow price will change, specifically because slack (or surplus) will be created. Therefore, the sensitivity range for the constraint quantity value is the range over which the shadow price is valid.

The sensitivity range for a constraint quantity value is also the range over which the shadow price is valid.

The shadow price of $16 for 1 hour of labor is not necessarily what the manager would pay for an hour of labor. This depends on how the objective function is defined. In the Beaver Creek Pottery Company example, we are assuming that all the resources available, 40 hours of labor and 120 pounds of clay, are already paid for. Even if the company does not use all the resources, it still must pay for them. These are sunk costs. Therefore, the individual profit values in the objective function for each product are not affected by how much of a resource is actually used; the total profit is independent of the resources used. In this case, the shadow prices are the maximum amounts the manager would pay for additional units of resource. The manager would pay up to $16 for 1 extra hour of labor and up to $6 for an extra pound of clay.

Alternatively, if each hour of labor and each pound of clay were purchased separately, and thus were not sunk costs, profit would be a function of the cost of the resources. In this case, the shadow price would be the additional amount, over and above the original cost of the resource, that would be paid for one more unit of the resource.

Solved problems

The Xecko Tool Company is considering bidding on a job for two airplane wing parts. Each wing part must be processed through three manufacturing stagesstamping, drilling, and finishingfor which the company has limited available hours. The linear programming model to determine how many of part 1 (x_1) and part 2 (x_2) the company should produce in order to maximize its profit is as follows:

- Maximize $Z = \$650x_1 + 910x_2$

  subject to:

  $4x_1 + 7.5x_2 \leq 105$ (stamping,hr)

  $6.2x_1 + 4.9x_2 \leq 90$ (drilling, hr)

  $9.1x_1 + 4.1x_2 \leq 110$ (finishing, hr)

  $x_1, x_2 \geq 0$

A. Solve the model graphically.
B. Indicate how much slack resource is available at the optimal solution point.
C. Determine the sensitivity ranges for the profit for wing part 1 and the stamping hours available.
D. Solve this model by using QM.

Solution:

- Two airplane parts: no.1 and no. 2.
- Three manufacturing stages: stamping, drilling, milling.
- Decision variables: x_1 (number of part no.1 to produce)

  x_2 (number of part no.2 to produce)

- Model: Maximize $Z = \$650x_1 + 910x_2$

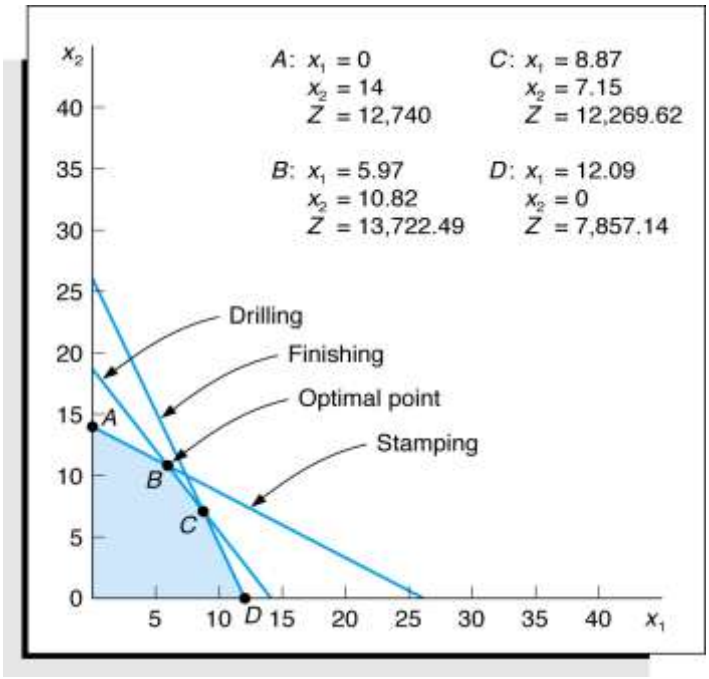
  subject to:

  $4x_1 + 7.5x_2 \leq 105$ (stamping,hr)

  $6.2x_1 + 4.9x_2 \leq 90$ (drilling, hr)

  $9.1x_1 + 4.1x_2 \leq 110$ (finishing, hr)

  $x_1, x_2 \geq 0$



A. We draw.
B. The slack at point B, where $x_1 = 5.97$ and $x_2 = 10.82$, is computed as follows:

$4(5.97) + 7.5(10.82) + s_1 = 105$ (stamping, hr.)

$s_1 = 0$ hr.

$6.2(5.97) + 4.9(10.82) + s_2 = 90$ (drilling, hr.)

$s_2 = 0$ hr.

$9.1(5.97) + 4.1(10.82) + s_3 = 110$ (finishing, hr.)

$s_3 = 11.35$ hr.

c. The sensitivity range for the profit for part 1 is determined by observing the graph of the model and computing how much the slope of the objective function must increase to make the optimal point move from B to C. This is the upper limit of the range and is determined by computing the value of c_1 that will make the slope of the objective function equal with the slope of the constraint line for drilling, $6.2x_1 + 4.9x_2 = 90$:

$485.33 \leq c_1 \leq 1,151.43$

$137.76 \leq q_1 \leq 89.10$

d. Lab

Problems

1. A company produces two products, A and B, which have profits of $9 and $7, respe assembly lines, where the required production times are as follows.

Hours/Unit		
Product	**Line 1**	**Line 2**
A	12	4
B	4	8
Total hours	60	40

a. Formulate a linear programming model to determine the optimal product mix that will maximize profit.
b. Transform this model into standard form.

2. Solve Problem 1 graphically.

a. Identify the amount of unused resources (i.e., slack) at each of the graphical extreme points.
b. What would be the effect on the optimal solution if the production time on line 1 were reduced to 40 hours?
c. What would be the effect on the optimal solution if the profit for product B were increased from $7 to $15? to $20?

3. For the linear programming model formulated in Problem 1 and solved graphically in Problem 2:

a. Determine the sensitivity ranges for the objective function coefficients, using graphical analysis.
b. Verify the sensitivity ranges determined in (a) by using the computer.
c. Using the computer, determine the shadow prices for additional hours of production time on line 1 and line 2 and indicate whether the company would prefer additional line 1 or line 2 hours.

4. The Bradley family owns 410 acres of farmland in North Carolina on which they grow corn and tobacco. Each acre of corn costs $105 to plant, cultivate, and harvest;

each acre of tobacco costs $210. The Bradleys have a budget of $52,500 for next year. The government limits the number of acres of tobacco that can be planted to 100. The profit from each acre of corn is $300; the profit from each acre of tobacco is $520. The Bradleys want to know how many acres of each crop to plant in order to maximize their profit.

Formulate a linear programming model for this problem.

5. Solve the linear programming model formulated in Problem 4 for the Bradley family farm graphically.

a. How many acres of farmland will not be cultivated at the optimal solution? Do the Bradleys use the entire 100-acre tobacco allotment?
b. What would the profit for corn have to be for the Bradleys to plant only corn?
c. If the Bradleys can obtain an additional 100 acres of land, will the number of acres of corn and tobacco they plan to grow change?
d. If the Bradleys decide not to cultivate a 50-acre section as part of a crop recovery program, how will it affect their crop plans?

6. Solve the linear programming model formulated in Problem 4 for the Bradley farm by using the computer.

a. The Bradleys have an opportunity to lease some extra land from a neighbor. The neighbor is offering the land to them for $110 per acre. Should the Bradleys lease the land at that price? What is the maximum price the Bradleys should pay their neighbor for the land, and how much land should they lease at that price?
b. The Bradleys are considering taking out a loan to increase their budget. For each dollar they borrow, how much additional profit would they make? If they borrowed an additional $1,000, would the number of acres of corn and tobacco they plant change?

CHAPTER 4. LINEAR PROGRAMMING: SIMPLEX METHOD

The *simplex method,* is a general mathematical solution technique for solving linear programming problems. In the simplex method, the model is put into the form of a table, and then a number of mathematical steps are performed on the table. These mathematical steps in effect replicate the process in graphical analysis of moving from one extreme point on the solution boundary to another. However, unlike the graphical method, in which we could simply search through *all* the solution points to find the best one, the simplex method moves from one *better* solution to another until the best one is found, and then it stops.

The manual solution of a linear programming model using the simplex method can be a lengthy and tedious process. Years ago, manual application of the simplex method was the only means for solving a linear programming problem. Now computer solution is certainly preferred. However, knowledge of the simplex method can greatly enhance one's understanding of linear programming. Computer software programs like QM for Windows or Excel spreadsheets provide solutions to linear programming problems, but they do not convey an in-depth understanding of how those solutions are derived. To a certain extent, graphical analysis provides an understanding of the solution process, and knowledge of the simplex method further expands on that understanding. In fact, computer solutions are usually derived using the simplex method. As a result, much of the terminology and notation used in computer software comes from the simplex method. Thus, for those students of management science who desire a more in-depth knowledge of linear programming, it is beneficial to study the simplex solution method as provided here.

Converting the Model into Standard Form

The first step in solving a linear programming model manually with the simplex method is to convert the model into standard form. At the Beaver Creek Pottery Company Native American artisans produce bowls (x_1) and mugs (x_2) from labor and clay. The linear programming model is formulated as

$$\text{maximize } Z = \$40x_1 + 50x_2 \text{ (profit)}$$

subject to

$$\begin{aligned} x_1 + 2x_2 &\leq 40 \text{ (labor, hr)} \\ 4x_1 + 3x_2 &\leq 120 \text{ (clay, lb)} \\ x_1, x_2 &\geq 0 \end{aligned}$$

We convert this model into *standard form* by adding slack variables to each constraint as follows.

$$\text{maximize } Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$$

subject to

$$x_1 + 2x_2 + s_1 = 40$$
$$4x_1 + 3x_2 + s_2 = 120$$
$$x_1, x_2, s_1, s_2 \geq 0$$

The slack variables, s_1 and s_2, represent the amount of unused labor and clay, respectively. For example, if no bowls and mugs are produced, and $x_1 = 0$ and $x_2 = 0$, then the solution to the problem is

$$x_1 + 2x_2 + s_1 = 40$$
$$0 + 2(0) + s_1 = 40$$
$$s_1 = 40 \text{ hr of labor}$$

and

$$4x_1 + 3x_2 + s_2 = 120$$
$$4(0) + 3(0) + s_2 = 120$$
$$s_2 = 120 \text{ lb of clay}$$

In other words, when we start the problem and nothing is being produced, all the resources are unused. Since unused resources contribute nothing to profit, the profit is zero.

$$Z = \$40x_1 + 50x_2 + 0s_1 + 0s_2$$
$$= 40(0) + 50(0) + 0(40) + 0(120)$$
$$Z = \$0$$

It is at this point that we begin to apply the simplex method. The model is in the required form, with the inequality constraints converted to equations for solution with the simplex method.

The Solution of Simultaneous Equations

Once both model constraints have been transformed into equations, the equations should be solved simultaneously to determine the values of the variables at every possible solution point. However, notice that our example problem has *two* equations and *four* unknowns (i.e., two decision variables and two slack variables), a situation that makes direct simultaneous solution impossible. The simplex method alleviates this problem by assigning some of the variables a value of zero. The number of variables assigned values of zero is $n \quad m$, where n equals the number of variables and m equals the number of constraints (excluding the nonnegativity constraints). For this model, $n = 4$ variables and $m = 2$ constraints; therefore, two of the variables are assigned a value of zero (i.e., $4 \quad 2 = 2$).

For example, letting $x_1 = 0$ and $s_1 = 0$ results in the following set of equations.

$$x_1 + 2x_2 + s_1 = 40$$
$$4x_1 + 3x_2 + s_2 = 120$$

and

$$0 + 2x_2 + 0 = 40$$
$$0 + 3x_2 + s_2 = 120$$

First, solve for x_2 in the first equation:

$$2x_2 = 40$$
$$x_2 = 20$$

Then, solve for s_2 in the second equation:

$$\begin{aligned} 3x_2 + s_2 &= 120 \\ 3(20) + s_2 &= 120 \\ s_2 &= 60 \end{aligned}$$

This solution corresponds with point A in Figure 4-1. The graph in Figure A-1 shows that at point $A, x_1 = 0, x_2 = 20, s_1 = 0$, and $s_2 = 60$, the exact solution obtained by solving simultaneous equations. This solution is referred to as a *basic feasible solution.* A feasible solution is any solution that satisfies the constraints. A *basic* feasible solution satisfies the constraints and contains as many variables with nonnegative values as there are model constraints—that is, m variables with nonnegative values and $n \quad m$ values set equal to zero. Typically, the m variables have positive nonzero solution values; however, when one of the m variables equals zero, the basic feasible solution is said to be *degenerate* .

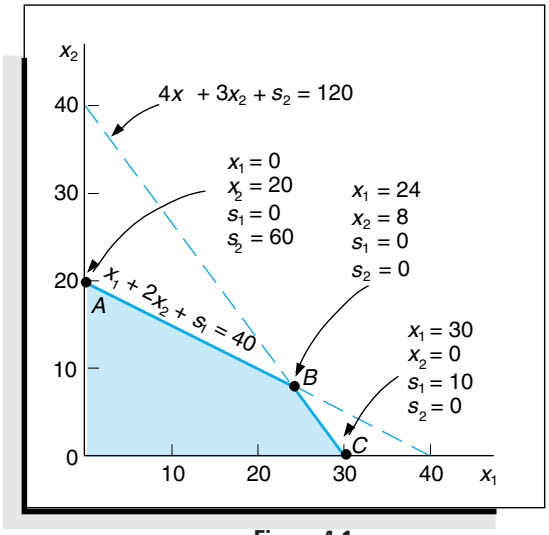


Figure 4-1

Consider a second example where $x_2 = 0$ and $s_2 = 0$. These values result in the following set of equations.

$$\begin{aligned} x_1 + 2x_2 + s_1 &= 40 \\ 4x_1 + 3x_2 + s_2 &= 120 \end{aligned}$$

and

$$\begin{aligned} x_1 + 0 + s_1 &= 40 \\ 4x_1 + 0 + 0 &= 120 \end{aligned}$$

Solve for x_1:

$$\begin{aligned} 4x_1 &= 120 \\ x_1 &= 30 \end{aligned}$$

Then solve for s_1:

$$\begin{aligned} 30 + s_1 &= 40 \\ s_1 &= 10 \end{aligned}$$

This basic feasible solution corresponds to point C in Figure A-1, where $x_1 = 30, x_2 = 0$, $s_1 = 10$, and $s_2 = 0$.

Finally, consider an example where $s_1 = 0$ and $s_2 = 0$. These values result in the following set of equations.

$$\begin{aligned} x_1 + 2x_2 + s_1 &= 40 \\ 4x_1 + 3x_2 + s_2 &= 120 \end{aligned}$$

and

$$\begin{aligned} x_1 + 2x_2 + 0 &= 40 \\ 4x_1 + 3x_2 + 0 &= 120 \end{aligned}$$

These equations can be solved using *row operations*. In row operations, the equations can be multiplied by constant values and then added or subtracted from each other without changing the values of the decision variables. First, multiply the top equation by 4 to get

$$4x_1 + 8x_2 = 160$$

and then subtract the second equation:

$$\begin{aligned} 4x_1 + 8x_2 &= 160 \\ 4x_1 \quad 3x_2 &= -120 \\ 5x_2 &= 40 \\ x_2 &= 8 \end{aligned}$$

Next, substitute this value of x_2 into either one of the constraints.

$$\begin{aligned} x_1 + 2(8) &= 40 \\ x_1 &= 24 \end{aligned}$$

This solution corresponds to point B on the graph, where $x_1 = 24$, $x_2 = 8$, $s_1 = 0$, and $s_2 = 0$, which is the optimal solution point.

All three of these example solutions meet our definition of *basic feasible solutions.* However, two specific questions are raised by the identification of these solutions.

1. In each example, how was it known which variables to set equal to zero?
2. How is the optimal solution identified?

The answers to both of these questions can be found by using the simplex method. The simplex method is a set of mathematical steps that determines at each step which variables should equal zero and when an optimal solution has been reached.

The Simplex Method

The steps of the simplex method are carried out within the framework of a table, or *tableau*. The tableau organizes the model into a form that makes applying the mathematical steps easier. The Beaver Creek Pottery Company example will be used again to demonstrate the simplex tableau and method.

$$\text{maximize } Z = \$40x_1 + 50x_2 + 0s_1 + 0s_2$$

subject to

$$\begin{aligned} x_1 + 2x_2 + s_1 &= 40 \text{ hr} \\ 4x_1 + 3x_2 + s_2 &= 120 \text{ lb} \\ x_1, x_2, s_1, s_2 &\geq 0 \end{aligned}$$

The initial simplex tableau for this model, with the various column and row headings, is shown in Table 4-1.

c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
	z_j					
	$c_j - z_j$					

Table 4-1

The first step in filling in Table 4-1 is to record the model variables along the second row from the top. The two decision variables are listed first, in order of their subscript magnitude, followed by the slack variables, also listed in order of their subscript magnitude. This step produces the row with x_1, x_2, s_1, and s_2 in Table 4-1.

The next step is to determine a basic feasible solution. In other words, which two variables will form the basic feasible solution and which will be assigned a value of zero? Instead of arbitrarily selecting a point (as we did with points *A*, *B*, and *C* in the previous section), the simplex method selects the origin as the initial basic feasible solution because the values of the decision variables at the origin are always known in all linear programming problems. At that point $x_1 = 0$ and $x_2 = 0$; thus, the variables in the basic feasible solution are s_1 and s_2

$$\begin{aligned} x_1 + 2x_2 + s_1 &= 40 \\ 0 + 2(0) + s_1 &= 40 \\ s_1 &= 40 \text{ hr} \end{aligned}$$

and

$$\begin{aligned} 4x_1 + 3x_2 + s_2 &= 120 \\ 4(0) + 3(0) + s_2 &= 120 \\ s_2 &= 120 \text{ lb} \end{aligned}$$

In other words, at the origin, where there is no production, all resources are slack, or unused. The variables s_1 and s_2, which form the initial basic feasible solution, are listed in Table A-2 under the column "Basic Variables," and their respective values, 40 and 120, are listed under the column "Quantity."

c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
	s_1	40				
	s_2	120				
	z_j					
	$c_j - z_j$					

Table 4-2

The initial simplex tableau always begins with the solution at the origin, where x_1 and x_2 equal zero. Thus, the basic variables at the origin are the slack variables, s_1 and s_2. Since the quantity values in the initial solution always appear as the right-hand-side values of the constraint equations, they can be read directly from the original constraint equations.

The top two rows and bottom two rows are standard for all tableaus; however, the number of middle rows is equivalent to the number of constraints in the model. For example, this problem has two constraints; therefore, it has two middle rows corresponding to s_1 and s_2. (Recall that n variables minus m constraints equals the number of variables in the problem with values of zero. This also means that the number of basic variables with values other than zero will be equal to m constraints.)

Similarly, the three columns on the left side of the tableau are standard, and the remaining columns are equivalent to the number of variables. Since there are four variables in this model, there are four columns on the right of the tableau, corresponding to x_1, x_2, s_1, and s_2.

The next step is to fill in the c_j values, which are the objective function coefficients, representing the *contribution to profit* (or cost) for each variable x_j or s_j in the objective function. Across the top row the c_j values 40, 50, 0, and 0 are inserted for each variable in the model, as shown in Table 4-3.

			40	**50**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2
0	s_1	40				
0	s_2	120				
	z_j					
	$c_j \quad z_j$					

Table 4-3

The values for c_j on the left side of the tableau are the contributions to profit of only those variables in the basic feasible solution, in this case s_1 and s_2. These values are inserted at this location in the tableau so that they can be used later to compute the values in the z_j row.

The columns under each variable (i.e., x_1, x_2, s_1, and s_2) are filled in with the coefficients of the decision variables and slack variables in the model constraint equations. The s_1 row represents the first model constraint; thus, the coefficient for x_1 is 1, the coefficient for x_2 is 2, the coefficient for s_1 is 1, and the coefficient for s_2 is 0. The values in the s_2 row are the second constraint equation coefficients, 4, 3, 0, and 1, as shown in Table 4-4.

			40	**50**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j					
	$c_j \quad z_j$					

Table 4-4

This completes the process of filling in the initial simplex tableau. The remaining values in the z_j and $c_j \quad z_j$ rows, as well as subsequent tableau values, are computed mathematically using simplex formulas.

The following list summarizes the steps of the simplex method (for a maximization model) that have been presented so far.

1. First, transform all inequalities to equations by adding slack variables.
2. Develop a simplex tableau with the number of columns equaling the number of variables plus three, and the number of rows equaling the number of constraints plus four.
3. Set up table headings that list the model decision variables and slack variables.
4. Insert the initial basic feasible solution, which are the slack variables and their quantity values.
5. Assign c_j values for the model variables in the top row and the basic feasible solution variables on the left side.
6. Insert the model constraint coefficients into the body of the table.

Computing the Zj and cj- Zj

So far the simplex tableau has been set up using values taken directly from the model. From this point on the values are determined by computation. First, the values in the z_j row are computed by multiplying each c_j *column value* (on the left side) by each column value under Quantity, x_1, x_2, s_1, and s_2 and then summing each of these sets of values. The z_j values are shown in Table 4-5.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j	0	0	0	0	0
	$c_j \quad z_j$					

Table 4-5

For example, the value in the z_j row under the quantity column is found as follows.

c_j ***Quantity***

$0 \times 40 = 0$
$0 \times 120 = 0$
$z_q = 0$

The value in the z_j row under the x_1 column is found similarly.

$c_j \quad x_1$

$0 \times 1 = 0$
$0 \times 4 = 0$
$z_j = 0$

All of the other z_j row values *for this tableau* will be zero when they are computed using this formula.

Now the $c_j \quad z_j$ row is computed by subtracting the z_j row values from the c_j (top) row values. For example, in the x_1 column the $c_j \quad z_j$ row value is computed as $40 \quad 0 = 40$.

This value as well as other $c_j - z_j$ values are shown in Table 4-6, which is the complete initial simplex tableau with all values filled in. This tableau represents the solution at the origin, where $x_1 = 0, x_2 = 0$, $s_1 = 40$, and $s_2 = 120$. The profit represented by this solution (i.e., the Z value) is given in the z_j row under the quantity column—0 in Table 4-6. This solution is obviously not optimal because no profit is being made. Thus, we want to move to a solution point that will give a *better* solution. In other words, we want to produce either some bowls (x_1) or some mugs (x_2). One of the *nonbasic* variables (i.e., variables not in the present basic feasible solution) will enter the solution and become basic.

			40	**50**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j	0	0	0	0	0
	$c_j - z_j$		40	50	0	0

Table 4-6

The Entering Nonbasic Variable

As an example, suppose the pottery company decides to produce some bowls. With this decision x_1 will become a basic variable. For every unit of x_1 (i.e., each bowl) produced, profit will be increased by \$40 because that is the profit contribution of a bowl. However, when a bowl (x_1) is produced, some previously unused resources will be used. For example, if

$$x_1 = 1$$

then

$$\begin{aligned} x_1 + 2x_2 + s_1 &= 40 \text{ hr of labor} \\ 1 + 2(0) + s_{\text{ll}} &= 40 \\ \text{s} &= 39 \text{ hr of labor} \end{aligned}$$

and

$$\begin{aligned} 4x_1 + 3x_2 + s_2 &= 120 \text{ lb of clay} \\ 4(1) + 3(0) + s_2 &= 120 \\ s_2 &= 116 \text{ lb of clay} \end{aligned}$$

In the labor constraint we see that, with the production of one bowl, the amount of slack, or unused, labor is *decreased* by *1 hour.* In the clay constraint the amount of slack is *decreased* by *4 pounds.* Substituting these increases (for x_1) and decreases (for slack) into the objective function gives

$$\begin{aligned} & \quad c_j \qquad\qquad z_j \\ Z &= 40(1) + 50(0) + 0(\ \ 1) + 0(\ \ 4) \\ Z &= \$40 \end{aligned}$$

The first part of this objective function relationship represents the values in the c_j row; the second part represents the values in the z_j row. The function expresses the fact that to produce some bowls, we must give up some of the profit already earned from the items they replace. In this case the production of bowls replaced only slack, so no profit was lost. In general, the $c_j - z_j$ row values represent the net *increase per unit of entering a nonbasic*

variable into the basic solution. Naturally, we want to make as much money as possible, because the objective is to maximize profit. Therefore, we enter the variable that will give the greatest net increase in profit per unit. From Table 4-7, we select variable x_2 as the entering basic variable because it has the greatest net increase in profit per unit, \$50 — the highest positive value in the $c_j \quad z_j$ row.

			40	**50**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j	0	0	0	0	0
	$c_j \quad z_j$		40	50	0	0

Table 4-7

The x_2 column, highlighted in Table 4-7, is referred to as the *pivot column*. (The operations used to solve simultaneous equations are often referred to in mathematical terminology as *pivot operations.*)

The selection of the entering basic variable is also demonstrated by the graph in Figure4-2. At the origin nothing is produced. In the simplex method we move from one solution point to an *adjacent* point (i.e., *one* variable in the basic feasible solution is replaced with a variable that was previously zero). In Figure 4-2 we can move along either the x_1 axis or the x_2 axis in order to seek a better solution. Because an increase in x_2 will result in a greater profit, we choose x_2.

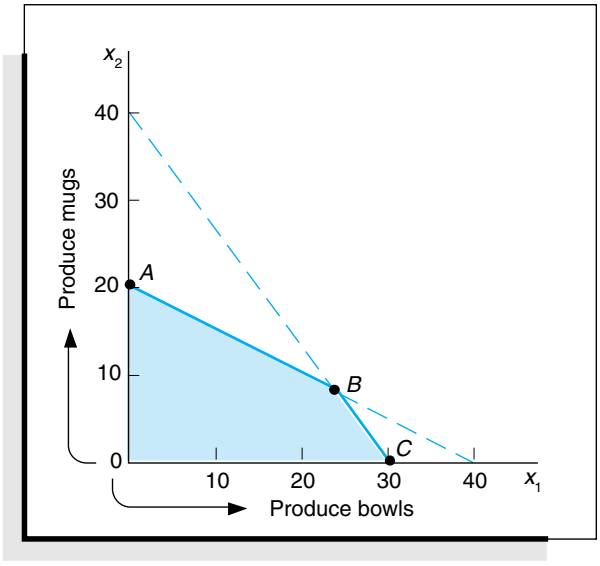


Figure 4-2

The Leaving Basic Variable

Since each basic feasible solution contains only two variables with nonzero values, one of the two basic variables present, s_1 or s_2, will have to leave the solution and become zero. Since we have decided to produce mugs (x_2), we want to produce as many as possible or, in other words, as many as our resources will allow. First, in the labor constraint we will use all

the labor to make mugs (because no bowls are to be produced, $x_1 = 0$; and because we will use all the labor possible and s_1 = unused labor resources, $s_1 = 0$ also).

$$\begin{aligned} 1x_1 + 2x_2 + s_1 &= 40 \text{ hr} \\ 1(0) + 2x_2 + 0 &= 40 \\ x_2 &= \frac{40 \text{ hr}}{2 \text{ hr/mug}} \\ &= 20 \text{ mugs} \end{aligned}$$

In other words, enough labor is available to produce 20 mugs. Next, perform the same analysis on the constraint for clay.

$$\begin{aligned} 4x_1 + 3x_2 + s_2 &= 120 \text{ lb} \\ 4(0) + 3x_2 + 0 &= 120 \\ x_2 &= \frac{120 \text{ lb}}{3 \text{ lb/mug}} \\ &= 40 \text{ mugs} \end{aligned}$$

This indicates that there is enough clay to produce 40 mugs. But there is enough labor to produce only 20 mugs. We are limited to the production of only 20 mugs because we do not have enough labor to produce any more than that. This analysis is shown graphically in Figure 4-3.

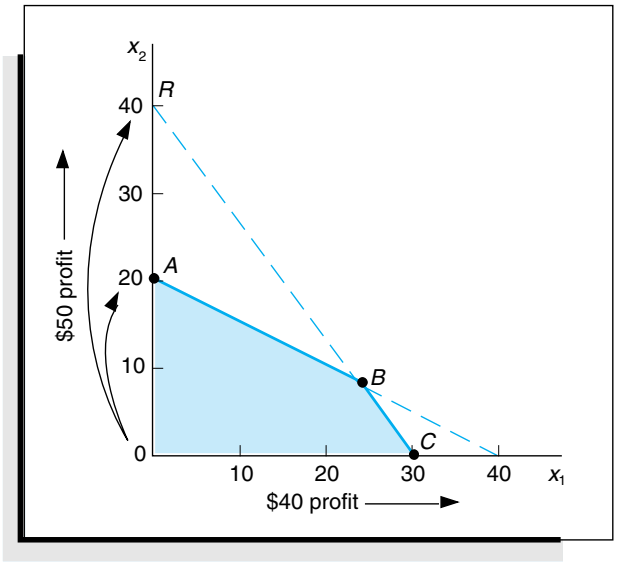


Figure 4-3

Because we are moving out the x_2 axis, we can move from the origin to either point *A* or point *R*. We select point *A* because it is the *most constrained* and thus feasible, whereas point *R* is infeasible.

This analysis is performed in the simplex method by dividing the quantity values of the basic solution variables by the pivot column values. For this tableau,

Basic Variables	***Quantity***	x_2
s_1	40	$\div 2 = 20$, the leaving basic variable
s_2	120	$\div 3 = 40$

The leaving basic variable is the variable that corresponds to the minimum nonnegative quotient, which in this case is 20. (Note that a value of zero would qualify as the minimum quotient and would be the choice for the leaving variable.) Therefore, s_1 is the leaving variable. (At point *A* in Figure 4-3, s_1 equals zero because all the labor is used to make the 20 mugs.) The s_1 row, highlighted in Table 4-8, is also referred to as the *pivot row*

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j	0	0	0	0	0
	$c_j - z_j$		40	50	0	0

Table 4-8

The value of 2 at the intersection of the pivot row and the pivot column is called the *pivot number.* The pivot number, row, and column are all instrumental in developing the next tableau. We are now ready to proceed to the second simplex tableau and a *better* solution.

Developing a New Tableau

Table A-9 shows the second simplex tableau with the new basic feasible solution variables of x_2 and s_2 and their corresponding c_j values.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
50	x_2					
0	s_2					
	z_j					
	$c_j - z_j$					

Table 4-9

The various row values in the second tableau are computed using several simplex formulas. First, the x_2 row, called the *new tableau pivot row,* is computed by dividing every value in the pivot row of the first (old) tableau by the pivot number. The formula for these computations is

$$\text{new tableau pivot row values} = \frac{\text{old tableau pivot row values}}{\text{pivot number}}$$

The new row values are shown in Table 4-10.

To compute all remaining row values (in this case there is only one other row), another formula is used.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
50	x_2	20	1/2	1	1/2	0
0	s_2					
	z_j					
	$c_j \quad z_j$					

Table A-10

$$\text{new tableau row values} = \text{old tableau row values} - \left(\begin{array}{c} \text{corresponding coefficients in pivot column} \\ \times \\ \text{corresponding new tableau pivot row value} \end{array} \right)$$

Thus, this formula requires the use of both the old tableau and the new one. The s_2 row values are computed in Table A-11.

Column	Old Tableau Row Value	−	Corresponding Coefficients in Pivot Column	×	New Tableau Pivot Row Value	=	New Tableau Row Value
Quantity	120		(3	×	20)	=	60
x_1	4		(3	×	1/2)	=	5/2
x_2	3		(3	×	1)	=	0
s_1	0		(3	×	1/2)	=	−3/2
s_2	1		(3	×	0)	=	1

Table A-11

These values have been inserted in the simplex tableau in Table A-12.

This solution corresponds to point A in the graph of this model in Figure A-3. The solution at this point is $x_1 = 0, x_2 = 20, s_1 = 0, s_2 = 60$. In other words, 20 mugs are produced and 60 pounds of clay are left unused. No bowls are produced and no labor hours remain unused.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
50	x_2	20	1/2	1	1/2	0
0	s_2	60	5/2	0	3/2	1
	z_j					
	$c_j \quad z_j$					

Table A-12

The second simplex tableau is completed by computing the z_j and $c_j \quad z_j$ row values the same way they were computed in the first tableau. The z_j row is computed by summing the products of the c_j column and all other column values.

Column	
Quantity	$z_j = (50)(20) + (0)(60) = 1000$
x_1	$z_1 = (50)(1/2) + (0)(5/2) = 25$
x_2	$z_2 = (50)(1) + (0)(0) = 50$
s_1	$z_3 = (50)(1/2) + (0)(\ 3/2) = 25$
s_2	$z_4 = (50)(0) + (0)(1) = 0$

The z_j row values and the c_j z_j row values are added to the tableau to give the completed second simplex tableau shown in Table A-13. The value of 1,000 in the z_j row is the value of the objective function (profit) for this basic feasible solution.

			40	**50**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2
50	x_2	20	1/2	1	1/2	0
0	s_2	60	5/2	0	3/2	1
	z_j	1,000	25	50	25	0
	c_j z_j		15	0	25	0

Table A-13

The computational steps that we followed to derive the second tableau in effect accomplish the same thing as *row operations* in the solution of simultaneous equations. These same steps are used to derive each subsequent tableau, called *iterations.*

The Optimal Simplex Tableau

The steps that we followed to derive the second simplex tableau are repeated to develop the third tableau. First, the pivot column or entering basic variable is determined. Because 15 in the c_j z_j row represents the greatest positive net increase in profit, x_1 becomes the entering nonbasic variable. Dividing the pivot column values into the values in the quantity column indicates that s_2 is the leaving basic variable *and* corresponds to the pivot row. The pivot row, pivot column, and pivot number are indicated in Table A-14.

At this point you might be wondering why the net increase in profit per bowl (x_1) is \$15 rather than the original profit of \$40. It is because the production of bowls (x_1) will require some of the resources previously used to produce mugs (x_2) only. Producing some bowls means not producing as many mugs; thus, we are giving up some of the profit gained from producing mugs to gain even more by producing bowls. This difference is the *net increase* of \$15.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
50	x_2	20	1/2	1	1/2	0
0	s_2	60	5/2	0	3/2	1
	z_j	1,000	25	50	25	0
	$c_j - z_j$		15	0	25	0

Table A-14

The new tableau pivot row (x_1) in the third simplex tableau is computed using the same formula used previously. Thus, all old pivot row values are divided through by 5/2, the pivot number. These values are shown in Table A-16. The values for the other row (x_2) are computed as shown in Table A-15.

Column	Old Tableau Row Value	−	Corresponding Coefficients in Pivot Column	×	New Tableau Pivot Row Value	=	New Tableau Row Value
Quantity	20		(1/2	×	24)	=	8
x_1	1/2		(1/2	×	1)	=	0
x_2	1		(1/2	×	0)	=	1
s_1	1/2		(1/2	×	−3/5)	=	4/5
s_2	0		(1/2	×	2/5)	=	−1/5

Table A-15

These new row values, as well as the new z_j row and $c_j - z_j$ row, are shown in the completed third simplex tableau in Table A-16.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
50	x_2	8	0	1	4/5	1/5
40	x_1	24	1	0	3/5	2/5
	z_j	1,360	40	50	16	6
	$c_j - z_j$		0	0	16	6

Table A-16

Observing the $c_j - z_j$ row to determine the entering variable, we see that a nonbasic variable would not result in a positive net increase in profit, as all values in the $c_j - z_j$ row are zero or negative. This means that the optimal solution has been reached. The solution is

$$x_1 = 24 \text{ bowls}$$
$$x_2 = 8 \text{ mugs}$$
$$Z = \$1{,}360 \text{ profit}$$

which corresponds to point B in Figure A-1.

An additional comment should be made regarding simplex solutions in general. Although this solution resulted in *integer* values for the variables (i.e., 24 and 8), it is possible to get a fractional solution for decision variables even though the variables reflect items that should be integers, such as airplanes, television sets, bowls, and mugs. To apply the simplex method, one must accept this limitation.

Summary of the Simplex Method

The simplex method demonstrated in the previous section consists of the following steps.

1. Transform the model constraint inequalities into equations.
2. Set up the initial tableau for the basic feasible solution at the origin and compute the z_j and $c_j - z_j$ row values.
3. Determine the pivot column (entering nonbasic solution variable) by selecting the column with the highest positive value in the $c_j - z_j$ row.
4. Determine the pivot row (leaving basic solution variable) by dividing the quantity column values by the pivot column values and selecting the row with the minimum nonnegative quotient.
5. Compute the new pivot row values using the formula

$$\text{new tableau pivot row values} = \frac{\text{old tableau pivot row values}}{\text{pivot number}}$$

6. Compute all other row values using the formula

$$\begin{matrix}\text{new tableau} \\ \text{row values}\end{matrix} = \begin{matrix}\text{old tableau} \\ \text{row values}\end{matrix} - \left(\begin{matrix}\text{corresponding} \\ \text{coefficients in} \\ \text{pivot Column}\end{matrix} \times \begin{matrix}\text{corresponding} \\ \text{new tableau} \\ \text{pivot row values}\end{matrix} \right)$$

7. Compute the new z_j and $c_j - z_j$ rows.
8. Determine whether or not the new solution is optimal by checking the $c_j - z_j$ row. If all $c_j - z_j$ row values are zero or negative, the solution is optimal. If a positive value exists, return to step 3 and repeat the simplex steps.

Simplex Solution of a Minimization Problem

In the previous section the simplex method for solving linear programming problems was demonstrated for a maximization problem. In general, the steps of the simplex method outlined at the end of this section are used for any type of linear programming problem. However, a minimization problem requires a few changes in the normal simplex process, which we will discuss in this section.

In addition, several exceptions to the typical linear programming problem will be presented later in this module. These include problems with mixed constraints ($=, \leq$, and $\geq$); problems with more than one optimal solution, no feasible solution, or an unbounded solution; problems with a tie for the pivot column; problems with a tie for the pivot row; and problems with constraints with negative quantity values. None of these kinds of problems require changes in the simplex method. They are basically unusual results in individual simplex tableaus that the reader should know how to interpret and work with.

Standard Form of a Minimization Model

Consider the following linear programming model for a farmer purchasing fertilizer.

$$\text{minimize } Z = \$6x_1 + 3x_2$$

subject to

$$2x_1 + 4x_2 \geq 16 \text{ lb of nitrogen}$$
$$4x_1 + 3x_2 \geq 24 \text{ lb of phosphate}$$

where

x_1 = bags of Super-gro fertilizer
x_2 = bags of Crop-quick fertilizer
Z = farmer's total cost ($) of purchasing fertilizer

This model is transformed into standard form by subtracting surplus variables from the two $\geq$ constraints as follows.

$$\text{minimize } Z = 6x_1 + 3x_2 + 0s_1 + 0s_2$$

subject to

$$2x_1 + 4x_2 - s_1 = 16$$
$$4x_1 + 3x_2 - s_1 = 24$$
$$x_1, x_2, s_1, s_2 \geq 0$$

The surplus variables represent the extra amount of nitrogen and phosphate that exceeded the minimum requirements specified in the constraints.

However, the simplex method requires that the initial basic feasible solution be at the origin, where $x_1 = 0$ and $x_2 = 0$. Testing these solution values, we have

$$\begin{aligned} 2x_1 + 4x_2 \quad & s_1 = 16 \\ 2(0) + 4(0) \quad & s_1 = 16 \\ & s_1 = -16 \end{aligned}$$

The idea of "negative excess pounds of nitrogen" is illogical and violates the nonnegativity restriction of linear programming. The reason the surplus variable does not work is shown in Figure A-4. The solution at the origin is outside the feasible solution space.

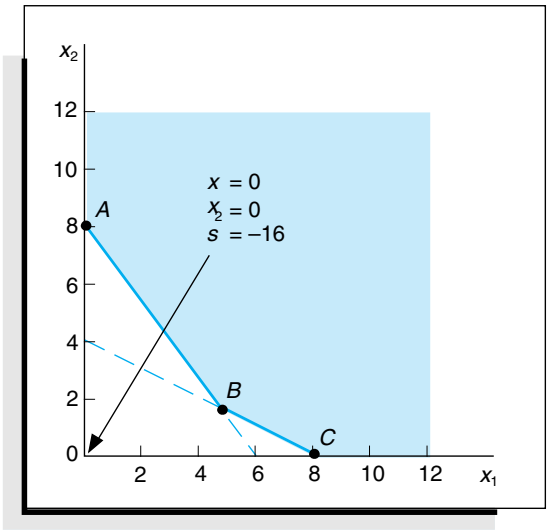


Figure A-4

To alleviate this difficulty and get a solution at the origin, we add an *artificial variable* (A_1) to the constraint equation,

$$2x_1 + 4x_2 \quad s_1 + A_1 = 16$$

The artificial variable, A_1, does not have a meaning as a slack variable or a surplus variable does. It is inserted into the equation simply to give a positive solution at the origin; we are artificially creating a solution.

$$\begin{aligned} 2x_1 + 4x_2 \quad & s_1 + A_1 = 16 \\ 2(0) + 4(0) \quad & 0 + A_1 = 16 \\ & A_1 = 16 \end{aligned}$$

The artificial variable is somewhat analogous to a booster rocket its purpose is to get us off the ground; but once we get started, it has no real use and thus is discarded. The artificial solution helps get the simplex process started, but we do not want it to end up in the optimal solution, because it has no real meaning.

When a surplus variable is subtracted and an artificial variable is added, the phosphate constraint becomes

$$4x_1 + 3x_2 \quad s_2 + A_2 = 24$$

The effect of surplus and artificial variables on the objective function must now be considered. Like a slack variable, a surplus variable has no effect on the objective function in terms of increasing or decreasing cost. For example, a surplus of 24 pounds of nitrogen does not contribute to the cost of the objective function, because the cost is determined solely by the number of bags of fertilizer purchased (i.e., the values of x_1 and x_2). Thus, a coefficient of 0 is assigned to each surplus variable in the objective function.

By assigning a "cost" of \$0 to each surplus variable, we are not prohibiting it from being in the final optimal solution. It would be quite realistic to have a final solution that showed some surplus nitrogen or phosphate. Likewise, assigning a cost of \$0 to an artificial variable in the objective function would not prohibit it from being in the final optimal solution. However, if the artificial variable appeared in the solution, it would render the final solution meaningless. Therefore, we must ensure that an artificial variable is *not* in the final solution.

As previously noted, the presence of a particular variable in the final solution is based on its relative profit or cost. For example, if a bag of Super-gro costs \$600 instead of \$6 and Crop-quick stayed at \$3, it is doubtful that the farmer would purchase Super-gro (i.e., x_1 would not be in the solution). Thus, we can prohibit a variable from being in the final solution by assigning it a very *large cost*. Rather than assigning a dollar cost to an artificial variable, we will assign a value of M, which represents a large positive cost (say, \$1,000,000). This operation produces the following objective function for our example:

$$\text{minimize } Z = 6x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$$

The completely transformed minimization model can now be summarized as

$$\text{minimize } Z = 6x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$$

subject to

$$\begin{aligned} 2x_1 + 4x_1 \quad & s_1 + A_1 = 16 \\ 4x_1 + 3x_2 \quad & s_2 + A_2 = 24 \\ x_1, x_2, s_1, s_2, A_1, A_2 & \geq 0 \end{aligned}$$

The Simplex Tableau for a Minimization Problem

The initial simplex tableau for a minimization model is developed the same way as one for a maximization model, except for one small difference. Rather than computing $c_j - z_j$ in the bottom row of the tableau, we compute $z_j - c_j$, which represents the *net per unit decrease in cost*, and the largest positive value is selected as the entering variable and pivot column. (An alternative would be to leave the bottom row as $c_j - z_j$ and select the largest *negative* value as the pivot column. However, to maintain a consistent rule for selecting the pivot column, we will use $z_j - c_j$.)

The initial simplex tableau for this model is shown in Table A-17. Notice that A_1 and A_2 form the initial solution at the origin, because that was the reason for inserting them in the first place to get a solution at the origin. This is not a basic feasible solution, since the origin is not in the feasible solution area, as shown in Figure A-4. As indicated previously, it is an artificially created solution. However, the simplex process will move toward feasibility in subsequent tableaus. Note that across the top the decision variables are listed first, then surplus variables, and finally artificial variables.

			6	**3**	**0**	**0**	***M***	***M***
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	A_1	A_2
M	A_1	16	2	4	1	0	1	0
M	A_2	24	4	3	0	1	0	1
	z_j	$40M$	$6M$	$7M$	M	M	M	M
	$z_j - c_j$		$6M - 6$	$7M - 3$	M	M	0	0

Table A-17

In Table A-17 the x_2 column was selected as the pivot column because $7M - 3$ is the largest positive value in the $z_j - c_j$ row. A_1 was selected as the leaving basic variable (and pivot row) because the quotient of 4 for this row was the minimum positive row value.

The second simplex tableau is developed using the simplex formulas presented earlier. It is shown in Table A-18. Notice that the A_1 column has been eliminated in the second simplex tableau. Once an artificial variable leaves the basic feasible solution, it will never return because of its high cost, M. Thus, like the booster rocket, it can be eliminated from the tableau. However, artificial variables are the *only* variables that can be treated this way.

			6	**3**	**0**	**0**	***M***
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	A_1
3	x_2	4	1/2	1	1/4	0	0
M	A_2	12	5/2	0	3/4	1	1
	z_j	$12M + 12$	$5M/2 + 3/2$	3	$3/4 + 3/M4$	M	M
	$z_j - c_j$		$5M/2 - 9/2$	0	$3/4 + 3/M4$	M	0

Table A-18

The third simplex tableau, with x_1 replacing A_2, is shown in Table A-19. Both the A_1 and A_2 columns have been eliminated because both variables have left the solution. The x_1 row

is selected as the pivot row because it corresponds to the minimum positive ratio of 16. In selecting the pivot row, the 4 value for the x_2 row was not considered because the minimum *positive* value *or zero* is selected. Selecting the x_2 row would result in a negative quantity value for s_1 in the fourth tableau, which is not feasible.

c_j	Basic Variables	Quantity	6 x_1	3 x_2	0 s_1	0 s_2
3	x_2	8/5	0	1	2/5	1/5
6	x_1	24/5	1	0	3/10	2/5
	z_j	168/5	6	3	3/5	9/5
	$z_j \quad c_j$		0	0	3/5	9/5

Table A-19

The fourth simplex tableau, with s_1 replacing x_1, is shown in Table A-20. Table A-20 is the optimal simplex tableau because the $z_j \quad c_j$ row contains no positive values. The optimal solution is

$x_1 = 0$ bags of Super-gro
$s_1 = 16$ extra lb of nitrogen
$x_2 = 8$ bags of Crop-quick
$s_2 = 0$ extra lb of phosphate
$Z = \$24$, total cost of purchasing fertilizer

c_j	Basic Variables	Quantity	6 x_1	3 x_2	0 s_1	0 s_2
3	x_2	8	4/3	1	0	1/3
0	s_1	16	10/3	0	1	4/3
	z_j	24	4	3	0	1
	$z_j \quad c_j$		2	0	0	1

Table A-20

Simplex Adjustments for a Minimization Problem

To summarize, the adjustments necessary to apply the simplex method to a minimization problem are as follows:

1. Transform all $\geq$ constraints to equations by subtracting a surplus variable and adding an artificial variable.
2. Assign a c_j value of M to each artificial variable in the objective function.
3. Change the $c_j \quad z_j$ row to $z_j \quad c_j$.

Although the fertilizer example model we just used included only $\geq$ constraints, it is possible for a minimization problem to have $\leq$ and $=$ constraints in addition to $\geq$ constraints. Similarly, it is possible for a maximization problem to have $\geq$ and $=$ constraints in addition to $\leq$ constraints. Problems that contain a combination of different types of inequality constraints are referred to as *mixed constraint problems.*

A Mixed Constraint Problem

So far we have discussed maximization problems with all $\leq$ constraints and minimization problems with all $\geq$ constraints. However, we have yet to solve a problem with a mixture of $\leq, \geq$, and $=$ constraints. Furthermore, we have not yet looked at a maximization problem with a $\geq$ constraint. The following is a maximization problem with $\leq$, $\geq$, and $=$ constraints.

A leather shop makes custom-designed, hand-tooled briefcases and luggage. The shop makes a \$400 profit from each briefcase and a \$200 profit from each piece of luggage. (The profit for briefcases is higher because briefcases require more hand tooling.) The shop has a contract to provide a store with exactly 30 items per month. A tannery supplies the shop with at least 80 square yards of leather per month. The shop must use at least this amount but can order more. Each briefcase requires 2 square yards of leather; each piece of luggage requires 8 square yards of leather. From past performance, the shop owners know they cannot make more than 20 briefcases per month. They want to know the number of briefcases and pieces of luggage to produce in order to maximize profit.

This problem is formulated as

$$\text{maximize } Z = \$400x_1 + 200x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &= 30 \text{ contracted items} \\ 2x_1 + 8x_2 &\geq 80 \text{ yd}^2 \text{ of leather} \\ x_1 &\leq 20 \text{ briefcases} \\ x_1, x_2 &\geq 0 \end{aligned}$$

where $x_1 =$ briefcases and $x_2 =$ pieces of luggage.

The first step in the simplex method is to transform the inequalities into equations. The first constraint for the contracted items is already an equation; therefore, it is not necessary to add a slack variable. There can be no slack in the contract with the store because exactly 30 items must be delivered. Even though this equation already appears to be in the necessary form for simplex solution, let us test it at the origin to see if it meets the starting requirements.

$$\begin{aligned} x_1 + x_2 &= 30 \\ 0 + 0 &= 30 \\ 0 &\neq 30 \end{aligned}$$

Because zero does not equal 30, the constraint is not feasible in this form. Recall that a $\geq$ constraint did not work at the origin either in an earlier problem. Therefore, an artificial variable was added. The same thing can be done here.

$$x_1 + x_2 + A_1 = 30$$

Now at the origin, where $x_1 = 0$ and $x_2 = 0$,

$$\begin{aligned} 0 + 0 + A_1 &= 30 \\ A_1 &= 30 \end{aligned}$$

Any time a constraint is initially an equation, an artificial variable is added. However, the artificial variable cannot be assigned a value of M in the objective function of a maximization problem. Because the objective is to maximize profit, a positive M value would represent a large positive *profit* that would definitely end up in the final solution. Because an artificial variable has no real meaning and is inserted into the model merely to create an initial

solution at the origin, its existence in the final solution would render the solution meaningless. To prevent this from happening, we must give the artificial variable a large *cost* contribution, or M.

The constraint for leather is a $\geq$ inequality. It is converted to equation form by subtracting a surplus variable and adding an artificial variable:

$$2x_1 + 8x_2 \quad s_1 + A_2 = 80$$

As in the equality constraint, the artificial variable in this constraint must be assigned an objective function coefficient of M.

The final constraint is a $\leq$ inequality and is transformed by adding a slack variable:

$$x_1 + s_2 = 20$$

The completely transformed linear programming problem is as follows:

$$\text{maximize } Z = 400x_1 + 200x_2 + 0s_1 + 0s_2 \quad MA_1 \quad MA_2$$

subject to

$$\begin{aligned} x_1 + x_2 + A_1 &= 30 \\ 2x_1 + 8x_2 \quad s_1 + A_2 &= 80 \\ x_1 + s_2 &= 20 \\ x_1, x_2, s_1, s_2, A_1, A_2 &\geq 0 \end{aligned}$$

The initial simplex tableau for this model is shown in Table A-21. Notice that the basic solution variables are a mix of artificial and slack variables. Note also that the third-row quotient for determining the pivot row $(20 \div 0)$ is an undefined value, or ∞. Therefore, this row would never be considered as a candidate for the pivot row. The second, third, and optimal tableaus for this problem are shown in Tables A-22, A-23, and A-24.

c_j	Basic Variables	Quantity	400 x_1	200 x_2	0 s_1	0 s_2	M A_1	M A_2
M	A_1	30	1	1	0	0	1	0
M	A_2	80	2	8	1	0	0	1
0	s_2	20	1	0	0	1	0	0
	z_j	$110M$	$3M$	$9M$	M	0	M	M
	$c_j \quad z_j$		$400 + 3M$	$200 + 9M$	M	0	0	0

Table A-21

c_j	Basic Variables	Quantity	400 x_1	200 x_2	0 s_1	0 s_2	M A_1
M	A_1	20	3/4	0	1/8	0	1
200	x_2	10	1/4	1	1/8	0	0
0	s_2	20	1	0	0	1	0
	z_j	$2{,}000 \quad 20M$	$50 \quad 3M/4$	200	$25 - M/8$	0	M
	$c_j \quad z_j$		$350 + 3M/4$	0	$25 + M/8$	0	0

Table A-22

c_j	Basic Variables	Quantity	400 x_1	200 x_2	0 s_1	0 s_2	M A_1
M	A_1	5	0	0	1/8	3/4	1
200	x_2	5	0	1	1/8	1/4	0
400	x_1	20	1	0	0	1	0
	z_j	9,000 $5M$	400	200	25 $M/8$	350 + $3M/4$	M
	c_j z_j		0	0	25 + $M/8$	350 $3M/4$	0

Table A-23

c_j	Basic Variables	Quantity	400 x_1	500 x_2	0 s_1	0 s_2
0	s_1	40	0	0	1	6
200	x_2	10	0	1	0	1
400	x_1	20	1	0	0	1
	z_j	10,000	400	200	0	200
	c_j z_j		0	0	0	200

Table A-24

The solution for the leather shop problem is (see Table A-24):

x_1 = 20 briefcases
x_2 = 10 pieces of luggage
s_1 = 40 extra yd^2 of leather
Z = \$10,000 profit per month

It is now possible to summarize a set of rules for transforming all three types of model constraints.

Constraint	Adjustment	Objective Function Coefficient: Maximization	Objective Function Coefficient: Minimization
$\leq$	Add a slack variable	0	0
$=$	Add an artificial variable	M	M
$\geq$	Subtract a surplus variable	0	0
	and add an artificial variable	M	M

Irregular Types of Linear Programming Problems

The basic simplex solution of typical maximization and minimization problems has been shown in this module. However, there are several special types of atypical linear programming problems. Although these special cases do not occur frequently, they will be described within the simplex framework so that you can recognize them when they arise.

These special types include problems with more than one optimal solution, infeasible problems, problems with unbounded solutions, problems with ties for the pivot column or ties for the pivot row, and problems with constraints with negative quantity values.

Multiple Optimal Solutions

Consider the Beaver Creek Pottery Company example with the objective function changed as follows.

$$Z = 40x_1 + 50x_2$$

to

$$Z = 40x_1 + 30x_2$$

$$\text{maximize } Z = 40x_1 + 30x_2$$

subject to

$$x_1 + 2x_2 \leq 40$$
$$4x_1 + 3x_2 \leq 120$$
$$x_1, x_2 \geq 0$$

The graph of this model is shown in Figure A-5. The slight change in the objective function makes it now *parallel* to the constraint line, $4x_1 + 3x_2 = 120$. Therefore, as the objective function edge moves outward from the origin, it touches the whole line segment BC rather than a single extreme corner point before it leaves the feasible solution area. The endpoints of this line segment, B and C, are typically referred to as the *alternate optimal solutions.* It is understood that these points represent the endpoints of a range of optimal solutions.

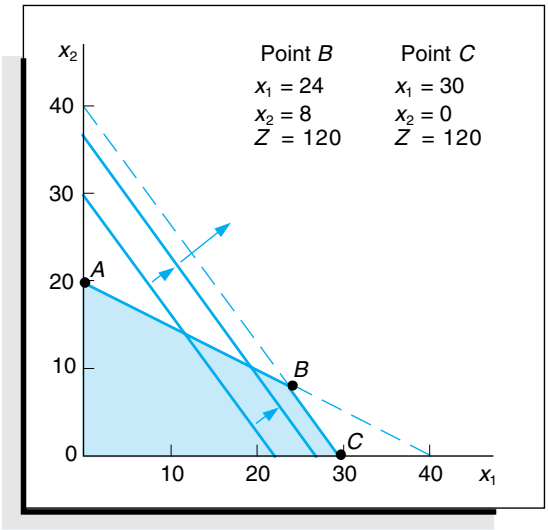


Figure A-5

The optimal simplex tableau for this problem is shown in Table A-25. This corresponds to point C in Figure A-5.

The fact that this problem contains multiple optimal solutions can be determined from the $c_j - z_j$ row. Recall that the $c_j - z_j$ row values are the net increases in profit per unit for the variable in each column. Thus, $c_j - z_j$ values of zero indicate no net increase in profit *and* no net loss in profit. We would expect the basic variables, s_1 and x_1, to have zero $c_j - z_j$ values because they are part of the basic feasible solution; they are already in the solution so they cannot be entered again. However, the x_2 column has a $c_j - z_j$ value of zero and it is

not part of the basic feasible solution. This means that if some mugs (x_2) were produced, we would have a new product mix but the same total profit. Thus, a multiple optimal solution is indicated by a $c_j - z_j$ (or $z_j - c_j$) row value of zero for a nonbasic variable.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
0	s_1	10	0	5/4	1	1/4
40	x_1	30	1	3/4	0	1/4
	z_j	1,200	40	30	0	10
	$c_j - z_j$		0	0	0	10

Table A-25

To determine the alternate endpoint solution, let x_2 be the entering variable (pivot column) and select the pivot row as usual. This selection results in the s_1 row being the pivot row. The alternate solution corresponding to point B in Figure A-5 is shown in Table A-26.

c_j	Basic Variables	Quantity	40 x_1	50 x_2	0 s_1	0 s_2
40	x_2	8	0	1	4/5	1/5
30	x_1	24	1	0	3/5	2/5
	z_j	1,200	40	30	0	10
	$c_j - z_j$		0	0	0	10

Table A-26

An Infeasible Problem

Another linear programming irregularity is the case where a problem has no feasible solution area; thus, there is no basic feasible solution to the problem.

An example of an infeasible problem is formulated next and depicted graphically in Figure A-6.

$$\text{maximize } Z = 5x_1 + 3x_2$$

subject to

$$\begin{aligned} 4x_1 + 2x_2 &\leq 8 \\ x_1 &\geq 4 \\ x_2 &\geq 6 \\ x_1, x_2 &\geq 0 \end{aligned}$$

The three constraints do not overlap to form a feasible solution area. Because no point satisfies all three constraints simultaneously, there is no solution to the problem. The final simplex tableau for this problem is shown in Table A-27.

The tableau in Table A-27 has all zero or negative values in the $c_j - z_j$ row, indicating that it is optimal. However, the solution is $x_2 = 4$, $A_1 = 4$, and $A_2 = 2$. Because the existence of artificial variables in the final solution makes the solution meaningless, this is not a real

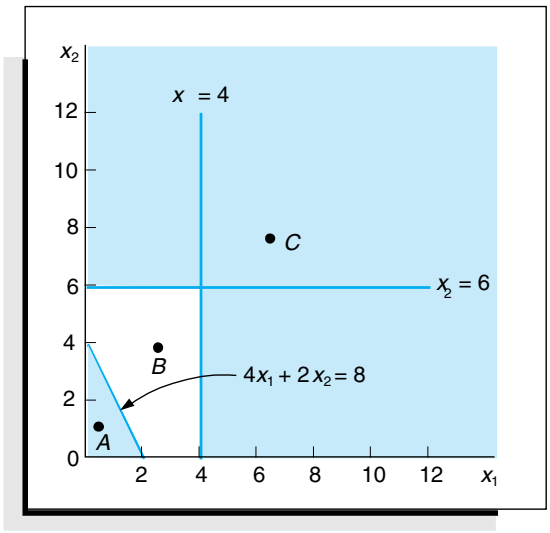


Figure A-6

c_j	Basic Variables	Quantity	5 x_1	3 x_2	0 s_1	0 s_2	0 s_3	M A_1	M A_2
3	x_2	4	2	1	1/2	0	0	0	0
M	A_1	4	1	0	0	1	0	1	0
M	A_2	2	2	0	1/2	0	1	0	1
	z_j	12 $6M$	$6 + M$	3	$3/2 + M/2$	M	M	M	M
	$c_j \quad z_j$		$1 - M$	0	$3/2 \quad M/2$	M	M	0	0

Table A-27

solution. In general, any time the $c_j \quad z_j$ (or $z_j \quad c_j$) row indicates that the solution is optimal but there are artificial variables in the solution, the solution is infeasible. Infeasible problems do not typically occur, but when they do they are usually a result of errors in defining the problem or in formulating the linear programming model.

An Unbounded Problem

In some problems the feasible solution area formed by the model constraints is not closed. In these cases it is possible for the objective function to increase indefinitely without ever reaching a maximum value because it never reaches the boundary of the feasible solution area.

An example of this type of problem is formulated next and shown graphically in Figure A-7.

$$\text{maximize } Z = 4x_1 + 2x_2$$

subject to

$$\begin{aligned} x_1 &\geq 4 \\ x_2 &\leq 2 \\ x_1, x_2 &\geq 0 \end{aligned}$$

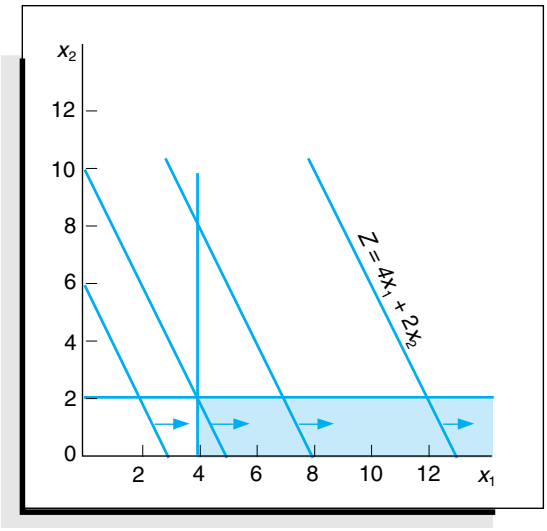


Figure A-7

In Figure A-7 the objective function is shown to increase without bound; thus, a solution is never reached.

The second tableau for this problem is shown in Table A-28. In this simplex tableau, s_1 is chosen as the entering nonbasic variable and pivot column. However, there is no pivot row or leaving basic variable. One row value is 4 and the other is undefined. This indicates that a "most constrained" point does not exist and that the solution is unbounded. In general, a solution is unbounded if the row value ratios are all negative or undefined.

			4	**2**	**0**	**0**	
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	
4	x_1	4	0	0	1	0	$4 \div -1 = 4$
0	s_2	2	1	1	0	1	$2 \div 0 = \infty$
	z_j	16	4	0	4	0	
	$c_j \quad z_j$		0	2	4	0	

Table A-28

Unlimited profits are not possible in the real world; an unbounded solution, like an infeasible solution, typically reflects an error in defining the problem or in formulating the model.

Tie for the Pivot Column

Sometimes when selecting the pivot column, you may notice that the greatest positive $c_j \quad z_j$ (or $z_j \quad c_j$) row values are the same; thus, there is a tie for the pivot column. When this happens, one of the two tied columns should be selected arbitrarily. Even though one choice may require fewer subsequent iterations than the other, there is no way of knowing this beforehand.

Tie for the Pivot Row -Degeneracy

It is also possible to have a tie for the pivot row (i.e., two rows may have identical lowest nonnegative values). Like a tie for a pivot column, a tie for a pivot row should be broken arbitrarily. However, after the tie is broken, the basic variable that was the *other* choice for

the leaving basic variable will have a quantity value of zero in the next tableau. This condition is commonly referred to as *degeneracy* because theoretically it is possible for subsequent simplex tableau solutions to degenerate so that the objective function value never improves and optimality never results. This occurs infrequently, however.

In general, tableaus with ties for the pivot row should be treated normally. If the simplex steps are carried out as usual, the solution will evolve normally.

The following is an example of a problem containing a tie for the pivot row.

$$\text{maximize } Z = 4x_1 + 6x_2$$

subject to

$$\begin{aligned} 6x_1 + 4x_2 &\leq 24 \\ x_2 &\leq 3 \\ 5x_1 + 10x_2 &\leq 40 \\ x_1, x_2 &\geq 0 \end{aligned}$$

For the sake of brevity we will skip the initial simplex tableau for this problem and go directly to the second simplex tableau in Table A-29, which shows a tie for the pivot row between the s_1 and s_3 rows.

			4	**6**	**0**	**0**	**0**		
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	s_3		
0	s_1	12	6	0	1	4	0	$12 \div 6 = 2$	*Tie*
6	x_2	3	0	1	0	1	0		
0	s_3	10	5	0	0	10	1	$10 \div 5 = 2$	*Tie*
	z_j	18	0	6	0	6	0		
	$c_j \quad z_j$		4	0	0	6	0		

Table A-29

The s_3 row is selected arbitrarily as the pivot row, resulting in the third simplex tableau, shown in Table A-30.

			4	**6**	**0**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	s_3
0	s_1	0	0	0	1	8	6/5
6	x_2	3	0	1	0	1	0
4	x_1	2	1	0	0	2	1/5
	z_j	26	4	6	0	2	4/5
	$c_j \quad z_j$		0	0	0	2	4/5

Table A-30

Note that in Table A-30 a quantity value of zero now appears in the s_1 row, representing the degenerate condition resulting from the tie for the pivot row. However, the simplex process should be continued as usual: s_2 should be selected as the entering basic variable and the s_1 row should be selected as the pivot row. (Recall that the pivot row value of zero is the minimum nonnegative quotient.) The final optimal tableau is shown in Table A-31.

c_j	Basic Variables	Quantity	4 x_1	6 x_2	0 s_1	0 s_2	0 s_3
0	s_2	0	0	0	1/8	1	3/20
6	x_2	3	0	1	1/8	0	3/20
4	x_1	2	1	0	1/4	0	1/10
	z_j	26	4	6	1/4	0	1/2
	$c_j \quad z_j$		0	0	1/4	0	1/2

Table A-31

Notice that the optimal solution did not change from the third to the optimal simplex tableau. The graphical analysis of this problem shown in Figure A-8 reveals the reason for this.

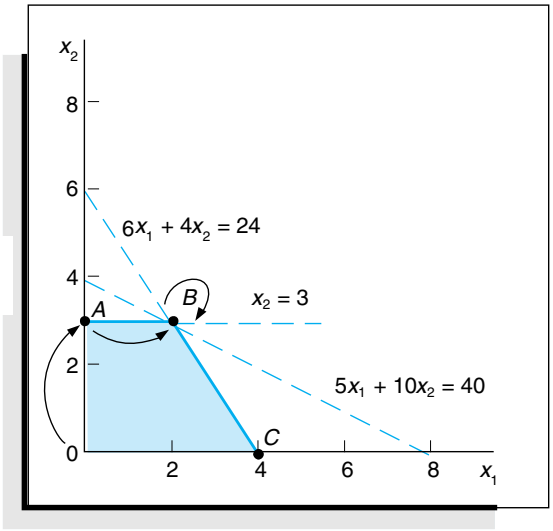


Figure A-8

Notice that in the third tableau (Table A-30) the simplex process went to point B, where all three constraint lines intersect. This is, in fact, what caused the tie for the pivot row and the degeneracy. Subsequently, the simplex process stayed at point B in the optimal tableau (Table A-31). The two tableaus represent two different basic feasible solutions corresponding to two different sets of model constraint equations.

Negative Quantity Values

Occasionally a model constraint is formulated with a negative quantity value on the right side of the inequality sign for example,

$$6x_1 + 2x_2 \geq -30$$

This is an improper condition for the simplex method, because for the method to work, all quantity values must be positive or zero.

This difficulty can be alleviated by multiplying the inequality by 1, which also changes the direction of the inequality.

$$\begin{array}{l} (\ \ 1)\,(\ \ 6x_1 + 2x_2 \geq -30) \\ \qquad\ \ 6x_1 \quad 2x_2 \leq 30 \end{array}$$

Now the model constraint is in proper form to be transformed into an equation and solved by the simplex method.

Summary of Simplex

Multiple optimal solutions are identified by $c_j - z_j$ (or $z_j - c_j$) $= 0$ for a nonbasic variable. To determine the alternate solution(s), enter the nonbasic variable(s) with a $c_j - z_j$ value equal to zero.

An infeasible problem is identified in the simplex procedure when an optimal solution is achieved (i.e., when all $c_j - z_j \leq 0$) and one or more of the basic variables are artificial.

An unbounded problem is identified in the simplex procedure when it is not possible to select a pivot row that is, when the values obtained by dividing the quantity values by the corresponding pivot column values are negative or undefined.

The Dual

Every linear programming model has two forms: the *primal* and the *dual.* The original form of a linear programming model is called the primal. All the examples in this module are primal models. The dual is an alternative model form derived completely from the primal. The dual is useful because it provides the decision maker with an alternative way of looking at a problem. Whereas the primal gives solution results in terms of the amount of profit gained from producing products, the dual provides information on the *value* of the constrained resources in achieving that profit.

The following example will demonstrate how the dual form of a model is derived and what it means. The Hickory Furniture Company produces tables and chairs on a daily basis. Each table produced results in $160 in profit; each chair results in $200 in profit. The production of tables and chairs is dependent on the availability of limited resources labor, wood, and storage space. The resource requirements for the production of tables and chairs and the total resources available are as follows.

	Resource Requirements		
Resource	TABLE	CHAIR	**Total Available per Day**
Labor (hr)	2	4	40
Wood (bd ft)	18	18	216
Storage (ft^2)	24	12	240

The company wants to know the number of tables and chairs to produce per day to maximize profit. The model for this problem is formulated as follows.

$$\text{maximize } Z = \$160x_1 + 200x_2$$

subject to

$$\begin{aligned} 2x_1 + 4x_2 &\leq 40 \text{ hr of labor} \\ 18x_1 + 18x_2 &\leq 216 \text{ bd ft of wood} \\ 24x_1 + 12x_2 &\leq 240 \text{ ft}^2 \text{ of storage space} \\ x_1, x_2 &\geq 0 \end{aligned}$$

where

x_1 = number of tables produced
x_2 = number of chairs produced

This model represents the primal form. For a primal maximization model, the dual form is a minimization model. The dual form of this example model is

$$\text{minimize } Z = 40y_1 + 216y_2 + 240y_3$$

subject to

$$2y_1 + 18y_2 + 24y_3 \geq 160$$
$$4y_1 + 18y_2 + 12y_3 \geq 200$$
$$y_1, y_2, y_3 \geq 0$$

The specific relationships between the primal and the dual demonstrated in this example are as follows.

1. The dual variables, y_1, y_2, and y_3, correspond to the model constraints in the primal. For every constraint in the primal there will be a variable in the dual. For example, in this case the primal has *three constraints;* therefore, the dual has *three decision variables*.
2. The quantity values on the right-hand side of the primal inequality constraints are the objective function coefficients in the dual. The constraint quantity values in the primal, 40, 216, and 240, form the dual objective function: $Z = 40y_1 + 216y_2 + 240y_3$.
3. The model constraint coefficients in the primal are the decision variable coefficients in the dual. For example, the labor constraint in the primal has the coefficients 2 and 4. These values are the y_1 variable coefficients in the model constraints of the dual: $2y_1$ and $4y_1$
4. The objective function coefficients in the primal, 160 and 200, represent the model constraint requirements (quantity values on the right-hand side of the constraint) in the dual.
5. Whereas the maximization primal model has $\leq$ constraints, the minimization dual model has $\geq$ constraints.

The primal dual relationships can be observed by comparing the two model forms shown in Figure A-9.

Now that we have developed the dual form of the model, the next step is determining what the dual means. In other words, what do the decision variables y_1, y_2, and y_3 mean, what do the $\geq$ model constraints mean, and what is being minimized in the dual objective function?

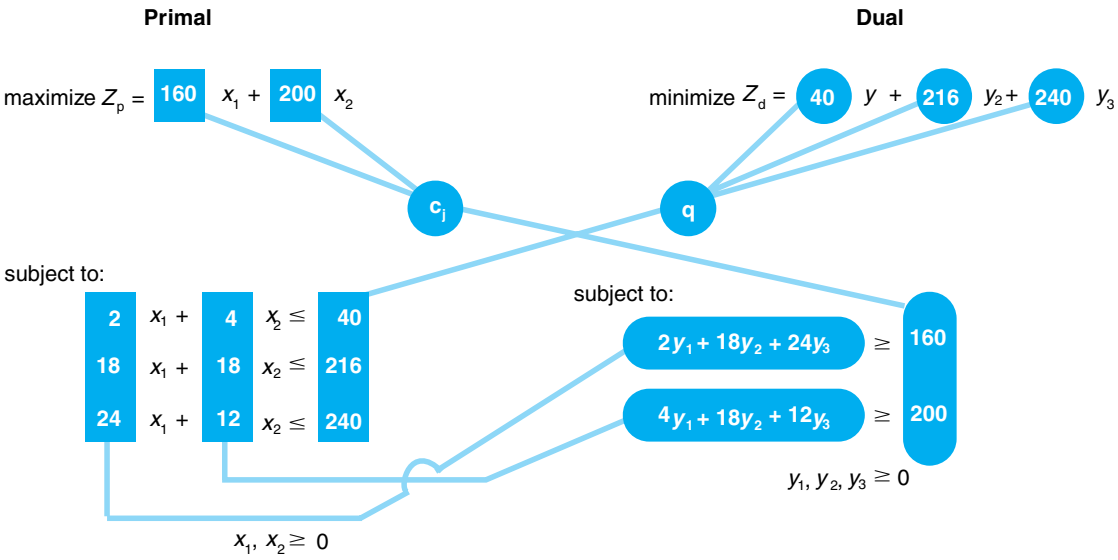
Interpreting the Dual Model

The dual model can be interpreted by observing the simplex solution to the primal form of the model. The simplex solution to the primal model is shown in Table A-32.

Interpreting this primal solution, we have

x_1 = 4 tables
x_2 = 8 chairs
s_3 = 48 ft^2 of storage space
Z = \$2,240 profit

Figure A-9



c_j	Basic Variables	Quantity	160 x_1	200 x_2	0 s_1	0 s_2	0 s_3
200	x_2	8	0	1	1/2	1/18	0
160	x_1	4	1	0	1/2	1/9	0
0	s_3	48	0	0	6	2	1
	z_j	2,240	160	200	20	20/3	0
	$c_j - z_j$		0	0	20	20/3	0

Table A-32

This optimal primal tableau also contains information about the dual. In the $c_j - z_j$ row of Table A-32, the negative values of 20 and 20/3 under the s_1 and s_2 columns indicate that if one unit of either s_1 or s_2 were entered into the solution, profit would *decrease* by \$20 or \$6.67 (i.e., 20/3), respectively.

Recall that s_1 represents unused labor and s_2 represents unused wood. In the present solution s_1 and s_2 are not basic variables, so they both equal zero. This means that all of the material and labor are being used to make tables and chairs, and there are no excess (slack) labor hours or board feet of material left over. Thus, if we enter s_1 or s_2 into the solution, then s_1 or s_2 no longer equals zero, we would be decreasing the use of labor or wood. If, for example, one unit of s_1 is entered into the solution, then one unit of labor previously used is not used, and profit is reduced by \$20.

Let us assume that one unit of s_1 has been entered into the solution so that we have one hour of unused labor ($s_1 = 1$). Now let us remove this unused hour of labor from the solution so that all labor is again being used. We previously noted that profit was decreased by \$20 by entering one hour of unused labor; thus, it can be expected that if we take this hour back (and use it again), profit will be increased by \$20. This is analogous to saying that if we could get one more hour of labor, we could increase profit by \$20. Therefore, if we could purchase one hour of labor, we would be willing to pay up to \$20 for it because that is the amount by which it would increase profit.

The negative $c_j - z_j$ row values of \$20 and \$6.67 are the *marginal values* of labor (s_1) and wood (s_2), respectively. These dual values are also often referred to as *shadow prices,* since they reflect the maximum "price" one would be willing to pay to obtain one more unit of the resource.

What happened to the third resource, storage space? The answer can be seen in Table A-32. Notice that the $c_j - z_j$ row value for s_3 (which represents unused storage space) is zero. This means that storage space has a marginal value of zero; that is, we would not be willing to pay anything for an extra foot of storage space.

The reason more storage space has no marginal value is because storage space was not a limitation in the production of tables and chairs. Table A-32 shows that 48 square feet of storage space were left unused (i.e., $s_3 = 48$) after the 4 tables and 8 chairs were produced. Since the company already has 48 square feet of storage space left over, an extra square foot would have no additional value; the company cannot even use all of the storage space it has available.

We need to consider one additional aspect of these marginal values. In our discussion of the marginal value of these resources, we have indicated that the marginal value (or *shadow price*) is the *maximum* amount that would be paid for additional resources. The marginal value of \$60 for one hour of labor is not necessarily what the Hickory Furniture Company would *pay* for an hour of labor. This depends on how the objective function is defined. In this example we are assuming that all of the resources available, 40 hours of labor, 216 board feet of wood, and 240 square feet of storage space, are already paid for. Even if the company does not use all the resources, it still must pay for them. They are *sunk* costs. In other words, the cost of any additional resources secured are included in the objective function coefficients. As such, the profit values in the objective function for each product are unaffected by how much of a resource is actually used; the profit is independent of the resources used. If the cost of the resources is not included in the profit function, then securing additional resources would reduce the marginal value.

Continuing our analysis, we note that the profit in the primal model was shown to be \$2,240. For the furniture company, the value of the resources used to produce tables and chairs must be in terms of this profit. In other words, the value of the labor and wood resources is determined by their contribution toward the \$2,240 profit. Thus, if the company wanted to assign a *value* to the resources it used, it could not assign an amount greater than the profit earned by the resources. Conversely, using the same logic, the total *value* of the resources must also be at least as much as the profit they earn. Thus, the value of all the resources must *exactly equal* the profit earned by the optimal solution.

Now let us look again at the dual form of the model.

$$\text{minimize } Z_d = 40y_1 + 16y_2 + 240y_3$$

subject to

$$\begin{aligned} 2y_1 + 18y_2 + 24y_3 &\geq 160 \\ 4y_1 + 18y_2 + 12y_3 &\geq 200 \\ y_1, y_2, y_3, &\geq 0 \end{aligned}$$

Given the previous discussion on the value of the model resources, we can now define the decision variables of the dual, y_1, y_2, and y_3, to represent the marginal values of the resources:

y_1 = marginal value of 1 hr of labor = \$20
y_2 = marginal value of 1 bd ft of wood = \$6.67
y_3 = marginal value of 1 ft^2 of storage space = \$0

Use of the Dual

The importance of the dual to the decision maker lies in the information it provides about the model resources. Often the manager is less concerned about profit than about the use of resources because the manager often has more control over the use of resources than over the accumulation of profits. The dual solution informs the manager of the value of the resources, which is important in deciding whether or not to secure more resources and how much to pay for these additional resources.

If the manager secures more resources, the next question is "How does this affect the original solution?" The feasible solution area is determined by the values forming the model constraints, and if those values are changed, it is possible for the feasible solution area to change. The effect on the solution of changes to the model is the subject of sensitivity analysis, the next topic to be presented here.

Sensitivity Analysis

In this section we will show how sensitivity ranges are mathematically determined using the simplex method. While this is not as efficient or quick as using the computer, close examination of the simplex method for performing sensitivity analysis can provide a more thorough understanding of the topic.

Changes in Objective Function Coefficients

To demonstrate sensitivity analysis for the coefficients in the objective function, we will use the Hickory Furniture Company example developed in the previous section. The model for this example was formulated as

$$\text{maximize } Z = \$160x_1 + 200x_2$$

subject to

$$\begin{aligned} 2x_1 + 4x_2 &\leq 40 \text{ hr of labor} \\ 18x_1 + 18x_2 &\leq 216 \text{ bd ft of wood} \\ 24x_1 + 12x_2 &\leq 240 \text{ ft}^2 \text{ of storage space} \\ x_1, x_2 &\geq 0 \end{aligned}$$

where

x_1 = number of tables produced
x_2 = number of chairs produced

The coefficients in the objective function will be represented symbolically as c_j (the same notation used in the simplex tableau). Thus, $c_1 = 160$ and $c_2 = 200$. Now, let us consider a change in one of the c_j values by an amount Δ. For example, let us change $c_1 = 160$ by $\Delta = 90$. In other words, we are changing c_1 from \$160 to \$250. The effect of this change on the solution of this model is shown graphically in Figure A-10.

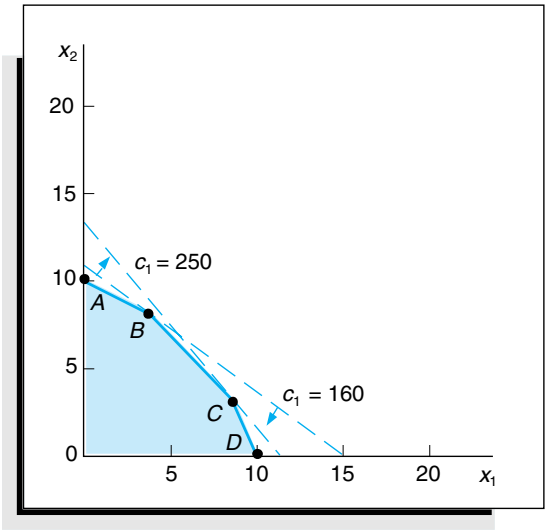


Figure A-10

Originally, the solution to this problem was located at point B in Figure A-10, where $x_1 = 4$ and $x_2 = 8$. However, increasing c_1 from \$160 to \$250 shifts the slope of the objective function so that point C ($x_1 = 8, x_2 = 4$) becomes the optimal solution. This demonstrates that a change in one of the coefficients of the objective function can change the optimal solution. Therefore, sensitivity analysis is performed to determine the range over which c_j can be changed without altering the optimal solution.

The range of c_j that will maintain the optimal solution can be determined directly from the optimal simplex tableau. The optimal simplex tableau for our furniture company example is shown in Table A-33.

			160	**200**	**0**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	s_3
200	x_2	8	0	1	1/2	1/18	0
160	x_1	4	1	0	1/2	1/9	0
0	s_3	48	0	0	6	2	1
	z_j	2,240	160	200	20	20/3	0
	$c_j - z_j$		0	0	20	20/3	0

Table A-33

First, consider a Δ change for c_1. This will change the c_1 value from $c_1 = 160$ to $c_1 = 160 + \Delta$, as shown in Table A-34. Notice that when c_1 is changed to $160 + \Delta$, the new value is included not only in the top c_j row but also in the left-hand c_j column. This is because x_1 is a basic solution variable. Since $160 + \Delta$ is in the left-hand column, it becomes

a multiple of the column values when the new z_j row values and the subsequent $c_j - z_j$ row values, also shown in Table A-34, are computed.

c_j	Basic Variables	Quantity	**160 + Δ** x_1	**200** x_2	**0** s_1	**0** s_2	**0** s_3
200	x_2	8	0	1	1/2	1/18	0
160 + Δ	x_1	4	1	0	1/2	1/9	0
0	s_3	48	0	0	6	−2	1
	z_j	2,240 + 4Δ	160 + Δ	200	20 − Δ/2	20/3 + Δ/9	0
	$c_j - z_j$		0	0	-20 + Δ/2	20/3 − Δ/9	0

Table A-34

The solution shown in Table A-34 will remain optimal as long as the $c_j - z_j$ row values *remain negative*. (If $c_j \quad z_j$ becomes positive, the product mix will change, and if it becomes zero, there will be an alternative solution.) Thus, for the solution to remain optimal,

$$20 + \Delta/2 \leq 0$$

and

$$20/3 - \Delta/9 \leq 0$$

Both of these inequalities must be solved for Δ.

$$\begin{aligned} -20 + \Delta/2 &\leq 0 \\ \Delta/2 &\leq 20 \\ \Delta &\leq 40 \end{aligned}$$

and

$$\begin{aligned} 20/3 - \Delta/9 &\leq 0 \\ -\Delta/9 &\leq 20/3 \\ -\Delta &\leq 60 \\ \Delta &\geq -60 \end{aligned}$$

Thus, $\Delta \leq 40$ and $\Delta \geq -60$. Now recall that $c_1 = 160 + \Delta$; therefore, $\Delta = c_1 - 160$. Substituting the amount $c_1 \quad 160$ for Δ in these inequalities,

$$\begin{aligned} \Delta &\leq 40 \\ c_1 - 160 &\leq 40 \\ c_1 &\leq 200 \end{aligned}$$

and

$$\begin{aligned} \Delta &\geq -60 \\ c_1 - 160 &\geq -60 \\ c_1 &\geq 100 \end{aligned}$$

Therefore, the range of values for c_1 over which the solution basis will remain optimal (although the value of the objective function may change) is

$$100 \leq c_1 \leq 200$$

Next, consider a Δ change in c_2 so that $c_2 = 200 + \Delta$. The effect of this change in the final simplex tableau is shown in Table A-35.

c_j	Basic Variables	Quantity	160 x_1	200 + Δ x_2	0 s_1	0 s_2	0 s_3
200 + Δ	x_2	8	0	1	1/2	1/18	0
160	x_1	4	1	0	1/2	1/9	0
0	s_3	48	0	0	6	2	1
	z_j	2,240 + 8Δ	160	200 + Δ	20 + Δ/2	20/3 − Δ/18	0
	$c_j - z_j$		0	0	-20 − Δ/2	- 20/3 + Δ/18	0

Table A-35

As before, the solution shown in Table A-35 will remain optimal as long as the $c_j \quad z_j$ row values remain negative or zero. Thus, for the solution to remain optimal, we must have

$$- 20 - \Delta/2 \leq 0$$

and

$$- 20/3 + \Delta/18 \leq 0$$

Solving these inequalities for Δ gives

$$\begin{aligned} - 20 - \Delta/2 &\leq 0 \\ -\Delta/2 &\leq 20 \\ \Delta &\geq -40 \end{aligned}$$

and

$$\begin{aligned} - 20/3 + \Delta/18 &\leq 0 \\ \Delta/18 &\leq 20/3 \\ \Delta &\leq 120 \end{aligned}$$

Thus, $\Delta \geq -40$ and $\Delta \leq 120$. Since $c_2 = 200 + \Delta$, we have $\Delta = c_2 \quad 200$. Substituting this value for Δ in the inequalities yields

$$\begin{aligned} \Delta &\geq -40 \\ c_2 \quad 200 &\geq -40 \\ c_2 &\geq 160 \end{aligned}$$

and

$$\begin{aligned} \Delta &\leq 120 \\ c_2 \quad 200 &\leq 120 \\ c_2 &\leq 320 \end{aligned}$$

Therefore, the range of values for c_2 over which the solution will remain optimal is

$$160 \leq c_2 \leq 320$$

The ranges for both objective function coefficients are as follows.

$$\begin{aligned} 100 \leq c_1 &\leq 200 \\ 160 \leq c_2 &\leq 320 \end{aligned}$$

However, these ranges reflect a possible change in either c_1 or c_2, not simultaneous changes in both c_1 and c_2. Both of the objective function coefficients in this example were for basic solution variables. Determining the c_j sensitivity range for a decision variable that is *not basic* is much simpler. Because it is not in the basic variable column, the Δ change

does not become a multiple of the z_j row. Thus, the Δ change will show up in only one column in the $c_j \quad z_j$ row.

Changes in Constraint Quantity Values

To demonstrate the effect of a change in the quantity values of the model constraints, we will again use the Hickory Furniture Company example.

$$\text{maximize } Z = 160x_1 + 200x_2$$

subject to

$$\begin{aligned} 2x_1 + 4x_2 &\leq 40 \text{ hr of labor} \\ 18x_1 + 18x_2 &\leq 216 \text{ bd ft of wood} \\ 24x_1 + 12x_2 &\leq 240 \text{ ft}^2 \text{ of storage space} \\ x_1, x_2 &\geq 0 \end{aligned}$$

The quantity values 40, 216, and 240 will be represented symbolically as q_i. Thus, $q_1 = 40$, $q_2 = 216$, and $q_3 = 240$. Now consider a Δ change in q_2. For example, let us change $q_2 = 216$ by $\Delta = 18$. In other words, q_2 is changed from 216 board feet to 234 board feet. The effect of this change is shown graphically in Figure A-11.

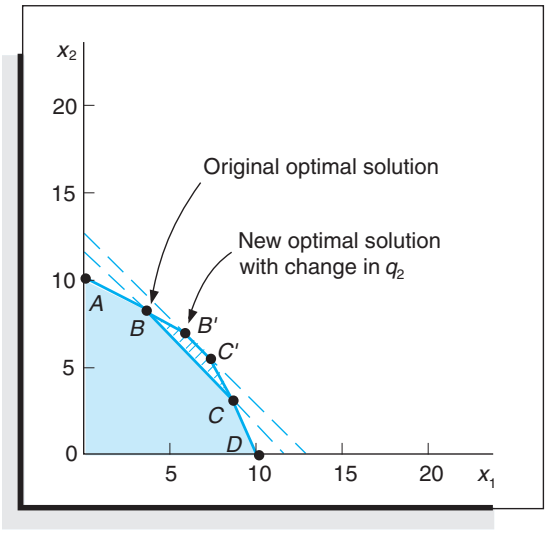


Figure A-11

In Figure A-11 a change in q_2 is shown to have the effect of changing the feasible solution area from $0ABCD$ to $0AB'C'D$. Originally, the optimal solution point was B; however, the change in q_2 causes B' to be the new optimal solution point. At point B the optimal solution is

$$\begin{aligned} x_1 &= 4 \\ x_2 &= 8 \\ s_3 &= 48 \\ s_1 \text{ and } s_2 &= 0 \\ Z &= \$2{,}240 \end{aligned}$$

At point B' the new optimal solution is

$$\begin{aligned} x_1 &= 6 \\ x_2 &= 7 \end{aligned}$$

$$s_3 = 12$$
$$s_1 \text{ and } s_2 = 0$$
$$Z = \$2{,}360$$

Thus, a change in a q_i value can change the values of the optimal solution. At some point an increase or decrease in q_i will change the variables in the optimal solution mix, including the slack variables. For example, if q_2 increases to 240 board feet, then the optimal solution point will be at

$$x_1 = 6.67$$
$$x_2 = 6.67$$
$$s_3 = 0$$
$$s_1 \text{ and } s_2 = 0$$
$$Z = \$2{,}400$$

Notice that s_3 has left the solution; thus, the optimal solution mix has changed. At this point, where $q_2 = 240$, which is the upper limit of the sensitivity range for q_2, the shadow price will also change. Therefore, the purpose of sensitivity analysis is to determine the range for q_i over which the optimal variable mix will remain the same and the shadow price will remain the same.

As in the case of the c_j values, the range for q_i can be determined directly from the optimal simplex tableau. As an example, consider a Δ increase in the number of labor hours. The model constraints become

$$2x_1 + 4x_2 \leq 40 + 1\Delta$$
$$18x_1 + 18x_2 \leq 216 + 0\Delta$$
$$24x_1 + 12x_2 \leq 240 + 0\Delta$$

Notice in the initial simplex tableau for our example (Table A-36) that the changes in the quantity column are the same as the coefficients in the s_1 column.

			160	**200**	**0**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	s_1	s_2	s_3
0	s_1	$40 + 1\Delta$	2	4	1	0	0
0	s_2	$216 + 0\Delta$	18	18	0	1	0
0	s_3	$240 + 0\Delta$	24	12	0	0	1
	z_j	0	0	0	0	0	0
	$c_j - z_j$		160	200	0	0	0

Table A-36

This duplication will carry through each subsequent tableau, so the s_1 column values will duplicate the Δ changes in the quantity column in the final tableau (Table A-37).

In effect, the Δ changes form a separate column identical to the s_1 column. Therefore, to determine the Δ change, we need only observe the slack (s_i) column corresponding to the model constraint quantity (q_i) being changed.

Recall that a requirement of the simplex method is that the quantity values not be negative. If any q_i value becomes negative, the current solution will no longer be *feasible* and a new variable will enter the solution. Thus, the inequalities

c_j	Basic Variables	Quantity	160 x_1	200 x_2	0 s_1	0 s_2	0 s_3
200	x_2	$8 + \Delta/2$	0	1	1/2	1/18	0
160	x_1	$4 - \Delta/2$	1	0	1/2	1/9	0
0	s_3	$48 + 6\Delta$	0	0	6	2	1
	z_j	$2{,}240 + 20\Delta$	160	200	20	20/3	0
	$c_j - z_j$		0	0	20	20/3	0

Table A-37

$$8 + \Delta/2 \geq 0$$
$$4 - \Delta/2 \geq 0$$
$$48 + 6\Delta \geq 0$$

are solved for Δ:

$$8 + \Delta/2 \geq 0$$
$$\Delta/2 \geq -8$$
$$\Delta \geq -16$$

and

$$4 - \Delta/2 \geq 0$$
$$-\Delta/2 \geq -4$$
$$\Delta \leq 8$$

and

$$48 + 6\Delta \geq 0$$
$$6\Delta \geq -48$$
$$\Delta \geq -8$$

Since

$$q_1 = 40 + \Delta$$

then

$$\Delta = q_1 - 40$$

These values are substituted into the inequalities $\Delta \geq -16$, $\Delta \leq 8$, and $\Delta \geq -8$ as follows.

$$\Delta \geq -16$$
$$q_1 - 40 \geq -16$$
$$q_1 \geq 24$$
$$\Delta \leq 8$$
$$q_1 - 40 \leq 8$$
$$q_1 \leq 48$$
$$\Delta \geq -8$$
$$q_1 - 40 \geq -8$$
$$q_1 \geq 32$$

Summarizing these inequalities, we have

$$24 \leq 32 \leq q_1 \leq 48$$

The value of 24 can be eliminated, since q_1 must be greater than 32; thus,

$$32 \leq q_1 \leq 48$$

As long as q_1 remains in this range, the present basic solution variables will remain the same and feasible. However, the quantity values of those basic variables may change. In other words, although the variables in the basis remain the same, their values can change.

To determine the range for q_2 (where $q_2 = 216 + \Delta$), the s_2 column values are used to develop the Δ inequalities.

$$\begin{aligned} 8 &- \Delta/18 \geq 0 \\ 4 &+ \Delta/9 \geq 0 \\ 48 &\quad 2\Delta \geq 0 \end{aligned}$$

The inequalities are solved as follows.

$$\begin{aligned} 8 - \Delta/18 &\geq 0 \\ -\Delta/18 &\geq -8 \\ \Delta &\leq 144 \end{aligned}$$

$$\begin{aligned} 4 + \Delta/9 &\geq 0 \\ \Delta/9 &\geq -4 \\ \Delta &\geq -36 \end{aligned}$$

$$\begin{aligned} 48 \quad 2\Delta &\geq 0 \\ 2\Delta &\geq -48 \\ \Delta &\leq 24 \end{aligned}$$

Since

$$q_2 = 216 + \Delta$$

we have

$$\Delta = q_2 - 216$$

Substituting this value into the inequalities $\Delta \leq 144$, $\Delta \geq -36$, and $\Delta \leq 24$ gives a range of possible values for q_2:

$$\begin{aligned} \Delta &\leq 144 \\ q_2 - 216 &\leq 144 \\ q_2 &\leq 360 \end{aligned}$$

$$\begin{aligned} \Delta &\geq -36 \\ q_2 - 216 &\geq -36 \\ q_2 &\geq 180 \end{aligned}$$

$$\begin{aligned} \Delta &\leq 24 \\ q_2 - 216 &\leq 24 \\ q_2 &\leq 240 \end{aligned}$$

That is,

$$180 \leq q_2 \leq 240 \leq 360$$

The value 360 can be eliminated, because q_2 cannot exceed 240. Thus, the range over which the basic solution variables will remain the same is

$$180 \leq q_2 \leq 240$$

The range for q_3 is

$$192 \leq q_3 \leq \infty$$

The upper limit of ∞ means that q_3 can increase indefinitely (without limit) without changing the optimal variable solution mix in the shadow price.

Sensitivity analysis of constraint quantity values can be used in conjunction with the dual solution to make decisions regarding model resources. Recall from our analysis of the dual solution of the Hickory Furniture Company example that

$$\begin{aligned} y_1 &= \$20, \text{ marginal value of labor} \\ y_2 &= \$6.67, \text{ marginal value of wood} \\ y_3 &= \$0, \text{ marginal value of storage space} \end{aligned}$$

Because the resource with the greatest marginal value is labor, the manager might desire to secure some additional hours of labor. How many hours should the manager get? Given that the range for q_1 is $32 \leq q_1 \leq 48$, the manager could secure up to an additional 8 hours of labor (i.e., 48 total hours) before the solution basis changes and the shadow price also changes. If the manager did purchase 8 more hours, the solution values could be found by observing the quantity values in Table A-37.

$$\begin{aligned} x_2 &= 8 + \Delta/2 \\ x_1 &= 4 - \Delta/2 \\ s_3 &= 48 + 6\Delta \end{aligned}$$

Since $\Delta = 8$,

$$\begin{aligned} x_2 &= 8 + (8)/2 \\ &= 12 \\ x_1 &= 4 \quad (8)/2 \\ &= 0 \\ s_3 &= 48 + 6(8) \\ &= 96 \end{aligned}$$

Total profit will be increased by \$20 for each extra hour of labor.

$$\begin{aligned} Z &= \$2{,}240 + 20\Delta \\ &= 2{,}240 + 20(8) \\ &= 2{,}240 + 160 \\ &= \$2{,}400 \end{aligned}$$

In this example for the Hickory Furniture Company, we considered only $\leq$ constraints in determining the sensitivity ranges for q_i values. To compute the q_i sensitivity range, we observed the slack column, s_i, since a Δ change in q_i was reflected in the s_i column. However, recall that with a $\geq$ constraint we subtract a surplus variable rather than adding a slack variable to form an equality (in addition to adding an artificial variable). Thus, for a $\geq$ constraint we must consider a $-\Delta$ change in q_i in order to use the s_i (surplus) column to perform sensitivity analysis. In that case sensitivity analysis would be performed exactly as shown in this example, except that the value of $q_i - \Delta$ would be used instead of $q_i + \Delta$ when computing the sensitivity range for q_i.

Problems

1. Following is a simplex tableau for a linear programming model.

c_j	Basic Variables	Quantity	10 x_1	2 x_2	6 x_3	0 s_1	0 s_2	0 s_3
2	x_1	10	0	1	2	1	1/2	0
10	x_2	40	1	0	2	0	1/2	0
0	s_3	30	0	0	8	3	3/2	1
	z_j	420	10	2	16	2	4	0
	$c_j \quad z_j$		0	0	10	2	4	0

   a. What is the solution given in this tableau?
   b. Is the solution in this tableau optimal? Why?
   c. What does x_3 equal in this tableau? s_2?
   d. Write out the original objective function for the linear programming model, using only decision variables.
   e. How many constraints are in the linear programming model?
   f. Explain briefly why it would have been difficult to solve this problem graphically.

2. The following is a simplex tableau for a linear programming model.

c_j	Basic Variables	Quantity	6 x_1	20 x_2	12 x_3	0 s_1	0 s_2
6	x_1	20	1	1	0	0	0
12	x_3	10	0	1/3	1	0	1/6
0	s_1	10	0	1/3	0	1	1/6
	z_j	240	6	10	12	0	2
	$z_j \quad c_j$		0	10	0	0	2

   a. Is this a maximization or a minimization problem? Why?
   b. What is the solution given in this tableau?
   c. Is the solution given in this tableau optimal? Why?

d. Write out the original objective function for the linear programming model using only decision variables.
e. How many constraints are in the linear programming model?
f. Were any of the constraints originally equations? Why?
g. What is the value of x_2 in this tableau?

3. The following is a simplex tableau for a linear programming problem.

			60	**50**	**45**	**50**	**0**	**0**	**0**	**0**
c_j	**Basic Variables**	**Quantity**	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4
0	s_1	20	0	1	0	0	1	0	0	0
50	x_4	15	0	0	0	1	0	1	0	0
60	x_1	12	1	1/2	0	0	0	0	1/10	0
0	s_4	45	0	0	8	6	0	6	0	1
	z_j	1,470	60	30	0	50	0	50	6	0
	$c_j \quad z_j$		0	20	45	0	0	50	6	0

a. Is this a maximization or a minimization problem?
b. What are the values of the decision variables in this tableau?
c. What are the values of the slack variables in this tableau?
d. What does the $c_j - z_j$ value of "20" in the "x_2" column mean?
e. Is this solution optimal? Why? If the solution is not optimal, determine the optimal solution.

4. The following is a simplex tableau for a linear programming problem.

			8	**10**	**4**	**0**	**0**	**0**	M	M
c_j	**Basic Variables**	**Quantity**	x_1	x_2	x_3	s_1	s_2	s_3	A_1	A_2
M	A_1	30	2/3	0	1	1	1/6	0	0	0
10	x_2	10	1/3	1	0	0	1/6	0	1	0
M	A_3	100	0	0	1	0	0	1	0	1
	z_j	$130M + 100$	$2M/3 + 10/3$	10	$2M$	M	$M/6 - 5/3$	M	M	M
	$z_j \quad c_j$		$2M/3 - 14/3$	0	$2M - 4$	M	$M/6 - 5/3$	M	0	0

a. Is this a maximization or a minimization problem?
b. What is the value of x_3 in this tableau?
c. What does the value "$M/6 - 5/3$" in the "s_2" column of the $z_j - c_j$ row mean?
d. What is the minimum number of additional simplex iterations that this problem must go through to determine a feasible optimal solution?
e. Is this solution optimal? Why? If the solution is not optimal, compute the optimal solution.

CHAPTER 5. TRANSPORTATION, TRANSSHIPMENT, AND ASSIGNMENT PROBLEMS

In this chapter, we examine three special types of linear programming model formulations transportation, transshipment, and assignment problems. They are part of a larger class of linear programming problems known as network flow problems.

We are considering these problems in a separate chapter because they represent a popular group of linear programming applications.

These problems have special mathematical characteristics that have enabled management scientists to develop very efficient, unique mathematical solution approaches to them.

These solution approaches are variations of the traditional simplex solution procedure. As in previous chapters, we will focus on model formulation and solution by using the computer, specifically by using QM for Windows.

The Transportation Model

The **transportation model** is formulated for a class of problems with the following unique characteristics:

(1) A product is transported from a number of sources to a number of destinations at the minimum possible cost; and

(2) each source is able to supply a fixed number of units of the product, and each destination has a fixed demand for the product.

Although the general transportation model can be applied to a wide variety of problems, it is this particular application to the transportation of goods that is most familiar and from which the problem draws its name.

In a transportation problem, items are allocated from sources to destinations at a minimum cost.

Lecturer note:

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Computer Solution of a Transportation Problem

Lecturer note:

The Transshipment Model

The transshipment model is an extension of the transportation model in which intermediate transshipment points are added between the sources and destinations.

An example of a transshipment point is a distribution center or warehouse located between plants and stores.

In a transshipment problem, items may be transported from sources through transshipment points on to destinations, from one source to another, from one transshipment point to another, from one destination to another, or directly from sources to destinations, or some combination of these alternatives.

The transshipment model includes intermediate points between sources and destinations.

Lecturer note:

The Assignment Model

The **assignment model** is a special form of a linear programming model that is similar to the transportation model.

There are differences, however. In the assignment model, the supply at each source and the demand at each destination are each limited to one unit.

An assignment model is for a special form of transportation problem in which all supply and demand values equal one.

Lecturer note:

Solved problems

1. A concrete company transports concrete from three plants to three construction sites. The supply capacities of the three plants, the demand requirements at the three sites, and the transportation costs per ton are as follows:

Construction Site

Plant	A	B	C	Supply (tons)
1	$ 8	$ 5	$ 6	120
2	15	10	12	80
3	3	9	10	80
Demand (tons)	150	70	100	280

Determine the linear programming model formulation for this problem and solve it by using QM.

Minimize $Z = \$8x_{1A} + 5x_{1B} + 6x_{1C} + 15x_{2A} + 10x_{2B} + 12x_{2C} + 3x_{3A} + 9x_{3B} + 10x_{3C}$

subject to:

$$x_{1A} + x_{1B} + x_{1C} = 120$$

$$x_{2A} + x_{2B} + x_{2C} = 80$$

$$x_{3A} + x_{3B} + x_{3C} = 80$$

$$x_{1A} + x_{2A} + x_{3A} \leq 150$$

$$x_{1B} + x_{2B} + x_{3B} \leq 70$$

$$x_{1C} + x_{2C} + x_{3C} \leq 100$$

$$x_{ij} \geq 0$$

T#1

* Given a transportation problem with the following costs, supply, and demand, find the optimal solution by using the computer:

To (cost)				
From	1	2	3	Supply
A	$6	$7	$4	100
B	5	3	6	180
C	8	5	7	200
Demand	135	175	170	

*Consider the following transportation problem:

To (cost)				
From	1	2	3	Supply
A	$ 6	$9	$M	130
B	12	3	5	70
C	4	8	11	100
Demand	80	110	60	

Formulate this problem as a linear programming model and solve it by using the computer.

*The Sav-Us Rental Car Agency has six lots in Nashville, and it wants to have a certain number of cars available at each lot at the beginning of each day for local rental. The agency would like a model it could quickly solve at the end of each day that would tell it how to redistribute the cars among the six lots in the minimum total time. The times required to travel between the six lots are as follows:

To (min.)

From	1	2	3	4	5	6
1		12	17	18	10	20
2	14		10	19	16	15
3	14	10		12	8	9
4	8	16	14		12	15

5	11	21	16	18		10
6	24	12	9	17	15	

*The agency would like the following number of cars at each lot at the end of the day. Also shown is the number of available cars at each lot at the end of a particular day. Determine the optimal reallocation of rental cars by using any initial solution approach and any solution method:

Lot (cars)

Cars	1	2	3	4	5	6
Available	37	20	14	26	40	28
Desired	30	25	20	40	30	20

*Bayville has built a new elementary school, increasing the town's total to four schoolsAddison, Beeks, Canfield, and Daley. Each has a capacity of 400 students. The school board wants to assign children to schools so that their travel time by bus is as short as possible. The school board has partitioned the town into five districts conforming to population densitynorth, south, east, west, and central. The average bus travel time from each district to each school is shown as follows:

Travel Time (min.)

District	Addison	Beeks	Canfield	Daley	Student Population
North	12	23	35	17	250
South	26	15	21	27	340
East	18	20	22	31	310
West	29	24	35	10	210
Central	15	10	23	16	290

Determine the number of children that should be assigned from each district to each school to minimize total student travel time.

Solution of the Transportation Model

The following example was used in chapter 6 of the text to demonstrate the formulation of the transportation model. Wheat is harvested in the Midwest and stored in grain elevators in three different cities—Kansas City, Omaha, and Des Moines. These grain elevators supply three flour mills, located in Chicago, St. Louis, and Cincinnati. Grain is shipped to the mills in railroad cars, each car capable of holding one ton of wheat. Each grain elevator is able to supply the following number of tons (i.e., railroad cars) of wheat to the mills on a monthly basis.

Grain Elevator	*Supply*
1. Kansas City	150
2. Omaha	175
3. Des Moines	275
Total	600 tons

Each mill demands the following number of tons of wheat per month.

Mill	*Demand*
A. Chicago	200
B. St. Louis	100
C. Cincinnati	300
Total	600 tons

The cost of transporting one ton of wheat from each grain elevator (source) to each mill (destination) differs according to the distance and rail system. These costs are shown in the following table. For example, the cost of shipping one ton of wheat from the grain elevator at Omaha to the mill at Chicago is $7.

	Mill		
Grain Elevator	*A. Chicago*	*B. St. Louis*	*C. Cincinnati*
1. Kansas City	$6	$ 8	$10
2. Omaha	7	11	11
3. Des Moines	4	5	12

The problem is to determine how many tons of wheat to transport from each grain elevator to each mill on a monthly basis in order to minimize the total cost of transportation.

The linear programming model for this problem is formulated in the equations that follow.

$$\text{minimize } Z = \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C}$$

subject to

$$\begin{aligned}
x_{1A} + x_{1B} + x_{1C} &= 150 \\
x_{2A} + x_{2B} + x_{2C} &= 175 \\
x_{3A} + x_{3B} + x_{3C} &= 275 \\
x_{1A} + x_{2A} + x_{3A} &= 200 \\
x_{1B} + x_{2B} + x_{3B} &= 100 \\
x_{1C} + x_{2C} + x_{3C} &= 300 \\
x_{ij} &\geq 0
\end{aligned}$$

In this model the decision variables, x_{ij}, represent the number of tons of wheat transported from each grain elevator, i (where $i = 1, 2, 3$), to each mill, j (where $j =$ A, B, C). The objective function represents the total transportation cost for each route. Each term in the objective function reflects the cost of the tonnage transported for one route. For example, if 20 tons are transported from elevator 1 to mill A, the cost of \$6 is multiplied by x_{1A} ($=20$), which equals \$120.

The first three constraints in the linear programming model represent the supply at each elevator; the last three constraints represent the demand at each mill. As an example, consider the first supply constraint, $x_{1A} + x_{1B} + x_{1C} = 150$. This constraint represents the tons of wheat transported from Kansas City to all three mills: Chicago (x_{1A}), St. Louis (x_{1B}), and Cincinnati (x_{1C}). The amount transported from Kansas City is limited to the 150 tons available. Note that this constraint (as well as all others) is an equation ($=$) rather than a $\leq$ inequality because all of the tons of wheat available will be needed to meet the *total demand* of 600 tons. In other words, the three mills demand 600 total tons, which is the exact amount that can be supplied by the three grain elevators. Thus, all that *can* be supplied *will be,* in order to meet demand. This type of model, in which supply exactly equals demand, is referred to as a *balanced transportation* model. The balanced model will be used to demonstrate the solution of a transportation problem.

Transportation models are solved manually within the context of a *tableau,* as in the simplex method. The tableau for our wheat transportation model is shown in Table B-1.

From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600

Table B-1

Each cell in the tableau represents the amount transported from one source to one destination. Thus, the amount placed in each cell is the value of a decision variable for that cell. For example, the cell at the intersection of row 1 and column A represents the decision variable x_{1A}. The smaller box within each cell contains the unit transportation cost for that route. For example, in cell 1A the value, \$6, is the cost of transporting one ton of wheat from Kansas City to Chicago. Along the outer rim of the tableau are the supply and demand constraint quantity values, which are referred to as *rim requirements*.

The two methods for solving a transportation model are the *stepping-stone method* and the *modified distribution method* (also known as *MODI*). In applying the simplex method, an initial solution had to be established in the initial simplex tableau. This same condition must be met in solving a transportation model. In a transportation model, an initial feasible solution can be found by several alternative methods, including the *northwest corner method,* the *minimum cell cost method,* and *Vogel's approximation model.*

The Northwest Corner Method

With the northwest corner method, an initial allocation is made to the cell in the upper left-hand corner of the tableau (i.e., the "northwest corner"). The amount allocated is the most possible, *subject to* the supply and demand constraints for that cell. In our example, we first allocate as much as possible to cell 1A (the northwest corner). This amount is 150 tons, since that is the maximum that can be supplied by grain elevator 1 at Kansas City, even though 200 tons are demanded by mill A at Chicago. This initial allocation is shown in Table B-2.

We next allocate to a cell *adjacent* to cell 1A, in this case either cell 2A or cell 1B. However, cell 1B no longer represents a feasible allocation, because the total tonnage of wheat available at source 1 (i.e., 150 tons) has already been allocated. Thus, cell 2A represents the only feasible alternative, and as much as possible is allocated to this cell. The amount allocated at 2A can be either 175 tons, the supply available from source 2 (Omaha), or 50 tons, the amount *now* demanded at destination A. (Recall that 150 of the 200 tons demanded at A have already been supplied.) Because 50 tons is the most constrained amount, it is allocated to cell 2A, as shown in Table B-2.

From \ To	A	B	C	Supply
1	6: 150	8	10	150
2	7: 50	11: 100	11: 25	175
	4	5	12: 275	275
	200	100	300	600

Table B-2

The third allocation is made in the same way as the second allocation. The only feasible cell adjacent to cell 2A is cell 2B. The most that can be allocated is either 100 tons (the amount demanded at mill B) or 125 tons (175 tons minus the 50 tons allocated to cell 2A). The smaller (most constrained) amount, 100 tons, is allocated to cell 2B, as shown in Table B-2.

The fourth allocation is 25 tons to cell 2C, and the fifth allocation is 275 tons to cell 3C, both of which are shown in Table B-2. Notice that all of the row and column allocations add up to the appropriate rim requirements.

The transportation cost of this solution is computed by substituting the cell allocations (i.e., the amounts transported),

$$\begin{aligned} x_{1A} &= 150 \\ x_{2A} &= 50 \\ x_{2B} &= 100 \\ x_{2C} &= 25 \\ x_{3C} &= 275 \end{aligned}$$

into the objective function.

$$\begin{aligned} Z &= \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C} \\ &= 6(150) + 8(0) + 10(0) + 7(50) + 11(100) + 11(25) + 4(0) + 5(0) + 12(275) \\ &= \$5{,}925 \end{aligned}$$

The steps of the northwest corner method are summarized here.

1. Allocate as much as possible to the cell in the upper left-hand corner, subject to the supply and demand constraints.
2. Allocate as much as possible to the next adjacent feasible cell.
3. Repeat step 2 until all rim requirements have been met.

The Minimum Cell Cost Method

With the minimum cell cost method, the basic logic is to allocate to the cells with the lowest costs. The initial allocation is made to the cell in the tableau having the lowest cost. In the transportation tableau for our example problem, cell 3A has the minimum cost of $4. As much as possible is allocated to this cell; the choice is either 200 tons or 275 tons. Even though 275 tons could be supplied to cell 3A, the most we can allocate is 200 tons, since only 200 tons are demanded. This allocation is shown in Table B-3.

From \ To	A	B	C	Supply
1	6	8	10	150
	7	11	11	175
	4	5	12	275
	100	300		600

Table B-3

Notice that all of the remaining cells in column A have now been eliminated, because all of the wheat was demanded at destination A, Chicago, has now been supplied by source 3, Des Moines.

The next allocation is made to the cell that has the minimum cost and also is feasible. This is cell 3B which has a cost of $5. The most that can be allocated is 75 tons (275 tons minus the 200 tons already supplied). This allocation is shown in Table B-4.

The third allocation is made to cell 1B, which has the minimum cost of $8. (Notice that cells with lower costs, such as 1A and 2A, are not considered because they were previously ruled out as infeasible.) The amount allocated is 25 tons. The fourth allocation of 125 tons is made to cell 1C, and the last allocation of 175 tons is made to cell 2C. These allocations, which complete the initial minimum cell cost solution, are shown in Table B-5.

From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4 200	5 75	12	275
Demand	200	100	300	600

Table B-4

From \ To	A	B	C	Supply
1	6	8 25	10 125	150
2	7	11	11 175	175
3	4 200	5 75	12	275
Demand	200	100	300	600

Table B-5

The total cost of this initial solution is $4,550, as compared to a total cost of $5,925 for the initial northwest corner solution. It is not a coincidence that a lower total cost is derived using the minimum cell cost method; it is a logical occurrence. The northwest corner method does not consider cost at all in making allocations—the minimum cell cost method does. It is therefore quite natural that a lower initial cost will be attained using the latter method. Thus, the initial solution achieved by using the minimum cell cost method is usually better in that, because it has a lower cost, it is closer to the optimal solution; fewer subsequent iterations will be required to achieve the optimal solution.

The specific steps of the minimum cell cost method are summarized next.

1. Allocate as much as possible to the feasible cell with the minimum transportation cost, and adjust the rim requirements.
2. Repeat step 1 until all rim requirements have been met.

Vogelís Approximation Model

The third method for determining an initial solution, Vogel's approximation model (also called VAM), is based on the concept of *penalty cost* or *regret*. If a decision maker incorrectly chooses from several alternative courses of action, a penalty may be suffered (and the decision maker may regret the decision that was made). In a transportation problem, the courses of action are the alternative routes, and a wrong decision is allocating to a cell that does not contain the lowest cost.

In the VAM method, the first step is to develop a penalty cost for each source and destination. For example, consider column A in Table B-6. Destination A, Chicago, can be

supplied by Kansas City, Omaha, and Des Moines. The best decision would be to supply Chicago from source 3 because cell 3A has the minimum cost of $4. If a wrong decision were made and the next higher cost of $6 were selected at cell 1A, a "penalty" of $2 per ton would result (i.e., $6 4 = $2). This demonstrates how the penalty cost is determined for each row and column of the tableau. The general rule for computing a penalty cost is to subtract the minimum cell cost from the next higher cell cost in each row and column. The penalty costs for our example are shown at the right and at the bottom of Table B-6.

From \ To	A	B	C	Supply	
1	6	8	10	150	2
	7	11	11	175	4
	4	5	12	275	1
	200	100	300	600	
	2	3	1		

Table B-6

The initial allocation in the VAM method is made in the row or column that has the highest penalty cost. In Table B-6, row 2 has the highest penalty cost of $4. We allocate as much as possible to the feasible cell in this row with the minimum cost. In row 2, cell 2A has the lowest cost of $7, and the most that can be allocated to cell 2A is 175 tons. With this allocation the greatest penalty cost of $4 has been avoided because the best course of action has been selected. The allocation is shown in Table B-7.

From \ To	A	B	C	Supply	
1	6	8	10	150	2
	7 175	11	11	175	
	4	5	12	275	1
	200	100	300	600	
	2	3	1		

Table B-7

After the initial allocation is made, *all* of the penalty costs must be recomputed. In some cases the penalty costs will change; in other cases they will not change. For example, the penalty cost for column C in Table B-7 changed from $1 to $2 (because cell 2C is no longer considered in computing penalty cost), and the penalty cost in row 2 was eliminated altogether (because no more allocations are possible for that row).

Next, we repeat the previous step and allocate to the row or column with the highest penalty cost, which is now column B with a penalty cost of $3 (see Table B-7). The cell in column B with the lowest cost is 3B, and we allocate as much as possible to this cell, 100 tons. This allocation is shown in Table B-8.

From \ To	A	B	C	Supply	
1	6	8	10	150	4
	7 175	11	11	175	
	4	5 100	12	275	8
[illegible]	200	100	300	600	
	2		2		

Table B-8

Note that all penalty costs have been recomputed in Table B-8. Since the highest penalty cost is now $8 for row 3 and since cell 3A has the minimum cost of $4, we allocate 25 tons to this cell, as shown in Table B-9.

From \ To	A	B	C	Supply
1	6	8	10	150
	7 175	11	11	175
	4 25	5 100	12	275
[illegible]	200	100	300	600
			2	

Table B-9

Table B-9 also shows the recomputed penalty costs after the third allocation. Notice that by now only column C has a penalty cost. Rows 1 and 3 have only one feasible cell, so a penalty does not exist for these rows. Thus, the last two allocations are made to column C. First, 150 tons are allocated to cell 1C because it has the lowest cell cost. This leaves only cell 3C as a feasible possibility, so 150 tons are allocated to this cell. Both of these allocations are shown in Table B-10.

The total cost of this initial Vogel's approximation model solution is $5,125, which is not as high as the northwest corner initial solution of $5,925. It is also not as low as the minimum cell cost solution of $4,550. Like the minimum cell cost method, VAM typically results in a lower cost for the initial solution than does the northwest corner method.

From \ To	A	B	C	Supply
1	6	8	10 150	150
2	7 175	11	11	175
	4 25	5 100	12 150	275
	200	100	300	600

Table B-10

The steps of Vogel's approximation model can be summarized in the following list.

1. Determine the penalty cost for each row and column by subtracting the lowest cell cost in the row or column from the next lowest cell cost in the same row or column.
2. Select the row or column with the highest penalty cost (breaking ties arbitrarily or choosing the lowest-cost cell).
3. Allocate as much as possible to the feasible cell with the lowest transportation cost in the row or column with the highest penalty cost.
4. Repeat steps 1, 2, and 3 until all rim requirements have been met.

The Stepping-Stone Solution Method

Once an initial *basic feasible solution* has been determined by any of the previous three methods, the next step is to solve the model for the optimal (i.e., minimum total cost) solution. There are two basic solution methods: the *stepping-stone solution method* and the modified distribution method (MODI). The stepping-stone solution method will be demonstrated first. Because the initial solution obtained by the minimum cell cost method had the lowest total cost of the three initial solutions, we will use it as the starting solution. Table B-11 repeats the initial solution that was developed from the minimum cell cost method.

From \ To	A	B	C	Supply
1	6	8 25	10 125	150
2	7	11	11 175	175
	4 200	5 75	12	275
	200	100	300	600

Table B-11

The basic solution principle in a transportation problem is to determine whether a transportation route not at present being used (i.e., an empty cell) would result in a lower total cost if it were used. For example, Table B-11 shows four empty cells (1A, 2A, 2B, 3C) representing unused routes. Our first step in the stepping-stone method is to evaluate these

empty cells to see whether the use of any of them would reduce total cost. If we find such a route, then we will allocate as much as possible to it.

First, let us consider allocating one ton of wheat to cell 1A. If one ton is allocated to cell 1A, cost will be increased by $6—the transportation cost for cell 1A. However, by allocating one ton to cell 1A, we increase the supply in row 1 to 151 tons, as shown in Table B-12.

From \ To	A	B	C	Supply	
1	6	8 25	10 125	150	151
2	7	11	11 175	175	
	4 200	5 75	12	275	
	200	100	300	600	

Table B-12

The constraints of the problem cannot be violated, and feasibility must be maintained. If we add one ton to cell 1A, we must subtract one ton from another allocation along that row. Cell 1B is a logical candidate because it contains 25 tons. By subtracting one ton from cell 1B, we now have 150 tons in row 1, and we have satisfied the supply constraint again. At the same time, subtracting one ton from cell 1B has reduced total cost by $8.

However, by subtracting one ton from cell 1B, we now have only 99 tons allocated to column B, where 100 tons are demanded, as shown in Table B-13. To compensate for this constraint violation, one ton must be added *to a cell that already has an allocation*. Since cell 3B has 75 tons, we will add one ton to this cell, which again satisfies the demand constraint of 100 tons.

From \ To	A	B	C	Supply
1	+1 6	−1 8 25	10 125	150
2	7	11	11 175	175
3	4 200	5 75	12	275
Demand	200	100	300	600
		99		

Table B-13

A requirement of this solution method is that units can only be added to and subtracted from cells that already have allocations. That is why one ton was added to cell 3B and not to cell 2B. It is from this requirement that the method derives its name. The process of adding and subtracting units from allocated cells is analogous to crossing a pond by stepping on stones (i.e., only allocated-to cells).

By allocating one extra ton to cell 3B we have increased cost by $5, the transportation cost for that cell. However, we have also increased the supply in row 3 to 276 tons, a violation of the supply constraint for this source. As before, this violation can be remedied by subtracting one ton from cell 3A, which contains an allocation of 200 tons. This satisfies the supply constraint again for row 3, and it also reduces the total cost by $4, the transportation cost for cell 3A. These allocations and deletions are shown in Table B-14.

From \ To	A	B	C	Supply
1	+1 6	−1 8	10	
		25	125	150
2	7	11	11	
			175	175
	4	+1 5	12	
	00	75		275
	00	100	300	600

Table B-14

Notice in Table B-14 that by subtracting one ton from cell 3A, we did not violate the demand constraint for column A, since we previously added one ton to cell 1A.

Now let us review the increases and reductions in costs resulting from this process. We initially increased cost by $6 at cell 1A, then reduced cost by $8 at cell 1B, then increased cost by $5 at cell 3B, and, finally, reduced cost by $4 at cell 3A.

$$\begin{array}{l} 1A \rightarrow 1B \rightarrow 3B \rightarrow 3A \\ +\$6 \quad 8 + 5 \quad 4 = -\$1 \end{array}$$

In other words, for each ton allocated to cell 1A (a route not at present used), total cost will be reduced by $1. This indicates that the initial solution is not optimal because a lower cost can be achieved by allocating additional tons of wheat to cell 1A (i.e., cell 1A is analogous to a pivot column in the simplex method). Our goal is to determine the cell or entering "variable" that will reduce cost the most. Another variable (empty cell) may result in an even greater decrease in cost than cell 1A. If such a cell exists, it will be selected as the entering variable; if not, cell 1A will be selected. To identify the appropriate entering variable, the remaining empty cells must be tested as cell 1A was.

Before testing the remaining empty cells, let us identify a few of the general characteristics of the stepping-stone process. First, we always start with an empty cell and form a *closed path* of cells that now have allocations. In developing the path, it is possible to skip over both unused and used cells. In any row or column there can be only *one* addition and *one* subtraction. (For example, in row 1, wheat is added at cell 1A and is subtracted at cell 1B.)

Let us test cell 2A to see if it results in a cost reduction. The stepping-stone closed path for cell 2A is shown in Table B-15. Notice that the path for cell 2A is slightly more complex than the path for cell 1A. Notice also that the path crosses itself at one point, which is perfectly acceptable. An allocation to cell 2A will reduce cost by $1, as shown in the computation in Table B-15. Thus, we have located another possible entering variable, although it is no better than cell 1A.

The remaining stepping-stone paths and the resulting computations for cells 2B and 3C are shown in Tables B-16 and B-17, respectively.

From \ To	A	B	C	Supply
1	6	– 8 25	+ 10 125	150
2	+ 7	11	– 11 175	175
3	– 4 200	+ 5 75	12	275
Demand	200	100	300	600

2A → 2C → 1C → 1B → 3B → 3A
+$7 11 + 10 8 + 5 4 = −$1

Table B-15

From \ To	A	B	C	Supply
1	6	+ 8 25	– 10 125	150
2	7	– 11	+ 11 175	175
3	4 200	5 75	12	275
Demand	200	100	300	600

2B → 2C → 1C → 1B
+$11 11 + 10 8 = +$2

Table B-16

From \ To	A	B	C	Supply
1	6	+ 8 25	– 10 125	150
2	7	11	11 175	175
3	4 200	– 5 75	+ 12	275
Demand	200	100	300	600

3C → 1C → 1B → 3B
+$12 10 + 8 5 = +$5

Table B-17

Notice that after all four unused routes are evaluated, there is a tie for the entering variable between cells 1A and 2A. Both show a reduction in cost of $1 per ton allocated to that route. The tie can be broken arbitrarily. We will select cell 1A (i.e., x_{1A}) to enter the solution.

Because the total cost of the model will be reduced by $1 for each ton we can reallocate to cell 1A, we naturally want to reallocate as much as possible. To determine how much to allocate, we need to look at the path for cell 1A again, as shown in Table B-18.

From \ To	A	B	C	Supply
1	+ 6	− 8	10	
		25	125	150
2	7	11	11	
			175	175
3	− 4	+ 5	12	
	200	75		275
Demand	200	100	300	600

Table B-18

The stepping-stone path in Table B-18 shows that tons of wheat must be subtracted at cells 1B and 3A to meet the rim requirements and thus satisfy the model constraints. Because we cannot subtract more than is available in a cell, we are limited by the 25 tons in cell 1B. In other words, if we allocate more than 25 tons to cell 1A, then we must subtract more than 25 tons from 1B, which is impossible because only 25 tons are available. Therefore, 25 tons is the amount we reallocate to cell 1A according to our path. That is, 25 tons are added to 1A, subtracted from 1B, added to 3B, and subtracted from 3A. This reallocation is shown in Table B-19.

From \ To	A	B	C	Supply
1	6	8	10	
	25		125	150
2	7	11	11	
			175	175
	4	5	12	
		100		275
		100	300	600

Table B-19

The process culminating in Table B-19 represents one *iteration* of the stepping-stone method. We selected x_{1A} as the entering variable, and it turned out that x_{1B} was the leaving variable (because it now has a value of zero in Table B-19). Thus, at each iteration one variable enters and one leaves (just as in the simplex method).

Now we must check to see whether the solution shown in Table B-19 is, in fact, optimal. We do this by plotting the paths for the unused routes (i.e., empty cells 2A, 1B, 2B, and 3C) that are shown in Table B-19. These paths are shown in Tables B-20 through B-23.

From \ To	A	B	C	Supply
1	− 6 25	8	+ 10 125	150
2	+ 7	11	− 11 175	175
3	4 175	5 100	12	275
Demand	200	100	300	600

3A → 2C → 1C → 1A
+ $7 11 + 10 6 = $0

Table B-20

From \ To	A	B	C	Supply
1	− 6 25	+ 8	10 125	150
2	7	11	11 175	175
3	+ 4 175	− 5 100	12	275
Demand	200	100	300	600

B → 3B → 3A → 1A
- $8 5 + 4 6 = +$1

Table B-21

From \ To	A	B	C	Supply
1	− 6 25	8	+ 10 125	150
2	7	+ 11	− 11 175	175
3	+ 4 175	− 5 100	12	275
Demand	200	100	300	600

B → 3B → 3A → 1A → 1C → 2C
-$11 5 + 4 6 + 10 11 = +$3

Table B-22

From \ To	A	B	C	Supply
1	+ 6 25	8	– 10 125	150
2	7	11	11 175	175
3	– 4 175	5 100	+ 12	275
Demand	200	100	300	600

3C → 3A → 1A → 1C
+$12 4 + 6 − 10 = +$4

Table B-23

Our evaluation of the four paths indicates no cost reductions; therefore, the solution shown in Table B-19 is optimal. The solution and total minimum cost are

$x_{1A} = 25$ tons $x_{2C} = 175$ tons $x_{3A} = 175$ tons
$x_{1C} = 125$ tons $x_{3B} = 100$ tons

$$Z = \$6(25) + 8(0) + 10(125) + 7(0) + 11(0) + 11(175) + 4(175) + 5(100) + 12(0)$$
$$= \$4{,}525$$

However, notice in Table B-20 that the path for cell 2A resulted in a cost change of $0. In other words, allocating to this cell would neither increase nor decrease total cost. This situation indicates that the problem has *multiple optimal solutions* of the text. Thus, x_{2A} could be entered into the solution and there would not be a change in the total minimum cost of $4,525. To identify the alternative solution, we would allocate as much as possible to cell 2A, which in this case is 25 tons of wheat. The alternative solution is shown in Table B-24.

From \ To	A	B	C	Supply
1	6	8	10 150	150
	7 25	11	11 150	175
	4 75	5 100	12	275
	00	100	300	600

Table B-24

The solution in Table B-24 also results in a total minimum cost of $4,525. The steps of the stepping-stone method are summarized here.

1. Determine the stepping-stone paths and cost changes for each empty cell in the tableau.
2. Allocate as much as possible to the empty cell with the greatest net decrease in cost.
3. Repeat steps 1 and 2 until all empty cells have positive cost changes that indicate an optimal solution.

The Modified Distribution Method

The *modified distribution method (MODI)* is basically a modified version of the stepping-stone method. However, in the *MODI* method the individual cell cost changes are determined mathematically, without identifying all of the stepping-stone paths for the empty cells.

To demonstrate *MODI,* we will again use the initial solution obtained by the minimum cell cost method. The tableau for the initial solution with the modifications required by *MODI* is shown in Table B-25.

	v_j	$v_A =$	$v_B =$	$v_C =$	
u_i	To / From	A	B	C	Supply
$u_1 =$	1	6	8 25	10 125	150
$u_2 =$	2	7	11	11 175	175
	3	4 200	5 75	12	275
	Demand	200	100	300	600

Table B-25

The extra left-hand column with the u_i symbols and the extra top row with the v_j symbols represent column and row values that must be computed in *MODI.* These values are computed for all *cells with allocations* by using the following formula.

$$u_i + v_j = c_{ij}$$

The value c_{ij} is the unit transportation cost for cell *ij.* For example, the formula for cell 1B is

$$u_1 + v_B = c_{1B}$$

and, since $c_{1B} = 8$,

$$u_1 + v_B = 8$$

The formulas for the remaining cells that presently contain allocations are

$$\begin{aligned} x_{1C}&: \quad u_1 + v_C = 10 \\ x_{2C}&: \quad u_2 + v_C = 11 \\ x_{3A}&: \quad u_3 + v_A = 4 \\ x_{3B}&: \quad u_3 + v_B = 5 \end{aligned}$$

Now there are five equations with six unknowns. To solve these equations, it is necessary to assign only one of the unknowns a value of zero. Thus, if we let $u_1 = 0$, we can solve for all remaining u_i and v_j values.

$$\begin{aligned} x_{1B}&: \quad u_1 + v_B = 8 \\ &\qquad 0 + v_B = 8 \\ &\qquad\quad v_B = 8 \\ x_{1C}&: \quad u_1 + v_C = 10 \\ &\qquad 0 + v_C = 10 \\ &\qquad\quad v_C = 10 \\ x_{2C}&: \quad u_2 + v_C = 11 \\ &\qquad u_2 + 10 = 11 \\ &\qquad\quad u_2 = 1 \\ x_{3B}&: \quad u_3 + v_B = 5 \\ &\qquad u_3 + 8 = 5 \\ &\qquad\quad u_3 = -3 \\ x_{3A}&: \quad u_3 + v_A = 4 \\ &\qquad 3 + v_A = 4 \\ &\qquad\quad v_A = 7 \end{aligned}$$

Notice that the equation for cell 3B had to be solved before the cell 3A equation could be solved. Now all the u_i and v_j values can be substituted into the tableau, as shown in Table B-26.

	v_j	$v_A = 7$	$v_B = 8$	$v_C = 10$	
u_i	To / From	A	B	C	Supply
$u_1 = 0$	1	6	8 25	10 125	150
$u_2 = 1$	2	7	11	11 175	175
	3	4 200	5 75	12	275
	Demand	200	100	300	600

Table B-26

Next, we use the following formula to evaluate all *empty cells:*

$$c_{ij} \quad u_i \quad v_j = k_{ij}$$

where k_{ij} equals the cost increase or decrease that would occur by allocating to a cell.

For the *empty cells* in Table B-26, the formula yields the following values:

$$\begin{array}{rlllll}
x_{1A}: & k_{1A} = c_{1A} & u_1 & v_A = 6 & 0 & 7 = -1 \\
x_{2A}: & k_{2A} = c_{2A} & u_2 & v_A = 7 & 1 & 7 = -1 \\
x_{2B}: & k_{2B} = c_{2B} & u_2 & v_B = 11 & 1 & 8 = +2 \\
x_{3C}: & k_{3C} = c_{3C} & u_3 & v_C = 12 & (\ 3) & 10 = +5
\end{array}$$

These calculations indicate that either cell 1A or cell 2A will decrease cost by $1 per allocated ton. Notice that those are exactly the same cost changes for all four empty cells as were computed in the stepping-stone method. That is, the same information is obtained by evaluating the paths in the stepping-stone method and by using the mathematical formulas of the MODI.

We can select either cell 1A or 2A to allocate to because they are tied at 1. If cell 1A is selected as the entering nonbasic variable, then the stepping-stone path for that cell must be determined so that we know how much to reallocate. This is the same path previously identified in Table B-18. Reallocating along this path results in the tableau shown in Table B-27 (and previously shown in Table B-19).

The u_i and v_j values for Table B-27 must now be recomputed using our formula for the allocated-to cells.

$$\begin{array}{rrl}
x_{1A}: & u_1 + v_A = 6 \\
& 0 + v_A = 6 \\
& v_A = 6 \\
x_{1C}: & u_1 + v_C = 10 \\
& 0 + v_C = 10 \\
& v_C = 10 \\
x_{2C}: & u_2 + v_C = 11 \\
& u_2 + 10 = 11 \\
& u_2 = 1
\end{array}$$

	v_j	$v_A =$	$v_B =$	$v_C =$	
u_i	To / From	A	B	C	Supply
$u_1 =$	1	6: 25	8	10: 125	150
	2	7	11	11: 175	175
	3	4: 175	5: 100	12	275
	Demand	200	100	300	600

Table B-27

$$
\begin{aligned}
x_{3A}: \quad u_3 + v_A &= 4 \\
u_3 + 6 &= 4 \\
u_3 &= -2 \\
x_{3B}: \quad u_3 + v_B &= 5 \\
2 + v_B &= 5 \\
v_B &= 7
\end{aligned}
$$

These new u_i and v_j values are shown in Table B-28.

	v_j	$v_A = 6$	$v_B = 7$	$v_C = 10$	
u_i	To / From	A	B	C	Supply
$u_1 = 0$	1	6: 25	8	10: 125	150
	2	7	11	11: 175	175
	3	4: 175	5: 100	12	275
	Demand	200	100	300	600

Table B-28

The cost changes for the empty cells are now computed using the formula $c_{ij} \quad u_i \quad v_j = k_{ij}$.

$$
\begin{aligned}
x_{1A}: \quad k_{1B} = c_{1B} \quad u_1 \quad v_B &= 8 \quad 0 \quad 7 = +1 \\
x_{2A}: \quad k_{2A} = c_{2A} \quad u_2 \quad v_A &= 7 \quad 1 \quad 6 = 0 \\
x_{2B}: \quad k_{2B} = c_{2B} \quad u_2 \quad v_B &= 11 \quad 1 \quad 7 = +3 \\
x_{3C}: \quad k_{3C} = c_{3C} \quad u_3 \quad v_C &= 12 \quad (\ 2) \quad 10 = +4
\end{aligned}
$$

Because none of these values is negative, the solution shown in Table B-28 is optimal. However, as in the stepping-stone method, cell 2A with a zero cost change indicates a multiple optimal solution.

The steps of the modified distribution method can be summarized as follows.

1. Develop an initial solution using one of the three methods available.
2. Compute u_i and v_j values for each row and column by applying the formula $u_i + v_j = c_{ij}$ to each cell that has an allocation.
3. Compute the cost change, k_{ij}, for each empty cell using $c_{ij} \quad u_i \quad v_j = k_{ij}$.
4. Allocate as much as possible to the empty cell that will result in the greatest net decrease in cost (most negative k_{ij}). Allocate according to the stepping-stone path for the selected cell.
5. Repeat steps 2 through 4 until all k_{ij} values are positive or zero.

The Unbalanced Transportation Model

Thus far, the methods for determining an initial solution and an optimal solution have been demonstrated within the context of a balanced transportation model. Realistically, however, an *unbalanced problem* is a more likely occurrence. Consider our example of transporting wheat. By changing the demand at Cincinnati to 350 tons, we create a situation in which total demand is 650 tons and total supply is 600 tons.

To compensate for this difference in the transportation tableau, a "dummy" row is added to the tableau, as shown in Table B-29. The dummy row is assigned a supply of 50 tons to balance the model. The additional 50 tons demanded, which cannot be supplied, will be allocated to a cell in the dummy row. The transportation costs for the cells in the dummy row are zero because the tons allocated to these cells are not amounts really transported but the amounts by which demand was not met. These dummy cells are, in effect, *slack variables.*

From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
	4	5	12	275
	0	0	0	50
Demand	200	100	350	650

Table B-29

Now consider our example with the supply at Des Moines increased to 375 tons. This increases total supply to 700 tons, while total demand remains at 600 tons. To compensate for this imbalance, we add a dummy column instead of a dummy row, as shown in Table B-30.

The addition of a dummy row or a dummy column has no effect on the initial solution methods or on the methods for determining an optimal solution. The dummy row or column cells are treated the same as any other tableau cell. For example, in the minimum cell cost method, three cells would be tied for the minimum cost cell, each with a cost of zero. In this case (or any time there is a tie between cells) the tie would be broken arbitrarily.

From \ To	A	B	C	Dummy	Supply
1	6	8	10	0	150
	7	11	11	0	175
	4	5	12	0	375
	200	100	300	100	700

Table B-30

Degeneracy

In all the tableaus showing a solution to the wheat transportation problem, the following condition was met.

$$m \text{ rows} + n \text{ columns} \quad 1 = \text{the number of cells with allocations}$$

For example, in any of the balanced tableaus for wheat transportation, the number of rows was three (i.e., $m = 3$) and the number of columns was three (i.e., $n = 3$); thus, $3 + 3 \quad 1 = 5$ cells with allocations.

These tableaus always had five cells with allocations; thus, our condition for normal solution was met. When this condition is not met and fewer than $m + n \quad 1$ cells have allocations, the tableau is said to be *degenerate.*

Consider the wheat transportation example with the supply values changed to the amounts shown in Table B-31. The initial solution shown in this tableau was developed using the minimum cell cost method.

From \ To	A	B	C	Supply
1	6	8 100	10 50	150
	7	11	11 250	250
	4 200	5	12	200
	200	100	300	600

Table B-31

The tableau shown in Table B-31 does not meet the condition

$$m + n \quad 1 = \text{the number of cells with allocations}$$
$$3 + 3 \quad 1 = 5 \text{ cells}$$

because there are only four cells with allocations. The difficulty resulting from a degenerate solution is that neither the stepping-stone method nor MODI will work unless the preceding condition is met (there is an appropriate number of cells with allocations). When the tableau is degenerate, a closed path cannot be completed for all cells in the stepping-stone

method, *and* not all the $u_i + v_j = c_{ij}$ computations can be completed in MODI. For example, a closed path cannot be determined for cell 1A in Table B-31.

To create a closed path, one of the empty cells must be artificially designated as a cell with an allocation. Cell 1A in Table B-32 is designated arbitrarily as a cell with artificial allocation of zero. (However, any symbol, such as ∅, could be used to signify the artificial allocation.) This indicates that this cell will be treated as a cell with an allocation in determining stepping-stone paths or MODI formulas, although there is no real allocation in this cell. Notice that the location of 0 was *arbitrary* because there is no general rule for allocating the artificial cell. Allocating zero to a cell does not guarantee that all of the stepping-stone paths can be determined.

From \ To	A	B	C	Supply
1	6 **0**	8 **100**	10 **50**	150
	7	11	11 **250**	250
	4 **200**	5	12	200
	200	100	300	600

Table B-32

For example, if zero had been allocated to cell 2B instead of to cell 1A, none of the stepping-stone paths could have been determined, even though technically the tableau would no longer be degenerate. In such a case, the zero must be reallocated to another cell and all paths determined again. This process must be repeated until an artificial allocation has been made that will enable the determination of all paths. In most cases, however, there is more than one possible cell to which such an allocation can be made.

The stepping-stone paths and cost changes for this tableau follow.

$$
\begin{array}{lll}
 & \text{2A} \quad \text{2C} \quad \text{1C} \quad \text{1A} & \\
x_{2A}: & 7 \quad 11 + 10 \quad 6 = 0 & \\
 & \text{2B} \quad \text{2C} \quad \text{1C} \quad \text{1B} & \\
x_{2B}: & 11 \quad 11 + 10 \quad 8 = +2 & \\
 & \text{3B} \quad \text{1B} \quad \text{1A} \quad \text{3A} & \\
x_{3B}: & 5 \quad 8 + 6 \quad 4 = -1 & \\
 & \text{3C} \quad \text{1C} \quad \text{1A} \quad \text{3A} & \\
x_{3C}: & 12 \quad 10 + 6 \quad 4 = +4 &
\end{array}
$$

Because cell 3B shows a $1 decrease in cost for every ton of wheat allocated to it, we will allocate 100 tons to cell 3B. This results in the tableau shown in Table B-33.

Notice that the solution in Table B-33 now meets the condition $m + n \quad 1 = 5$. Thus, in applying the stepping-stone method (or MODI) to this tableau, it is not necessary to make an artificial allocation to an empty cell. It is quite possible to begin the solution process with a normal tableau and have it become degenerate or begin with a degenerate tableau and have it become normal. If it had been indicated that the cell with the zero should have units subtracted from it, no actual units could have been subtracted. In that case the zero would have been moved to the cell that represents the entering variable. (The solution shown in Table B-33 is optimal; however, a multiple optimal solution exists at cell 2A.)

From \ To	A	B	C	Supply
1	6 100	8	10 50	150
2	7	11	11 250	250
3	4 100	5 100	12	200
Demand	200	100	300	600

Table B-33

Prohibited Routes

Sometimes one or more of the routes in the transportation model are *prohibited*. That is, units cannot be transported from a particular source to a particular destination. When this situation occurs, we must make sure that no units in the optimal solution are allocated to the cell representing this route. In our study of the simplex tableau, we learned that assigning a large relative cost or a coefficient of M to a variable would keep it out of the final solution. This same principle can be used in a transportation model for a prohibited route. A value of M is assigned as the transportation cost for a cell that represents a prohibited route. Thus, when the prohibited cell is evaluated, it will always contain a large positive cost change of M, which will keep it from being selected as an entering variable.

Solution of the Assignment Model

The assignment model is a special form of a linear programming model that is similar to the transportation model. There are differences, however. In the assignment model, the supply at each source and the demand at each destination are each limited to one unit.

The following example from the text will be used to demonstrate the assignment model and its special solution method. The Atlantic Coast Conference has four basketball games on a particular night. The conference office wants to assign four teams of officials to the four games in a way that will minimize the total distance traveled by the officials. The distances in miles for each team of officials to each game location are shown in Table B-34.

	Game Sites			
Officials	Raleigh	Atlanta	Durham	Clemson
A	210	90	180	160
B	100	70	130	200
C	175	105	140	170
D	80	65	105	120

Table B-34

The supply is always one team of officials, and the demand is for only one team of officials at each game. Table B-34 is already in the proper form for the assignment.

The first step in the assignment method of solution is to develop an *opportunity cost table*. We accomplish this by first subtracting the minimum value in each row from every value in the row. These computations are referred to as *row reductions*. We applied a similar principle in the VAM method when we determined penalty costs. In other words, the

best course of action is determined for each row, and the penalty or "lost opportunity" is developed for all other row values. The row reductions for this example are shown in Table B-35.

	Game Sites			
Officials	Raleigh	Atlanta	Durham	Clemson
A	120	0	90	70
B	30	0	60	130
C	70	0	35	65
D	15	0	40	55

Table B-35

Next, the minimum value in each column is subtracted from all column values. These computations are called *column reductions* and are shown in Table B-36. It represents the completed opportunity cost table for our example. Assignments can be made in this table wherever a zero is present. For example, team A can be assigned to Atlanta. An *optimal solution* results when each of the four teams can be uniquely assigned to a different game.

	Game Sites			
Officials	Raleigh	Atlanta	Durham	Clemson
A	105	0	55	15
B	15	0	25	75
C	55	0	0	10
D	0	0	5	0

Table B-36

Notice in Table B-36 that the assignment of team A to Atlanta means that no other team can be assigned to that game. Once this assignment is made, the zero in row B is infeasible, which indicates that there is not a unique optimal assignment for team B. Therefore, Table B-36 does not contain an optimal solution.

A test to determine if four unique assignments exist in Table B-36 is to draw the minimum number of horizontal or vertical lines necessary to cross out all zeros through the rows and columns of the table. For example, Table B-37 shows that three lines are required to cross out all zeros.

	Game Sites			
Officials	Raleigh	Atlanta	Durham	Clemson
A	105	0	55	15
B	15	0	25	75
C	35	0	0	10
D	0	0	5	0

Table B-37

The three lines indicate that there are only three unique assignments, whereas four are required for an optimal solution. (Note that even if the three lines could have been drawn differently, the subsequent solution method would not be affected.) Next, subtract the minimum value that is not crossed out from all other values not crossed out. Then add this minimum value to those cells where two lines intersect. The minimum value not crossed out in Table B-37 is 15. The second iteration for this model with the appropriate changes is shown in Table B-38.

	Game Sites			
Officials	Raleigh	Atlanta	Durham	Clemson
A	90	0	40	0
B	0	0	10	60
C	55	15	0	10
D	0	15	5	0

Table B-38

No matter how the lines are drawn in Table B-38, at least four are required to cross out all the zeros. This indicates that four unique assignments can be made and that an optimal solution has been reached. Now let us make the assignments from Table B-38.

First, team A can be assigned to either the Atlanta game or the Clemson game. We will assign team A to Atlanta first. This means that team A cannot be assigned to any other game, and no other team can be assigned to Atlanta. Therefore, row A and the Atlanta column can be eliminated. Next, team B is assigned to Raleigh. (Team B cannot be assigned to Atlanta, which has already been eliminated.) The third assignment is of team C to the Durham game. This leaves team D for the Clemson game. These assignments and their respective distances (from Table B-34) are summarized as follows.

Assignment	***Distance***
Team A → Atlanta	90
Team B → Raleigh	100
Team C → Durham	140
Team D → Clemson	120
	450 miles

Now let us go back and make the initial assignment of team A to Clemson (the alternative assignment we did not initially make). This will result in the following set of assignments.

Assignment	***Distance***
Team A → Clemson	160
Team B → Atlanta	70
Team C → Durham	140
Team D → Raleigh	80
	450 miles

These two assignments represent *multiple optimal solutions* for our example problem. Both assignments will result in the officials traveling a minimum total distance of 450 miles.

Like a transportation problem, an assignment model can be unbalanced when supply exceeds demand or demand exceeds supply. For example, assume that, instead of four teams of officials, there are five teams to be assigned to the four games. In this case a *dummy column* is added to the assignment tableau to balance the model, as shown in Table B-39.

	Game Sites				
Officials	Raleigh	Atlanta	Durham	Clemson	Dummy
A	210	90	180	160	0
B	100	70	130	200	0
C	175	105	140	170	0
D	80	65	105	120	0
E	95	115	120	100	0

Table B-39

In solving this model, one team of officials would be assigned to the dummy column. If there were five games and only four teams of officials, a *dummy row* would be added instead of a dummy column. The addition of a dummy row or column does not affect the solution method.

Prohibited assignments are also possible in an assignment problem, just as prohibited routes can occur in a transportation model. In the transportation model, an M value was assigned as a large cost for the cell representing the prohibited route. This same method is used for a prohibited assignment. A value of M is placed in the cell that represents the prohibited assignment.

The steps of the assignment solution method are summarized here.

1. Perform row reductions by subtracting the minimum value in each row from all row values.
2. Perform column reductions by subtracting the minimum value in each column from all column values.
3. In the completed opportunity cost table, cross out all zeros, using the minimum number of horizontal or vertical lines.
4. If fewer than m lines are required (where m = the number of rows or columns), subtract the minimum uncrossed value from all other uncrossed values, and add this same minimum value to all cells where two lines intersect. Leave all other values unchanged, and repeat step 3.
5. If m lines are required, the tableau contains the optimal solution and m unique assignments can be made. If fewer than m lines are required, repeat step 4.

Problems

1. Solve T#1 using MODI

2. A transportation problem involves the following costs, supply, and demand.

	To				
From	*1*	*2*	*3*	*4*	**Supply**
1	$500	$750	$300	$450	12
2	650	800	400	600	17
3	400	700	500	550	11
Demand	10	10	10	10	

   a. Find the initial solution using the northwest corner method, the minimum cell cost method, and Vogel's approximation model. Compute total cost for each.
   b. Using the VAM initial solution, find the optimal solution using the modified distribution method (MODI).

3. Consider the following transportation tableau and solution.

From \ To	A	B	C	Supply
1	12	10	6 600	600
2	4 400	15	3	400
	9 300	7	*M*	300
	11	8 500	6 300	800
Dummy	0 200	0	0	200
Demand	900	500	900	2,300

   a. Is this a balanced or an unbalanced transportation problem? Explain.
   b. Is this solution degenerate? Explain. If it is degenerate, show how it would be put into proper form.
   c. Is there a prohibited route in this problem?
   d. Compute the total cost of this solution.
   e. What is the value of x_{2B} in this solution?

4. Solve the following transportation problem.

	To			
From	*1*	*2*	*3*	**Supply**
1	$ 40	$ 10	$ 20	800
2	15	20	10	500
3	20	25	30	600
Demand	1,050	500	650	

5. Given a transportation problem with the following costs, supply, and demand, find the initial solution using the minimum cell cost method and Vogel's approximation model. Is the VAM solution optimal?

	To			
From	*1*	*2*	*3*	**Supply**
A	$ 6	$ 7	$ 4	100
B	5	3	6	180
C	8	5	7	200
Demand	135	175	170	

6. Consider the following transportation problem.

	To			
From	*1*	*2*	*3*	**Supply**
A	$ 6	9	*M*	130
B	12	3	5	70
C	4	8	11	100
Demand	80	110	60	

   a. Find the initial solution by using VAM and then solve it using the stepping-stone method.
   b. Formulate this problem as a general linear programming model.

7. Solve the following linear programming problem.

Student Workbook

Chapter 1
Management Science

TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

1) Management science involves the philosophy of approaching a problem in a subjective manner.

2) Management science techniques can be applied only to business and military organizations.

3) Once management scientist makes his or her decision and recommendation to management, then typically, his or her involvement with the problem is finished.

4) A variable is a value that is usually a coefficient of a parameter in an equation.

5) Parameters are known, constant values that are usually coefficients of variables in equations.

6) Data are pieces of information from the problem environment.

7) A model is a mathematical representation of a problem situation including variables, parameters, and equations.

8) A management science technique usually applies to a specific model type.

9) The first step of the management science process is to define the problem.

10) Management science modeling techniques provide results that are known with certainty.

11) The term sensitivity analysis refers to testing how a problem solution reacts to changes in one or more of the model parameters.

12) Fixed costs depend on the number of items produced.

13) Variable costs depend on the number of items produced.

14) Fixed cost is the difference between total cost and total variable cost.

15) The break-even point is the volume that equates total revenue with total cost.

16) In general, an increase in price increases the break even point if all costs are held constant.

17) If variable costs increase, but price and fixed costs are held constant, the break even point will decrease.

18) Managers utilize spreadsheets to conduct their own analyses in management science studies.

19) Management science techniques focus primarily on observation, model construction and implementation to find an appropriate solution to a problem.

Chapter 2
Linear Programming: Model Formulation and Graphical Solution

TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

1) In the term linear programming, the word programming comes from the phrase "computer programming."

2) One of the assumptions of LP is "simultaneity."

3) Any linear programming problem can be solved using the graphical solution procedure.

4) An LP formulation typically requires finding the maximum value of an objective while simultaneously maximizing usage of the resource constraints.

5) There are no limitations on the number of constraints or variables that can be graphed to solve an LP problem.

6) Resource restrictions are called constraints.

7) One of the assumptions of LP is "proportionality."

8) The set of solution points that satisfies all of a linear programming problem's constraints simultaneously is defined as the feasible region in graphical linear programming.

9) An objective function is necessary in a maximization problem but is not required in a minimization problem.

10) In some instances, an infeasible solution may be the optimum found by the corner point method.

11) The rationality assumption implies that solutions need not be in whole numbers (integers).

12) The solution to a linear programming problem must always lie on a constraint.

13) In a linear program, the constraints must be linear, but the objective function may be nonlinear.

14) Resource mix problems use LP to decide how much of each product to make, given a series of resource restrictions.

15) The existence of non-negativity constraints in a two-variable linear program implies that we are always working in the northwest quadrant of a graph.

16) In linear programming terminology, "dual price" and "sensitivity price" are synonyms.

17) Any time that we have an isoprofit line that is parallel to a constraint, we have the possibility of multiple solutions.

18) If the isoprofit line is not parallel to a constraint, then the solution must be unique.

19) When two or more constraints conflict with one another, we have a condition called unboundedness.

20) The addition of a redundant constraint lowers the isoprofit line.

21) Sensitivity analysis enables us to look at the effects of changing the coefficients in the objective function, one at a time.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

22) A widely used mathematical programming technique designed to help managers and decision making relative to resource allocation is called _____.
A) goal programming
B) linear programming
C) computer programming
D) constraint programming
E) None of the above

23) Typical resources of an organization include _____.
A) raw material usage
B) machinery usage
C) warehouse space utilization
D) labor volume
E) All of the above

24) Which of the following is <u>not</u> a property of all linear programming problems?
A) the presence of restrictions
B) a computer program
C) optimization of some objective
D) usage of only linear equations and inequalities
E) alternate courses of action to choose from

25) A feasible solution to a linear programming problem
A) need not satisfy all of the constraints, only the non-negativity constraints.
B) must be a corner point of the feasible region.
C) must give the maximum possible profit.
D) must satisfy all of the problem's constraints simultaneously.
E) must give the minimum possible cost.

26) Infeasibility in a linear programming problem occurs when
A) there is an infinite solution.
B) more than one solution is optimal.
C) the feasible region is unbounded.
D) a constraint is redundant.
E) there is no solution that satisfies all the constraints given.

27) In a maximization problem, when one or more of the solution variables and the profit can be made infinitely large without violating any constraints, the linear program has
A) an infeasible solution.
B) alternate optimal solutions.
C) an unbounded solution.
D) a redundant constraint.
E) None of the above

28) Which of the following is not a part of every linear programming problem formulation?
A) a set of constraints
B) a redundant constraint
C) maximization or minimization of a linear function
D) an objective function
E) non-negativity constraints

29) When appropriate, the optimal solution to a maximization linear programming problem can be found by graphing the feasible region and
A) finding the profit at every corner point of the feasible region to see which one gives the highest value.
B) locating the point that is highest on the graph.
C) moving the isoprofit lines towards the origin in a parallel fashion until the last point in the feasible region is encountered.
D) None of the above
E) All of the above

30) The mathematical theory behind linear programming states that an optimal solution to any problem will lie at a(n) _____ of the feasible region.
A) interior point or extreme point
B) maximum point or minimum point
C) corner point or extreme point
D) interior point or center
E) None of the above

31) Which of the following is not a property of linear programs?
A) alternative courses of action
B) one or more constraints
C) one objective function
D) at least two separate feasible regions
E) objective function and constraints are linear

32) The corner point solution method
A) will yield different results from the isoprofit line solution method.
B) requires that all corners created by all constraints be compared.
C) will not provide a solution at an intersection or corner where a non-negativity constraint is involved.
D) requires that the profit from all corners of the feasible region be compared.
E) will always provide one, and only one, optimum.

33) When a constraint line bounding a feasible region has the same slope as an isoprofit line,
A) there may be more than one optimum solution.
B) the problem involves redundancy.
C) a condition of infeasibility exists.
D) an error has been made in the problem formulation.
E) None of the above

34) The simultaneous equation method is
A) useful only in minimization methods.
B) an alternative to the corner point method.
C) useful only when more than two product variables exist in a product mix problem.
D) an algebraic means for solving the intersection of two or more constraint equations.
E) None of the above

35) Consider the following linear programming problem:

Maximize $12X + 10Y$
Subject to: $4X + 3Y \leq 480$
$2X + 3Y \leq 360$
all variables ≥ 0

The maximum possible value for the objective function is
A) 1520.
B) 360.
C) 1560.
D) 480.
E) None of the above

36) Consider the following linear programming problem:

Maximize $4X + 10Y$
Subject to: $3X + 4Y \leq 480$
$4X + 2Y \leq 360$
all variables ≥ 0

The feasible corner points are (48,84), (0,120), (0,0), (90,0). What is the maximum possible value for the objective function?
A) 1032
B) 360
C) 1600
D) 1200
E) None of the above

37) Consider the following linear programming problem:

Maximize 5X + 6Y
Subject to: 4X + 2Y ≤ 420
1X + 2Y ≤ 120
all variables ≥ 0

Which of the following points (X,Y) is not a feasible corner point?
A) (120,0)
B) (105,0)
C) (100,10)
D) (0,60)
E) None of the above

38) Consider the following linear programming problem:

Maximize 5X + 6Y
Subject to: 4X + 2Y ≤ 420
1X + 2Y ≤ 120
all variables ≥ 0

Which of the following points (X,Y) is not feasible?
A) (50,40)
B) (60,30)
C) (90,10)
D) (20,50)
E) None of the above

39) Two models of a product — Regular (X) and Deluxe (Y) — are produced by a company. A linear programming model is used to determine the production schedule. The formulation is as follows:

Maximize profit = 50X + 60 Y
Subject to:
8X + 10Y ≤ 800 (labor hours)
X + Y ≤ 120 (total units demanded)
4X + 5Y ≤ 500 (raw materials)
all variables ≥ 0

The optimal solution is X = 100, Y = 0.

How many units of the regular model would be produced based on this solution?
A) 100
B) 0
C) 50
D) 120
E) None of the above

Chapter 3
Linear Programming: Computer Solution and Sensitivity Analysis

TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

1) The reduced cost (shadow price) for a positive decision variable is 0.

2) When the right-hand sides of 2 constraints are both increased by 1 unit, the value of the objective function will be adjusted by the sum of the constraints' prices.

3) When a linear programming problem is solved using a computer package decision variables will always be integer and therefore decision variable values never need to be rounded.

4) Sensitivity ranges can be computed only for the right hand sides of constraints.

5) Sensitivity analysis determines how a change in a parameter affects the optimal solution.

6) The sensitivity range for an objective function coefficient is the range of values over which the current optimal solution point (product mix) will remain optimal.

7) The sensitivity range for an objective function coefficient is the range of values over which the profit does not change.

8) The sensitivity range for a constraint quantity value is the range over which the shadow price is valid.

9) If we change the constraint quantity to a value outside the sensitivity range for that constraint quantity, the shadow price will change.

10) The sensitivity range for a constraint quantity value is the range over which the optimal values of the decision variables do not change.

11) Linear programming problems are restricted to decisions in a single time period.

12) A maximization problem may be characterized by all greater than or equal to constraints.

13) A change in the value of an objective function coefficient will always change the value of the optimal solution.

14) The terms reduced cost, shadow price, and dual price all mean the same thing.

15) Sensitivity analysis can be used to determine the effect on the solution for changing several parameters at once.

16) For a profit maximization problem, if the allowable increase for a coefficient in the objective function is infinite, then profits are unbounded.

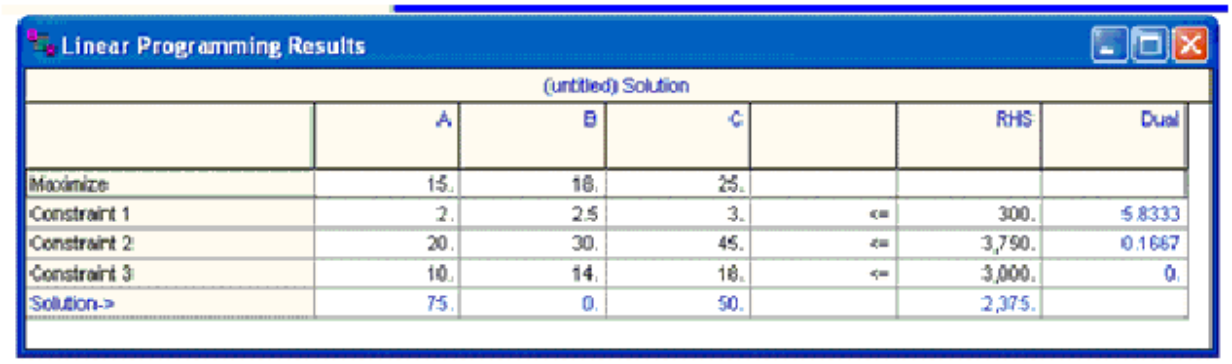
Billy's Blues sells 3 types of T-shirts: Astro, Bling, and Curious. Manufacturing Astros requires 2 minutes of machine time, 20 minutes of labor, and costs $10. Brand Bling requires 2..5 minutes of machine time, 30 minutes of labor, and costs $14 to produce. Brand Curious requires 3 minutes of machine time, 45 minutes of labor, and costs $18 to produce. There are 300 machining hours available per week, 3,750 labor hours, and he has a budget of $3,000. Brand Astro sells for $15, Brand Bling for $18, and Brand Curious for $25.

The LP formulation that maximizes week profit shown below.

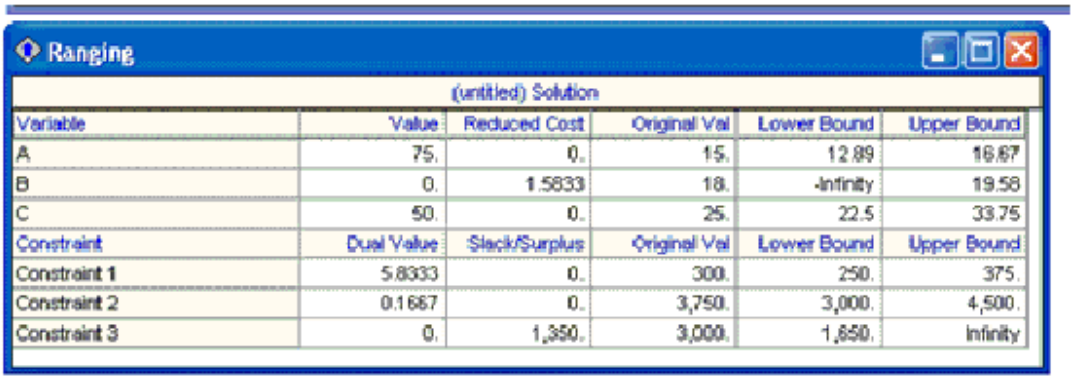
MAX 15A +18B + 25 C
s.t.

2A + 2.5B + 3C ≤ 300
20A + 30B + 45C ≤ 3,750
10A + 14B + 18C ≤ 3,000

The solution from QM for Windows is show below.



	A	B	C		RHS	Dual
Maximize	15.	18.	25.			
Constraint 1	2.	2.5	3.	<=	300.	5.8333
Constraint 2	20.	30.	45.	<=	3,750.	0.1667
Constraint 3	10.	14.	18.	<=	3,000.	0.
Solution->	75.	0.	50.		2,375.	



Variable	Value	Reduced Cost	Original Val	Lower Bound	Upper Bound
A	75.	0.	15.	12.89	16.67
B	0.	1.5833	18.	-Infinity	19.58
C	50.	0.	25.	22.5	33.75
Constraint	Dual Value	Slack/Surplus	Original Val	Lower Bound	Upper Bound
Constraint 1	5.8333	0.	300.	250.	375.
Constraint 2	0.1667	0.	3,750.	3,000.	4,500.
Constraint 3	0.	1,350.	3,000.	1,650.	Infinity

80) If Billy could acquire more of any resource, which would it be?
A) money B) buyers C) machining time D) labor time

81) If one of Billy's machines breaks down, it usually results in about 6 hours of downtime. When this happens, Billy's profits are reduced by
A) $35 B) 18 C) $25 D) $15

82) Billy's accountant made an error, and the budget has been reduced from $3000 to $2500. Billy's profit will go down by
A) $1350 B) $625 C) $1650 D) $0

83) Billy has decided that he can raise the price on the Curious t-shirt by 10% without losing sales. If he raises the price, his profits will
A) increase by $2.50 B) increase by 10% C) decrease by 10% D) increase by $125

Chapter 4
The Simplex Solution Method

TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

1) The simplex method cannot be used to solve quadratic programming problems.

2) The simplex method is a general mathematical solution technique for solving linear programming problems.

3) In the simplex method, the model is put into the form of a table, and then a number of mathematical steps are performed on the table.

4) The simplex method can be used to solve quadratic programming problems.

5) The simplex method is a general mathematical solution technique for solving nonlinear programming problems.

6) The simplex method moves from one better solution to another until the best one is found, and then it stops.

7) The mathematical steps in the simplex method replicate the process in graphical analysis of moving from one extreme point on the solution boundary to another.

8) The first step in solving a linear programming model manually with the simplex method is to convert the model into standard form.

9) The last step in solving a linear programming model manually with the simplex method is to convert the model into standard form.

10) Slack variables are added to constraints and represent unused resources.

11) Slack variables are added to constraints and represent unused resources.

12) A basic feasible solution satisfies the model constraints and has the same number of variables with non-negative values as there are constraints.

13) A basic feasible solution satisfies the model constraints and has the same number of variables with negative values as there are constraints.

14) Row operations are used to solve simultaneous equations where equations are multiplied by constants and added or subtracted from each other.

15) The basic feasible solution in the initial simplex tableau is the origin where all decision variables equal zero.

16) At the initial basic feasible solution at the origin, only slack variables have a value greater than zero.

17) At the initial basic feasible solution at the origin, only slack variables have a value greater than one.

18) In using the simplex method, the number of basic variables is equal to the number of constraints.

19) The simplex method does not guarantee an integer solution.

20) In solving a linear programming problem with simplex method, the number of basic variables is the same as the number of constraints n the original problem.

21) A change in the objective function coefficient of a basic variable cannot change the value of z_j for a non-basic variable in the final simplex tableau.

22) When solving a linear programming problem, a decision variable that leaves the basis in one iteration of the simplex method can return to the basis on a later iteration.

23) Final tableaus cannot be used to conduct sensitivity analysis.

24) The dual form of a linear program is used to determine how much one should pay for additional resources.

25) Multiple optimal solutions cannot be determined from the simplex method.

26) The theoretical limit on the number of decision variables that can be handled by the simplex method is 50.

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

27) The _____ column is the column corresponding to the entering variable.

28) The variable with the largest positive $c_j - z_j$ is the _____ variable.

29) _____ variables are added to constraints and represent unused resources.

30) The first step in solving a linear programming model manually with the simplex method is to convert the model into _____ form.

31) The _____ values are contribution to profit for each variable.

32) The _____ values are computed by multiplying the c_j column values by the variable column values and summing.

33) The _____ variable allows for an initial basic feasible solution, but it has no meaning. Therefore, after we get the simplex tableau started, they are discarded in later iterations.

34) In solving a minimization problem, artificial variables are assigned a _____ in the objective function to eliminate them form the final solution.

35) A(n) _____ maximization linear programming problem has an artificial variable in the final simplex tableau where all $c_j - z_j$ values are less than or equal to zero.

36) In using the simplex method, _____ optimal solutions are identified by $c_j - z_j = 0$ for a non basic variable.

37) A primal maximization model with ≤ constraints converts to a _____ minimization model with constraints.

38) The quantity values on the right-hand-side of the primal inequality constraints are the _____ coefficients in the dual.

39) If the primal problem has three constraints, then the corresponding dual problem will have three _____.

40) Whereas the maximization primal model has ≥ constraints, the _____ dual model has constraints.

41) _____ in linear programming is when a basic variable takes on a value of zero. (i.e. a zero in the right hand side of the constraints of the tableau)

42) In a _____ problem, artificial variables are assigned a very high cost.

43) A(n) _____ problem can be identified in the simplex procedure when it is not possible to select a pivot row.

44) The _____ form of a linear program is used to determine how much one should pay for additional resources.

45) To determine the sensitivity range for the coefficient of a variable in the objective function, calculations are performed such that all values in the $c_j - z_j$ row are _____.

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

46) Given the following linear programming problem:

maximize $4x_1 + 3x_2$

subject to $4x_1 + 3x_2 \leq 23$

$5x_1 \quad x_2 \leq 5$

$x_1, x_2 \geq 0$

What are the basic variables in the initial tableau?

47) Given the following linear programming problem:

maximize $4x_1 + 3x_2$

subject to $4x_1 + 3x_2 \leq 23$

$5x_1 - x_2 \leq 5$

$x_1, x_2 \geq 0$

What are the Cj values for the basic variables?

48) Given the following linear programming problem:

maximize $4x_1 + 3x_2$

subject to $4x_1 + 3x_2 \leq 23$

$5x1 - x2 \leq 5$

$x1, x2 \geq 0$

What is the $(C_j - Z_j)$ value for S_1 at the initial solution?

69) Consider the following linear programming problem:

$$\begin{aligned} \text{MAX} \quad & Z = 10x_1 + 30x_2 \\ s.t. \quad & 4x_1 + 6x_2 \le 12 \\ & 8x_1 + 4x_2 \le 16 \end{aligned}$$

Use the two tables below to create the initial tableau and perform 1 pivot.

c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
	z_j					
	$c_j - z_j$					

c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
	z_j					
	$c_j - z_j$					

70) Consider the following linear programming problem and the corresponding final tableau.

MAX $Z = 3x_1 + 5x_2$
s.t. $x_1 \le 4$
$2x_2 \le 12$
$3x_1 + 2x_2 \ge 18$

			3	5	0	0	0	-M
c_j	*Basic Variables*	Quantity	x_1	x_2	s_1	s_2	s_3	a_3
3	x_1	4	1	0	1	0	0	0
0	s_3	6	0	0	3	1	1	-1
5	x_2	6	0	1	0	½	0	0
	zj	42	3	5	3	5/2	0	0
	cj-zj		0	0	-3	-5/2	0	-M

What is the shadow price for each constraint?

71) Consider the following linear programming problem and the corresponding final tableau.

MAX $Z = 3x_1 + 5x_2$
s.t. $x_1 \le 4$
$2x_2 \le 12$
$3x_1 + 2x_2 \ge 18$

			3	5	0	0	0	-M
c_j	*Basic Variables*	Quantity	x_1	x_2	s_1	s_2	s_3	a_3
3	x_1	4	1	0	1	0	0	0
0	s_3	6	0	0	3	1	1	-1
5	x_2	6	0	1	0	½	0	0
	zj	42	3	5	3	5/2	0	0
	cj-zj		0	0	-3	-5/2	0	-M

What is the sensitivity range for the first constraint?

72) Write the dual form of the following linear program.

$$\begin{aligned} \text{MAX} \quad & Z = 3x_1 + 5x_2 \\ \textit{s.t.} \quad & x_1 \le 4 \\ & 2x_2 \le 12 \\ & 3x_1 + 2x_2 \ge 18 \end{aligned}$$

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

73) The simplex method _____ be used to solve quadratic programming problems.
A) cannot B) can C) may D) should

74) The simplex method is a general mathematical solution technique for solving _____ programming problems.
A) integer B) linear C) non-linear D) A, B, and C

75) Slack variables are added to _____ constraints and represent unused resources.
A) > B) ≥ C) < D) = E) ≤

76) The _____ step in solving a linear programming model manually with the simplex method is to convert the model into standard form.
A) only B) second C) first D) last

77) Row operations are used to solve simultaneous equations where equations are _____ by constants and added to or subtracted from each other.
A) converted B) divided C) restrained D) multiplied

78) The basic feasible solution in the initial simplex tableau is the origin where all decision variables equal _____.
A) 1 or -1 B) 1 C) 0 D) -1

79) At the initial basic feasible solution at the origin, only slack variables have a value greater than _____.
A) 1 or 1 B) 1 C) 0 D) 1

80) At the initial basic feasible solution at the origin, only _____ variables have a value greater than zero.
A) non linear B) linear C) integer D) slack

81) The leaving variable is determined by _____ the quantity values _____ the pivot column values and selecting the minimum possible value or zero.
A) adding, to B) dividing, by C) subtracting, from D) multiplying, by

82) The leaving variable is determined by dividing the quantity values by the pivot column values and selecting the _____.
A) maximum positive value B) minimum positive value
C) minimum negative value D) maximum negative value

83) The simplex method _____ guarantee integer solutions.
A) sometimes does B) may C) does not D) does

84) For a maximization linear programming problem, a _____ is _____ for a less-than-or-equal to constraint.
A) artificial, subtracted
B) artificial, added
C) slack, added
D) surplus, subtracted
E) surplus, added

85) The objective function coefficient of an artificial variable for a minimization linear programming problem is:
A) 0
B) -M
C) +M
D) 1
E) an arbitrary value between 0 and positive infinity

86) If a slack variable has a positive value (is basic) in the optimal solution to a linear programming problem, then the shadow price of the associated constraint:
A) is always zero
B) is always greater than zero
C) is always less than zero
D) could be any value (i.e. zero greater than zero or less than zero)

87) In the simplex procedure, if $c_j - z_j = 0$ for a nonbasic variable, this indicates that
A) the solution is infeasible
B) the formulation is incorrect
C) there are multiple optimal solutions
D) the solution is unbounded

88) In the simplex procedure, if it is not possible to select a pivot row, this indicates that
A) the solution is infeasible
B) the solution is unbounded
C) the formulation is incorrect
D) there are multiple optimal solutions

89) In the simplex procedure, if all $c_j - z_j \leq 0$ and one or more of the basic variables are artificial, this indicates that
A) the formulation is incorrect
B) there are multiple optimal solutions
C) the solution is infeasible
D) the solution is unbounded

90) The _____ form of a linear program is used to determine how much one should pay for additional resources.
A) standard B) feasible C) simplex D) dual E) primal

chapter 5
Transportation, Transshipment, and Assignment Problems

TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

1) Transportation and assignment problems are really linear programming techniques called network flow problems.

2) Transportation models may be used when a firm is trying to decide where to locate a new facility.

3) A typical transportation problem may ask the question, "How many of X should be shipped to point E from source A?"

4) The objective of a transportation problem solution is to schedule shipments from sources to destinations while minimizing total transportation and production costs.

5) In a transportation problem, each destination must be supplied by one and only one source.

6) In a transportation problem, a single source may supply something to all destinations.

7) In finding the maximum quantity that can be shipped on the least costly route using the stepping-stone method, one examines the closed path of plus and minus signs drawn and selects the smallest number found in those squares containing minus signs.

8) In using the stepping-stone method, the path can turn at any box or cell that is unoccupied.

9) Using the stepping-stone method to solve a maximization problem, we would choose the route with the largest positive improvement index.

10) One of the advantages of the stepping-stone method is that if, at a particular iteration, we accidentally choose a route that is not the best, the only penalty is to perform additional iterations.

11) A "balanced problem" exists in a transportation model when the optimal solution has the same amount being shipped over all paths that have any positive shipment.

12) It is possible to find an optimal solution to a transportation problem that is degenerate.

13) A solution to the transportation problem can become degenerate at any iteration.

14) The transportation algorithm can be used to solve both minimization problems and maximization problems.

15) Assignment problems involve determining the most efficient assignment of people to projects, salesmen to territories, contracts to bidders, and so on.

16) The objective of an assignment problem solution most often is to minimize the total costs or time of performing the assigned tasks.

17) In the assignment problem, the costs for a dummy row will be equal to the lowest cost of the column for each respective cell in that row.

18) The Hungarian method is designed to solve transportation problems efficiently.

19) Maximization assignment problems can easily be converted to minimization problems by subtracting each rating from the largest rating in the table.

20) In a transportation problem, a dummy source is given a zero cost, while in an assignment problem, a dummy source is given a very high cost.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Table 9-1

	To==>	1	2	3	Supply
From	A	| 3 20	| 6 20	| 4	40
	B	| 3	| 4 30	| 5	30
	C	| 5	| 7 20	| 6 10	30
	Demand	20	70	10	

21) What is the total cost represented by the solution shown in Table 9-1?
A) 500
B) 60
C) 2600
D) 2500
E) None of the above

22) What is the value of the improvement index for cell B1 shown in Table 9 1?
A) +1
B) +2
C) +3
D) 50
E) None of the above

Table 9-2

	To==>	1	2	3	Supply
From	A	| 3 20	| 6 30	| 3	50
	B	| 4	| 4 40	| 3	40
	C	| 5	| 7 10	| 6 15	25
	Demand	20	80	15	

23) In Table 9-2, cell A3 should be selected to be filled in the next solution. If this was selected as the cell to be filled, and the next solution was found using the appropriate stepping-stone path, how many units would be assigned to this cell?

A) 15
B) 30
C) 20
D) 10
E) None of the above

Table 9-3

From	To ==>	1	2	3	Dummy	Supply
	A	10	8	12	0	
			80		20	100
	B	6	7	4	0	
		120		30		150
	C	10	9	6	0	
				170	80	250
	Demand	120	80	200	100	

The following improvements are proved for Table 9-3:

Cell	Improvement Index
A1	+2
A3	+6
B2	+1
B- Dummy	+2
C1	+2
C2	+1

24) The cell improvement indices for Table 9-3 suggest that the optimal solution has been found. Based on this solution, how many units would actually be sent from source C?
A) 170
B) 250
C) 10
D) 180
E) None of the above

25) In Table 9-3, suppose shipping cost from source C to point 2 was 8, which below would be true?
A) There would be multiple optimal solutions.
B) The minimum possible total cost would decrease.
C) Another dummy column would be needed.
D) The minimum possible total cost would increase.
E) None of the above

26) Both transportation and assignment problems are members of a category of LP techniques called _____.
A) supply and demand problems
B) Hungarian problems
C) network flow problems
D) transshipment problems
E) source-destination problems